POLARIZING CORPORATIONS: DOES TALENT FLOW TO "GOOD" FIRMS?

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Abstract. We conduct a field experiment with the largest job platform in Brazil to study how environmental, social, and governance (ESG) practices of firms affect talent allocation. We find that job-seekers value ESG, with stronger preferences among highly educated, white, and politically liberal individuals. We combine these estimates with matched employer-employee data and a firm survey to estimate an equilibrium model of the labor market with endogenous ESG adoption. Skilled workers reallocate toward productive ESG-adopting firms, raising output while widening wage inequality. Worker welfare rises through the direct amenity benefits of firms' ESG adoption and equilibrium adjustments in wages.

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1. Introduction

The past decade has seen a tremendous rise in the pressure corporations face to do "good" not just to shareholders, but to society at large. This shift has brought the environmental, social, and governance (ESG) practices of corporations into the spotlight (Bénabou and Tirole, 2010; Hart and Zingales, 2017). Recent heightened attention to corporate social responsibility, such as that seen in the U.S. following the 2024 presidential election, further highlights a growing divide, as some firms publicly scale back their ESG commitments while others reaffirm them.[1] In parallel, a growing body of academic work examines how employees consider nonmonetary attributes—such as an organization's mission and values—when making employment decisions (Ashraf and Bandiera, 2018; Cassar and Meier, 2018; Mas and Pallais, 2020). These developments raise important questions about how ESG signals influence job-seeker preferences and labor market outcomes.

In this paper, we study whether and how corporate ESG practices affect talent allocation, and specifically whether different aspects of ESG are attractive to different types of job-seekers. Unlike other nonwage amenities, corporate ESG practices often extend beyond direct employee benefits to reflect broader organizational commitments to society. While there is a growing body of work showing that various stakeholders of corporations care about specific aspects of ESG (e.g., diversity) and that certain nonwage amenities (e.g., work-from-home policies) are valued by workers, little is known about which ESG practices matter to which types of job-seekers and, crucially, about the efficiency and distributional consequences of ESG on the allocation of talent in the labor market.

We study these questions in the context of Brazil, where we are able to combine: (i) a field experiment to estimate job-seekers' preferences for different job and firm characteristics, with an emphasis on ESG; (ii) matched employer-employee administrative data on the entire formal sector, combined with a new survey of firm-level ESG; and (iii) a structural model to evaluate the quantitative impact of ESG preferences on the labor market equilibrium.

We begin by motivating our analysis with a survey conducted among 1,067 firms of various sizes. The survey provides evidence on current and intended ESG practices of firms, key ESG drivers, and adoption barriers. Two important facts emerge from the survey. First, firm owners identify talent attraction and retention as an important benefit of ESG adoption, along with other factors such as alignment with firm values and regulatory compliance. Second, actual and intended ESG adoption is heterogeneous across firms. Larger, higher-paying firms, as well as those with a more educated workforce, are more likely to invest in ESG.

[1] See, for example, discussions on corporate diversity, equity, and inclusion (DEI) initiatives among major U.S. firms, as covered in "Trump's D.E.I. Order Creates 'Fear and Confusion' Among Corporate Leaders" (The New York Times, 2025) and "Google Kills Diversity Hiring Targets" (The Wall Street Journal, 2025).

Our firm survey suggests that workers may value the ESG practices of their employers. To study this question directly, we conducted a field experiment in collaboration with Catho, a leading job-matching platform in Brazil. Our experiment is inspired by recent developments in the literature on labor market discrimination, and specifically by the nondeceptive incentivized resume rating approach proposed by Kessler et al. (2019). We design a variation of that approach, where Catho invites job-seekers to rate a set of job postings under the real incentive that Catho will match them to a curated set of active job postings in their database using an artificial intelligence algorithm that filters the best matches based on their preferences. There is no deception, as the job postings that job-seekers rate are designed to appear realistic, but respondents are aware that they are synthetic postings created by our team. Importantly, this design gives us complete flexibility in creating job postings and allows us to orthogonalize ESG disclosures or certifications against other firm characteristics that a worker might value. Since job-seekers receive no other incentive to participate, this design provides a robust methodology to estimate individual preferences for ESG via simple regressions of the rating on an indicator for whether the job posting displays a positive ESG signal.

Our main result is that job-seekers have a strong preference for ESG. We find that on average job-seekers value the ESG signal as equivalent to 8.04% of average wages. Not surprisingly, our respondents also have a preference for higher-paying jobs, as well as for jobs offering more nonwage amenities (such as transportation allowances, among many others). In terms of relative magnitude, ESG signals are comparable to private pension plans and more important than many other nonwage amenities—including working for a multinational company, various food and medical allowances, and professional development programs, among others—in eliciting interest from job-seekers. ESG signals are about 60% as valued as work-from-home arrangements.

Our design also allows us to unpack what job-seekers respond to when they see ESG information. Because we vary both the form and the content of ESG signals, we can distinguish responses to firms' own ESG statements from responses to external certifications, and responses to environmental practices from responses to social and governance practices. We find that workers respond to both textual ESG descriptions and external ESG certifications, though the certification effects are larger. We also find that ESG preferences are driven primarily by *environmental* practices: signals of firms' *social* and *governance* practices have little impact on job-seeker preferences. Textual analysis of responses to an open-ended question reinforces this interpretation, showing that environmental concerns dominate what respondents associate with working for companies with ESG practices in place.

Central to our analysis, we also examine the heterogeneity in ESG preferences across the sociodemographic spectrum. We find that preferences for ESG are concentrated among

highly educated, white, and politically liberal or moderate individuals. In wage-equivalent terms, highly educated job-seekers value firm ESG activities at 9.25% of their mean wage, compared with 3.11% among less-educated job-seekers. We do not find meaningful differences in ESG preferences by gender or age.

We complement these results with a series of additional tests suggesting that job-seekers respond to ESG as a signal of firms' broader environmental and societal commitments, rather than as a proxy for financial strength, workplace culture, or coworker fit. On average, job-seekers do not respond to the financial-strength signals randomized in the postings, and the estimated ESG effects are not attenuated by the presence of financial-strength signals or by job attributes that proxy for workplace climate. Open-ended responses from the original experiment, together with evidence from a supplemental survey of Brazilian job-seekers, reinforce this interpretation: respondents primarily associate ESG with environmental and broader societal impact.

We also address several measurement concerns that could affect the interpretation of ESG preferences in wage-equivalent terms. These include how participants interpret missing ESG information, whether posted wages differ from expected negotiated wages, whether higher wages reduce perceived hiring probabilities, and whether ESG signals are unusually visually salient in the experimental postings. Evidence from the original experiment and the supplemental survey suggests that these concerns are unlikely to account for our results.

Motivated by the heterogeneity in workers' ESG preferences and firms' adoption decisions, we develop a structural equilibrium model of the labor market. Workers differ in their skills and in how much they value ESG, with skilled workers placing a substantially higher value on ESG activities than unskilled workers. They choose among employers based on wages, ESG certifications, other nonwage amenities, and idiosyncratic preferences for particular firms. Firms differ in their overall productivity, in the relative productivity of skilled and unskilled workers, and in the nonwage amenities they offer. Firms decide whether to obtain an ESG certification, incurring both fixed and scale-dependent costs, and subsequently set wages for each skill group. Because workers do not regard employers as perfect substitutes, firms face upward-sloping labor supply curves and have monopsony power, allowing them to pay wages below workers' marginal products.

An important implication of the model is that ESG can affect aggregate production even though ESG activities do not directly enter firms' production functions. In the absence of ESG preferences, the equilibrium allocation of labor need not maximize labor value added. Workers' preferences over employers and firms' monopsony power can generate differences in marginal products of labor across firms, creating scope for productivity gains from labor reallocation. If more productive firms are more likely to adopt ESG, the introduction of ESG preferences can attract workers toward these firms, particularly skilled workers who value

ESG more strongly. When these firms also have higher marginal products of skilled labor in the initial equilibrium, the resulting reallocation reduces existing marginal-product gaps and raises aggregate production. Whether this mechanism operates, and its quantitative importance, ultimately depends on the initial allocation of workers, the pattern of ESG adoption across firms, and the relative productivity of different worker groups.

We combine our experimental estimates with matched employer-employee administrative data to estimate the model's labor supply and demand parameters. The experiment identifies workers' valuation of ESG and the firm-level labor supply elasticity, illustrating how experimental evidence can help discipline quantitative equilibrium models. We then recover valuations of other nonwage amenities as structural residuals that rationalize observed firm employment shares. On the demand side, we calibrate returns to scale using prior work and use observed wages and employment to recover firm-level total factor productivity (TFP) and skilled-worker productivity multipliers.

Our estimates of worker preferences for firm ESG activities are highly consistent with our previous reduced-form results. Skilled workers value firm ESG activities as equivalent to a 0.152-point increase in the log wage, while unskilled workers value firm ESG activities as equivalent to a 0.019-point increase in the log wage. Consistent with previous studies, the estimation of our structural model also reveals a positive correlation between firm TFP and the firm-specific skilled productivity multiplier. Skilled workers are more productive at high-TFP firms, leading to equilibrium assortative matching between skilled workers and productive firms.

Using our structural estimates, we evaluate how ESG preferences reshape the labor market equilibrium and affect worker welfare. Our central counterfactual compares a baseline economy in which workers do not value firms' ESG activities to an economy in which worker ESG preferences are set equal to our experimental estimates. This exercise can be interpreted as capturing a shift in workers' valuations of ESG, potentially reflecting greater awareness of firms' environmental and societal practices. Given our estimates, this corresponds to introducing a substantial ESG amenity value for skilled workers, while leaving the valuation of unskilled workers close to zero. The resulting effects on worker welfare depend on the extent to which firms adopt ESG, as well as any associated equilibrium adjustments in wages and worker sorting across firms.

A key input to this exercise is firms' endogenous adoption of ESG. Since direct estimates of ESG adoption costs are unavailable, we discipline these costs using our firm survey. The survey elicits firms' reported likelihood of making future ESG investments, along with information on wages and employment for skilled and unskilled workers. We use these data to predict ESG adoption across the firm clusters in the structural model and calibrate the adoption cost parameters to match these predicted probabilities. The resulting adoption

pattern is heterogeneous: higher-TFP firms are more likely to adopt ESG, consistent with the patterns in the firm survey.

The counterfactual results imply that ESG preferences have economically meaningful equilibrium effects. Depending on the definition of ESG adoption, introducing ESG preferences raises annual labor value added and the total wage bill by 5–6 basis points. Because more productive firms are more likely to adopt ESG, skilled workers, who value ESG more strongly, sort toward these firms. In the estimated economy, this reallocation reduces existing differences in marginal products of labor across firms, improving allocative efficiency and increasing aggregate production. The resulting wage gains accrue disproportionately to skilled workers. Quantitatively, average skilled wages rise by 12–15 basis points, compared with 1–2 basis points for unskilled workers, increasing the skilled–unskilled wage differential by 29–36 basis points.

These wage effects reflect general equilibrium adjustments. While a partial-equilibrium intuition might suggest that firms offering a valued ESG amenity could lower skilled wages while continuing to attract skilled workers, equilibrium wage effects depend on how ESG adoption varies across firms. If all firms adopted ESG, the amenity would no longer differentiate employers, and ESG preferences would have no effect on labor allocation or equilibrium wages.

Depending on the definition of ESG adoption, average worker welfare is 0.61–0.83% higher in wage-equivalent terms in the economy with ESG preferences. These differences are unevenly distributed: welfare is 2.15–2.93% higher for skilled workers, compared with 0.20–0.28% for unskilled workers. These welfare differences reflect both the direct amenity value workers derive from ESG-adopting firms and the indirect effects of equilibrium adjustments in wages and worker sorting. Our decomposition attributes 95–96% of the total welfare difference to direct amenity benefits and the remaining 4–5% to equilibrium adjustments in wages and worker sorting.

Our findings contribute to three broad strands of literature. First, our study speaks to a growing literature on the role of organizational culture, mission, and values in shaping the workplace. Much of the work centers around the impact of pecuniary versus nonpecuniary incentives on applicant traits and subsequent performance within traditionally "mission-oriented" organizations, such as NGOs and public sector organizations (Ashraf et al., 2014; Spenkuch et al., 2023).[2] A recent wave of studies has explored the link between corporate and personal values and various worker and business outcomes in the private sector (Hussam

[2]Dal Bó et al. (2013) demonstrate that higher wages can help attract both high-ability and motivated applicants for civil service jobs in Mexico, while Deserranno (2019) examines the signaling effect of financial incentives on recruiting NGO workers in Uganda, finding that financial incentives deterred candidates with strong prosocial preferences from applying. Similarly, Ashraf et al. (2020) examine how emphasizing career prospects versus community contributions in job postings affects the selection and performance of healthcare workers in Zambia.

et al., 2022; Ashraf et al., 2023).[3] A subset of this literature studies the impact of organizational values on worker selection and sorting across firms.[4] Burbano (2016) and Burbano (2021) find that virtual workers on MTurk and Elance set lower reservation wages and are willing to do extra work for jobs at firms that provide information about their charitable activities. Hedblom et al. (2022) use a field experiment with data-entry workers in the U.S. and a structural model to discuss the labor selection and productivity effects of a firm's charitable practices.[5] We estimate preferences for ESG among a broad set of private-sector workers in a new setting in Brazil. Importantly, by combining a new firm survey, a randomized experimental survey of workers, a structural model, and rich matched employer-employee administrative microdata, we provide, to the best of our knowledge, the first quantitative estimates of how corporate engagement with social and environmental values affects labor market equilibrium and distributional outcomes. Along these lines, we build on the literature that uses experimental estimates to identify structural models and show how surveys can also be used to flexibly discipline endogenous decisions that are otherwise difficult to calibrate but play a key role in shaping equilibrium outcomes of interest (DellaVigna and Pope, 2018; Lagakos and Shu, 2023; Colonnelli et al., 2026).[6]

Second, we contribute to the rapidly growing literature on ESG (see Gillan et al. 2021 and Christensen et al. 2021 for reviews), which has predominantly focused on the relationship between ESG and investment decisions. Several studies have shown that investors take into account firms' ESG activities when making investment and fundraising decisions, in part due to the presence of nonpecuniary motives.[7] Several papers look at firms' decisions to become greener to access cheaper sources of capital (Broccardo et al., 2022; Edmans et al., 2023; Hartzmark and Shue, 2023; Oehmke and Opp, 2025) and to attract customers (see Leonidou et al. 2013 for a review), while work on how ESG impacts the actions of other stakeholders remains more limited (Kitzmueller and Shimshack, 2012; Colonnelli et al., 2024a). We

[3]See, among others, Edmans (2011), Guiso et al. (2015), Gartenberg et al. (2019), Li et al. (2021), Graham et al. (2022), Rice and Schiller (2022), Edmans et al. (2024) and Huang et al. (2026) for studies suggesting a positive correlation between organizational culture and business outcomes.

[4]Abraham and Burbano (2022) and Burbano et al. (2024) investigate the role of gender in preferences for meaning at work and related consequences for organizational structure. Colonnelli et al. (2025b) show that workers match with business owners sharing their same political views. Adrjan et al. (2023) study the impact of public announcements of socially and politically polarizing corporate policies on job-seeker interest and employee satisfaction. See Bond and Glode (2014) and Ferreira and Nikolowa (2025) for models of firms and workers with nonpecuniary preferences.

[5]See LaViers and Sandvik (2022) and Choi et al. (2023) for similar studies on the role of diversity.

[6]Examples range from work on charitable giving (DellaVigna et al., 2012), rural infrastructure and labor market integration (Brooks and Donovan, 2020), gift exchange and social preferences at work (DellaVigna et al., 2022), and rural–urban migration in low-income countries (Lagakos et al., 2023). See Whited (2023) for a discussion on structural methods in empirical finance.

[7]Examples include Hong and Kacperczyk (2009), Riedl and Smeets (2017), Hartzmark and Sussman (2019), Pedersen et al. (2021), Pastor et al. (2022), Pastor et al. (2021), Van der Beck (2026), Zhang (2022), and Gormsen et al. (2023).

provide a direct estimate of the effect of ESG on a key set of stakeholders: workers. We document the strongest effects for corporate environmental practices, which is consistent with recent evidence by Krueger et al. (2023) that workers in Sweden earn substantially lower wages in more environmentally sustainable firms. Our findings therefore provide direct causal evidence on a potential quantitatively meaningful motive behind firms' decisions to invest in ESG: to attract and retain talent. Our paper is related to, but distinct from, Colonnelli et al. (2025a), who use a different set of results from the same experiment to document that ESG practices are uniquely polarizing, exhibiting substantial heterogeneity in job-seeker preferences across sociodemographic groups relative to other nonwage amenities. In contrast, we study how ESG affects talent allocation from both sides of the labor market by combining evidence on job-seekers' preferences with firms' ESG adoption decisions to examine equilibrium outcomes. A key contribution of our work is to quantify the aggregate and distributional consequences of ESG for worker sorting, wage inequality, allocative efficiency, and worker welfare.

Finally, we speak to the labor literature on nonwage amenities dating back to the theoretical contributions by Rosen (1974, 1986) on compensating wage differentials.[8] Our structural modeling approach is based on work by Bhaskar et al. (2002), Manning (2013), Card et al. (2018), and Lamadon et al. (2022), who study how worker heterogeneity, nonwage amenities, and vertical and horizontal differentiation between employers impact firm monopsony power, earnings inequality, and employer rents in the U.S. A large body of empirical work, including field experiments, shows that employees value nonwage amenities such as work flexibility (Mas and Pallais, 2017; He et al., 2021; Maestas et al., 2023), job stability (Wiswall and Zafar, 2018), and fringe benefits (Eriksson and Kristensen, 2014).[9] After the COVID-19 pandemic, several studies find that employees value work-from-home policies (Barrero et al., 2021; Adrjan et al., 2021; Aksoy et al., 2022). We extend this literature by providing experimental evidence on job-seekers' valuation of ESG as a distinct nonwage amenity reflecting broader organizational commitments to sustainability, governance, and social responsibility, and by shedding new light on the role of sociodemographic heterogeneity.

This paper is organized as follows. Section 2 motivates our study using our survey on firms' ESG practices. Section 3 details our experimental design. Section 4 reports the experimental results. Section 5 introduces our model and structural estimation to quantify the impact of ESG. Section 6 concludes.

[8]See Mas (2025) for a review of the work on nonwage amenities. For a review of the literature on firms, earnings inequality, worker sorting, and compensating differentials, see Card et al. (2018), Sorkin (2018), and Taber and Vejlin (2020), among others.

[9]See Mas and Pallais (2020) for a review of the literature on alternative work arrangements.

2. A New Survey of Firm-Level ESG Adoption

To motivate and better contextualize our analysis, we first discuss the results of a new descriptive survey on ESG practices among Brazilian firms that we conducted in July 2023. The objective of the survey was to better understand firms' knowledge of ESG, current ESG practices, motivations for ESG adoption, primary adoption barriers, and their intentions for ESG adoption in the near future. We also leverage this survey subsequently in our structural work.

The survey begins with an introduction outlining its purpose, verifying whether the respondent is the firm owner, confirming firm size, and obtaining consent. We then collect data on the firm's industry, establishment year, and location. We focus on firms with more than 10 employees (which are typically the organizations that engage with ESG initiatives and advertise job openings on major job platforms), and we aim to be broadly representative of the Brazilian private sector in terms of location and industry. Next, we provide an overview of ESG and assess respondents' prior knowledge of ESG, the extent to which their firm has implemented ESG practices, perceived benefits and challenges of ESG adoption, and awareness of a major ESG certification such as B Corp. We then ask respondents to estimate the likelihood of their company implementing various ESG practices. The survey covers the three ESG categories—environmental, social, and governance practices—by asking respondents to select the two most relevant practices for businesses similar to their own within each category. For each selected practice, we outline specific criteria required to achieve strong ESG performance (see Appendix Tables H27, H28, and H29 for criteria by ESG category). The respondents then assess the likelihood that their company will make an investment to meet these criteria within the next 1–3 years. After that, we provide an overview of B Corp and similarly inquire about the probability of making an investment to attain the certification within the next 1–3 years. Finally, we conclude by gathering additional information on the firm's employment and wages for skilled and unskilled employees, among other questions.

Our sampling frame relied on the firm panel of the commercial market research company Dynata. We obtained a total of 1,067 responses from firm owners. The median company in our ESG survey has 100 employees, has been in business for 13 years, and employs 66% college graduates. The most common industries of our surveyed firms are information and communication (17.24%), manufacturing (11.15%), and retail (10.50%) (Appendix Table A1). The median time for respondents to complete the survey is 20 minutes.

We start by establishing that firms are knowledgeable about ESG and the B Corp certification. As shown in Figure 1 Panel A (under "Current ESG Adoption"), the median firm rates its self-reported knowledge and understanding of ESG a 4 out of 5. More specifically, 81% state they are currently implementing some form of ESG practices, and 41% indicate they are already "extensively" implementing ESG in their operations. As discussed in detail

later in this paper, our structural estimation benefits from an understanding of firms' future ESG investment plans. As shown in Figure 1 Panel A (under "Future ESG Implementation Likelihood"), across the four measures—environmental, social, governance, and B Corp—respondents on average signal a strong likelihood (around 70% or higher) of achieving high ESG standards within three years.

In Figure 1 Panel B, we show that respondents perceive the main barriers to ESG adoption to be competing priorities (40%), limited human capital (30%), and cost constraints (29%). In Panel C, we find respondents identify the primary benefits of ESG adoption to be alignment with firm values (59%), enhanced reputation and brand value (53%), regulatory compliance (40%), and—directly relevant to this paper—attracting and retaining talent (27%), which appears more relevant than access to finance and risk management considerations.

Next, we examine which types of firms adopt ESG practices. To analyze heterogeneity in ESG adoption, we estimate OLS regressions of the following form:

$$\begin{aligned} \textit{ESG Adoption}_j = \alpha &+ \beta_1 \ln(\textit{Number of Employees}_j) + \beta_2 \ln(\textit{Average Wage}_j) \\ &+ \beta_3 \textit{Employees with College Degree (\%)}_j + \textit{Industry FE} \\ &+ \textit{State FE} + \epsilon_j, \end{aligned} \tag{2.1}$$

where j denotes the jth firm. $\textit{ESG Adoption}_j$ is a measure of the firm's self-reported current ESG adoption status or the firm's stated likelihood of future ESG adoption. $\ln(\textit{Number of Employees})$ is the natural logarithm of the firm's total number of employees. $\ln(\textit{Average Wage})$ is the natural logarithm of the average wage at the firm. *Employees with College Degree (%)* is the percentage of employees who have a college degree. We also control for industry and state fixed effects and use robust standard errors (Abadie et al., 2023).

In Table 1, we report the corresponding regression results. Columns (1) and (2) correspond to firms' self-reported current ESG adoption status, where $\textit{ESG Adoption}_j$ is an indicator variable equal to one if the firm has already implemented ESG practices (Column (1)) or has extensively implemented ESG practices (Column (2)). For both measures, we find that firms with more employees, higher average wages, and a greater proportion of employees with a college degree are more likely to have adopted ESG practices. Columns (3) to (6) correspond to firms' expectations of future ESG adoption, where $\textit{ESG Adoption}_j$ represents the firm's stated likelihood (from 0% to 100%) of making a financial investment to meet ESG standards within the next 1–3 years. Once again, we find that larger, higher-paying firms with a more educated workforce are more likely to adopt good environmental (Column (3)),

social (Column (4)), and governance practices (Column (5)), as well as pursue a B Corp certification (Column (6)).[10]

Overall, our descriptive evidence highlights how firm owners in Brazil identify talent attraction and retention as an important benefit of adopting ESG practices and how adoption of ESG is heterogeneous across firms.

3. The Job Rating Experiment

In this section, we describe the experimental survey we conducted in collaboration with Catho to estimate job-seekers' preferences for firms' ESG practices. In Section 3.1, we provide an overview of the experimental design. In Section 3.2, we describe the key features of the synthetic job postings. In Section 3.3, we detail the two main questions we ask job-seekers in their evaluation of job postings. Section 3.4 describes the matching to real job vacancies.

3.1. **Experimental Survey Design.** Our experimental survey is inspired by the nondeceptive incentivized resume rating design proposed by Kessler et al. (2019), which aims to estimate preferences (in their case, employers' preferences for resume characteristics) while avoiding deception.[11] In our context, we collaborate with a leading job-matching platform in Brazil—Catho (https://www.catho.com.br)—to invite job-seekers to report their interest in a set of synthetic job postings, whose components—corporate ESG practices, wages, nonwage amenities, among others—are fully randomized by our research team. There is no deception involved, as respondents are aware that the job postings are hypothetical. Job-seekers have a strong incentive to respond truthfully because we inform them that their ratings will be used to match them to real job openings aligned with their preferences through a new artificial intelligence tool. For example, if a job-seeker sees a job posting indicating that the company is a certified B Corp, she would rate it highly—assuming all else equal—only if she is genuinely interested in working for B Corp-certified firms. Conversely, she would not rate a job posting highly for strong environmental practices if such factors were not important to her. In short, our incentive structure ensures that job-seekers provide accurate ratings, as doing so will maximize the value of the real job openings received.

3.1.1. *Recruitment.* Catho implemented the study. The survey targets only Catho customers, who do not receive any compensation for participating in the survey.[12] We present

[10]One potential limitation of the analysis in Table 1 is that it relies on survey responses, which may not accurately reflect firms' actual ESG practices. To address this concern, we reproduce the analysis using data we manually collected on B Corp and Great Place to Work certified companies in Brazil, which we link to our matched employer-employee administrative data (see Section 5.2). As shown in Appendix Table A2, we find similar results: larger, higher-paying firms with more educated workforces are more likely to be certified.

[11]See Macchi (2023); Colonnelli et al. (2024b); Low (2024) for applications of this design in a variety of settings, and Harrison and List (2004) for a broader discussion of "framed field experiments."

[12]We only target job-seekers labeled as "engaged" by Catho, namely those who had opened Catho's emails within the past 60 days of our experiment and who are actively looking for employment.

the survey tool as a new artificial intelligence solution designed to assist Catho in suggesting the most suitable jobs for every job-seeker. We report the email script used for recruitment in Appendix Figure A2.

From September to November 2022, Catho sent four waves of survey invitation emails to a subset of active users on their platform. We received 238, 255, 337, and 422 responses, respectively, for a total of 1,252 responses.[13] We excluded respondents who took fewer than 8 minutes or more than 2 hours to complete the survey, resulting in a final sample of 1,206 responses.[14] Our respondents are located across all areas of Brazil, as shown in Appendix Figure A1.

Column (1) of Table 2 reports summary statistics of the sociodemographic characteristics of the survey participants. The table shows that 42.95% of respondents are female, the median age is 42 years, 51.91% identify as white, the median monthly wage is BRL 2,750, and 56.3% have attained a four-year college degree or higher. Columns (2) and (3) of Table 2 present the same information using the latest available data from RAIS (2020) and PNAD (2022).[15] Overall, the demographics of our survey respondents broadly resemble those of the Brazilian formal labor market with respect to gender, race, and age. However, our survey sample is characterized by individuals with somewhat higher wages and higher levels of education. In our subsequent analysis, we address these differences in sample composition by re-weighting the experimental estimates.

3.1.2. *Survey Structure.* We document the structure of our experimental survey in Appendix Figure A3 and provide the exact survey text in Appendix D. The survey begins by outlining the purpose of the survey, incentives to participate, and confirming consent to proceed. We then provide instructions on how to evaluate job postings using the 1–7 scale rating system. We also provide a brief definition of ESG practices and mention that companies can signal their ESG practices in job postings.

We next ask respondents a set of "filtering" questions on their level of education and their preferred professional area. We use these responses to avoid showing job postings that do not elicit any interest from the respondent for mechanical reasons, such as unsuitable job titles

[13]For confidentiality reasons and since Catho was responsible for disseminating emails to their clients, we are unable to observe the pool of individuals who received the survey email but did not participate in the study. Catho informed us that the response rate was approximately 5%.

[14]In Appendix Table A3, we show our results are robust to using the entire raw sample without filtering responses based on completion time.

[15]The Brazilian Ministry of Labor's RAIS database provides employee-employer matched data from the formal sector economy. PNAD (*Pesquisa Nacional por Amostra de Domicílios*) is a large-scale, nationally representative survey conducted quarterly by the Brazilian Institute of Geography and Statistics (IBGE) (Instituto Brasileiro de Geografia e Estatística, 2022), designed to be representative of the entire Brazilian population. It provides detailed information on the socioeconomic characteristics of the respondents, including employment status, wage, education level, and other demographics. PNAD offers accurate information on both the formal and informal labor markets (Rocha et al., 2018).

or prerequisites. This prevents, for example, a respondent in the Commercial and Sales area from being shown a veterinary or architectural position, while preserving random assignment of job attributes within each group. We then ask job-seekers to rate 20 unique, synthetic job postings, whose contents are discussed in greater detail in the next subsection. Finally, following the job ratings, we ask respondents about their demographic and socioeconomic characteristics, as well as their views on working for companies with ESG practices.

3.2. **Creating Synthetic Job Postings.** To construct the synthetic job postings, we first conducted a structured manual review of 1,000 randomly selected real job postings from Catho's platform. We analyzed the components of typical job postings, focusing on their content and visual layout, and estimated the probability distribution of each component. With these components and probabilities defined, we then generated synthetic job postings by randomizing components based on their estimated inclusion probabilities.

Job postings typically consist of a few main categories: primary job characteristics, general firm characteristics, general job characteristics, job prerequisites, hiring stages, and nonwage amenities. ESG characteristics are typically included as part of the general firm characteristics, but we include and randomize them independently. In Appendix Figures A4, A5, A6, and A7, we provide examples of synthetic job postings. In the subsections below, we provide additional details on all job posting categories. A summary of job categories and components, with respective inclusion probabilities, is reported in Appendix Table A4. The full material used to create the synthetic job postings is reported in Appendix H.

3.2.1. *Primary Job Characteristics.* At the beginning of each job posting, we include four key components: (i) job title; (ii) location; (iii) wage; and (iv) contract type. Based on their responses to the filtering questions detailed in Section 3.1.2, respondents are shown relevant job titles and select at least one of interest. Respondents also select their preferred city and state for work, with an option to select additional cities. Job postings always display job title and location.[16] We always include the wage at the top of each job posting. To ensure realism, the wage follows different distributions based on respondents' education level and selected professional area (see Appendix Table H5 for details). We also include the type of contract (the Brazilian "work regime") with probability 1/2.

3.2.2. *General Firm Characteristics.* Following the primary job characteristics, we include a brief description of the employer. Drawing inspiration from actual employer descriptions found on firm websites and social media channels, we developed several versions of realistic employer profiles. We randomize the firm's sector, age, number of employees, number

[16]In Appendix Tables H1, H2, and H3, we tabulate the job titles available to respondents based on their level of education and professional area. In Appendix Table H4, we report the complete list of cities by state.

of countries in which it operates, and financial strength. We report details on each firm characteristic in Appendix Tables H6 to H11.

3.2.3. *Firm ESG Characteristics.* For a subset of job postings, we provide information about the firm's ESG activities in two independent ways: (i) ESG signaling sentences and (ii) third-party ESG certifications. There is a 26% unconditional probability that at least one ESG sentence is displayed (with an equal probability that one or two sentences are shown) and a 10% probability that an ESG certification is displayed. ESG signaling sentences highlight the company's efforts in relation to a specific ESG practice. We developed the signaling sentences based on real-world examples of firm ESG statements (e.g., on websites and job postings). We randomize firms' ESG sentences across the following three categories: (i) *environmental* practices, covering topics related to emissions, recycling, land footprint, waste, and energy; (ii) *social* practices, featuring diversity & inclusion and professional development; and (iii) *governance* practices, covering anti-lobbying, anti-bribery and anti-corruption, and whistleblowing. In Appendix Table H12, we provide a comprehensive list of ESG sentences shown to respondents.

For third-party ESG certifications, we include the certification logo and a descriptive sentence based on similar statements made by certified corporations. We randomly select one of three common real-world ESG certifications: *B Corp*, *Great Place to Work*, and *Green Business Bureau*. We provide additional details on these ESG certifications in Appendix Table H13. If a job posting does not include ESG information, we include an "auxiliary sentence" (that is, a filler sentence) to match the approximate length of job postings containing ESG information (Appendix Table H14). This ensures that our experimental estimates do not pick up spurious effects related to textual length.

3.2.4. *General Job Characteristics and Job Prerequisites.* Next, we independently randomize several general job characteristics: (i) on-the-job opportunities; (ii) on-the-job activities; (iii) workload; and (iv) work-from-home arrangements. We also always include a sentence about the job opening to provide additional structure to the job posting. We designed the job prerequisites to be sufficiently broad to not discourage respondents. When prerequisites are displayed, we also include major requirements for all respondents who have completed college.[17]

3.2.5. *Hiring Stages.* Next, we specify the hiring stages for the position. We select the hiring stages by randomizing from the following categories: (i) application, (ii) online assessments, (iii) other assessments, and (iv) final interview (see Appendix Table H25 for details).

[17]See Appendix Tables H15 to H24 for all these details.

3.2.6. *Nonwage Amenities.* Finally, we add several nonwage amenities to each job posting. We classify nonwage amenities into two categories: (i) "amenities," which includes all non-monetary nonwage amenities, such as wellness programs or office gyms; and (ii) "benefits," which consists of all monetary nonwage amenities, such as food or transportation allowances. We randomly draw between 2 and 4 nonwage amenities. For a description of all nonwage amenities, see Appendix Table H26.

3.3. **Rating Jobs.** We measure job-seekers' interest in specific job and firm characteristics by asking respondents to evaluate a random set of 20 synthetic job postings, randomly drawn from the pool of all possible job postings generated by our content randomization process. A key advantage of our experimental methodology is that we obtain more granular measures of job-seekers' preferences compared to correspondence study approaches that only rely on call-back rates (Kessler et al., 2019). We use a 7-point Likert scale to measure the rating, which allows us to observe job-seekers' preferences for the characteristics of inframarginal job postings. Our main dependent variable is the response to the question: *"How interested would you be in receiving an offer for this job position?"*

We measure the response on a scale of 1 to 7, where 1="Moderately interested" and 7="Dream job!" We indicate the responses to this question as *Interest*, which represents our main dependent variable, to capture how interested a job-seeker is in a given job posting. We also directly specify: "Imagine that the employer guarantees you a job offer—consider only your perception of the quality of the position." This allows us to isolate the job-seeker's interest in the job posting from their perceived hiring chances.

We then ask an additional question to further motivate job-seekers to focus strictly on their interest in the job posting of the given employer when answering the main question. Beyond its incentive effects, this additional question also allows us to explore job-seekers' perceptions about the likelihood of receiving an offer from an employer. The question asks *"How likely do you think it is that the company will offer you the position?"* We again measure the response on a scale of 1 to 7, where 1 = "Not likely" and 7 = "Extremely likely." We also specify: "Imagine that you applied for the job—consider only whether you think the employer would make you an offer based on your qualifications and experience."

3.4. **Match Delivery.** To deliver the incentive described to participants, we translated the experimentally estimated preferences into a recommendation rule for real vacancies on Catho's platform. We estimated preferences over job-posting attributes and converted statistically significant coefficients into wage-equivalent values. These values were added to the posted wages to form a group-specific score, which Catho received to rank the vacancies within the relevant city and professional area. Appendix G describes the scoring rule and the job characteristics used.

4. Estimating Preferences for ESG

In this section, we describe our experimental results. In Sections 4.1 and 4.2 we report our main empirical specifications and results. In Sections 4.3, 4.4, and 4.5 we report additional results on heterogeneity, mechanisms, magnitudes, and robustness.

4.1. **Estimating Equations.**

We run regressions of the following form:

$$\begin{aligned} Interest_{ij} = \alpha + \beta_1 ESG_{ij} + \beta_2 \ln(Wage_{ij}) + \beta_3 NWA_{ij} + \beta_4 FS_{ij} \\ + X_{ij}\Gamma + Strata\ FE + \epsilon_{ij}, \end{aligned} \tag{4.1}$$

where i indicates the job-seeker responding to the survey and j indicates the synthetic job posting that is evaluated. *Interest* is our main dependent variable, indicating the level of interest, rated from 1-7, that a respondent has in a given job posting. The main parameter of interest is β_1, which measures the impact of ESG information on job-seeker preferences. Specifically, *ESG* is a binary indicator equal to one if the job posting contains an ESG signaling sentence or a third-party ESG certification, as detailed in Section 3.2.3.

The parameters β_2, β_3, and β_4 capture the average preferences of job-seekers for wages, other nonwage amenities, and employer financial strength, respectively. The variable ln *(Wage)* is defined as the natural logarithm of the monthly wage shown on the job posting. *NWA* is the number of nonwage amenities. *FS* is a binary indicator equal to one if the job posting contains information that signals strong financial performance by the firm. We also include a vector X_{ij} of additional job-posting controls, which accounts for the number of on-the-job activities, number of on-the-job opportunities, employer industry, employer establishment year, number of job prerequisites, an indicator for whether the job is not located in the respondent's primary chosen city, an indicator for work-from-home arrangements, and an indicator for whether the employer is a multinational.

Since randomization probabilities vary by respondents' education level and preferred professional area, we also control for these strata fixed effects to account for variation in the randomization probabilities across respondents.[18]

4.2. **Average Job-Seeker Preferences.**

We report our main experimental results in Table 3. In particular, we show regression results where the dependent variable is *Interest*, which measures the job-seeker's interest in job postings on a scale of 1-7.[19] We use heteroskedasticity-robust standard errors for statistical inference.

[18]Specifically, we define strata as the interaction of education level and professional area and include fixed effects for each stratum. These fixed effects are not absorbed by the individual fixed effects, as each respondent can select multiple preferred professional areas.

[19]We report the distribution of interest scores in Appendix Figure A8.

We uncover the presence of a large *responsible firm premium*. In Column (1), we report a simple bivariate regression of job-seeker interest on ESG. The *ESG* coefficient is positive and statistically significant at the 1% level, indicating that job-seekers prefer to work for companies that signal their ESG practices in job postings. This result remains robust when we control for job-posting characteristics (Column (2)) and when we include individual fixed effects (Column (3)). Appendix Figure A14 shows our estimates are stable across all combinations of sociodemographic controls. Since our experimental sample is tilted toward more-educated workers, in Column (4) we present estimates re-weighted using the nationally representative PNAD household survey. The re-weighted ESG coefficient is modestly smaller in magnitude, consistent with the subsequent heterogeneity in preference we document, but remains statistically significant and economically meaningful. In Column (5), we quantify the average ESG preference in monetary terms by including the wage in *levels* (in BRL). We find that, on average, ESG signals elicit the same marginal interest in a job posting as approximately a BRL 346 increase in monthly salary, equivalent to 8.04% (12.59%) of the mean (median) monthly wage of our survey respondents. The same increase in wage is equivalent to 12.06% (18.67%) of the mean (median) monthly wage in the formal sector.

As expected, respondents also prefer higher-paying jobs and jobs with more nonwage amenities, as indicated by the positive coefficients on ln *(Wage)* and *NWA*. Furthermore, using the independent cross-randomization of other job attributes such as work-from-home policies, we can precisely benchmark the impact of ESG signals on job-seekers' interest to that of other nonwage amenities and firm characteristics.

In Panel A of Figure 2, we segment the point estimates into four categories: ESG signals, work-from-home arrangements, nonwage amenities, and multinational status. Panel B provides a more granular comparison of ESG signals and nonwage amenities. Our results underscore that, on average, job-seekers place greater value on ESG signals than on most other nonwage amenities. Work-from-home arrangements are highly attractive to job-seekers. ESG signals elicit about 60% as much interest among job-seekers as work-from-home arrangements. We also estimate that transportation allowances and partnerships with educational institutions elicit greater interest than certain ESG signals. ESG signals are comparable in magnitude to food allowances or private pension plans, but are valued more highly than the majority of other nonwage amenities, including working for a multinational company, medical and educational assistance, and professional development programs.[20]

4.3. **Heterogeneity Across Sociodemographic Groups.** While we have documented an economically meaningful average preference for ESG, an important goal of our paper is to

[20]Appendix Table A10 shows that the results are robust to excluding job postings with potentially uncommon combinations of attributes. Specifically, we exclude postings with extreme wages, implausible wage–location or firm-size combinations, and unusual ESG or amenity content.

understand the implications of ESG for talent allocation in the labor market. In Table 4, we therefore explore whether job-seekers' preferences for ESG vary across sociodemographic groups. Column (1) partitions respondents into those with a college degree and otherwise, Column (2) into white and non-white, Column (3) into politically liberal or moderate and otherwise, Column (4) into those above and below the age of 45, and Column (5) into male and female. We then interact *ESG* with the demographic indicators and control for strata and individual fixed effects.

In Columns (1) to (3) of Table 4, the baseline *ESG* coefficient is close to zero and statistically insignificant, indicating that job-seekers with lower levels of formal education, non-white individuals, and those who identify as politically conservative show little preference for firm ESG activities. Conversely, the coefficient on the interaction term is positive and statistically significant at the 5% level or lower, indicating that ESG preferences are strong among individuals who are highly educated, white, and who self-identify as politically liberal or moderate. In Columns (4) and (5), we find little evidence of differences in ESG preferences across age and gender, with the interaction term close to zero and statistically insignificant.

In Panels A and B of Appendix Table A11, we subsample based on demographic splits and report ESG preferences in BRL terms. Highly educated job-seekers value firm ESG activities at 542 BRL, equivalent to 9.25% of their mean wage. Less-educated respondents value ESG at 72 BRL, equivalent to 3.11% of their mean wage and statistically indistinguishable from zero. White (non-white) job-seekers value firm ESG activities at 481 (193) BRL. Politically liberal and moderate (conservative) respondents value ESG at 494 (212) BRL.[21]

4.4. **Certification Effects and Unpacking ESG.** We have shown that job-seekers value ESG signals on average, and that these preferences are especially pronounced among highly educated, white, and politically liberal or moderate respondents. These estimates treat ESG information as a single signal, however, even though our design varies both how ESG information is conveyed and what type of ESG practice it describes. We therefore first examine whether job-seekers respond differently to firms' ESG statements versus third-party certifications. Columns (1) and (2) of Appendix Table A5 show that job-seekers respond to both textual ESG sentences and ESG certifications. However, the estimated coefficient is somewhat larger for certifications, suggesting that job-seekers may respond more strongly to ESG signals validated by external organizations than to firms' own descriptions of ESG practices.

This distinction matters because "ESG" is not a single firm attribute. Rather, it is a broad label that groups together conceptually different practices: environmental commitments,

[21]In Appendix Table A17, we show that ESG preferences are larger among workers with higher predicted AKM individual fixed effects. We construct this index using RAIS employee-employer matched data by regressing AKM individual fixed effects on the demographic characteristics in our experimental job-seeker sample (see Appendix Table A16).

social initiatives such as diversity and community impact, and governance practices such as executive compensation and risk management. We therefore next unpack the ESG signal into its E, S, and G components to assess which dimensions drive job-seekers' interest.

We report this decomposition in Panel B of Figure 2 and Columns (3) and (4) of Appendix Table A5. We document meaningful heterogeneity in job-seekers' preferences. Only ESG sentences that emphasize firm environmental practices generate positive and statistically significant effects, while sentences that emphasize social and governance practices have coefficients close to zero.

These results are consistent with open-ended textual responses of our experiment's participants to a question posed after their job ratings. Specifically, respondents were asked "When you think of working for companies with Environmental, Social, and Governance (ESG) practices in place, what are the main considerations that come to mind?" In Appendix Figure A9, we generate word clouds for the most common words and bigrams. The word clouds show that "environment" and "care environment" dominate what comes to respondents' minds when thinking about ESG in this context.

4.5. **Alternative Interpretations, Measurement Concerns, and External Validity.** The evidence so far suggests that job-seekers value ESG signals, especially environmental signals, and that these preferences are concentrated among specific sociodemographic groups. We next address whether this interpretation reflects genuine preferences over firms' ESG activities, rather than alternative signals embedded in ESG language. We also address potential measurement concerns related to quantifying ESG preferences in wage-equivalent terms and concerns related to external validity.

Some of the evidence below comes from a new, qualitative survey we conducted in May 2026. We obtained responses from 497 Brazilian job-seekers through our commercial survey partner Dynata. The survey is described in more detail in Appendix E, and its main goal was to provide additional suggestive evidence on mechanisms.

4.5.1. *Alternative Interpretations.* We first address whether job-seekers respond to ESG information because of societal concerns that reflect their own values, or whether they instead use ESG as a proxy for other nonwage amenities they may care about. For example, job-seekers may wish to work for financially strong firms because they offer greater job security, and may infer that ESG activities are disproportionately undertaken by such firms. Workers might also infer from ESG activities that the firm has a healthier workplace culture or more desirable coworkers.

We conduct several tests that argue against such alternative interpretations of our results. First, as shown in Table 3 and Appendix Table A7, we do not find a statistically significant effect on respondents' job ratings for the financial strength indicator. Moreover, if respondents interpret ESG as a signal of firm financial stability, then ESG should add

less information when a financial strength signal is also displayed in the same posting, thus producing a negative interaction between ESG and financial strength. Appendix Table A13 reports this test in the full sample and separately by demographic groups. The interaction is small and statistically insignificant across all columns.

Next, if job-seekers use ESG to infer positive workplace culture, one might expect the Great Place to Work (GPTW) certification to have particularly strong effects. However, when we exclude all job postings that featured a GPTW certification and re-estimate our main specification, reported in Column (3) of Appendix Table A8 Panel B, the ESG point estimate of 0.074 is very similar in magnitude and statistical significance to the baseline. Appendix Table A9 further shows that the ESG effect is not attenuated by the presence in the job posting of attributes that proxy for workplace climate, including on-the-job opportunities related to company culture and wellness programs.

The open-ended responses from our main experiment point in the same direction: when respondents describe what comes to mind when thinking about ESG firms, they overwhelmingly mention environmental themes, with little reference to colleagues, workplace ideology, cultural fit, or financial stability. We further validate this interpretation in our May 2026 survey. Respondents were shown job postings from the same experimental pool and asked how they interpreted the ESG signals. As shown in Appendix Figure A10, the most common interpretation across ESG signals is that the firm has broader environmental or societal commitments, selected by 25–66% of respondents depending on the signal. By contrast, relatively few respondents interpret ESG as primarily conveying employees' shared values, coworkers, cultural fit, or financial strength. The GPTW certification provides a useful benchmark: unlike the other ESG signals, it is most often interpreted as indicating better benefits or employee treatment. Open-ended responses in the same survey reinforce this pattern. As shown in Appendix Figure A11, LLM classifications of respondents' explanations again show that ESG signals are most often interpreted as broader environmental or societal commitments, while references to coworkers or cultural fit are uncommon.

4.5.2. *Measurement Concerns.* We next discuss potential concerns related to the measurement of ESG preferences in wage-equivalent terms, which requires measuring the sensitivity of job-seekers to both ESG activities and wages correctly.

One concern is that applicants might assume that firms not mentioning ESG still follow such practices.[22] In that case, job-seekers may interpret the absence of an ESG signal as a noisy or incomplete signal rather than a true absence, implying that our estimates may be a lower bound on the strength of the true ESG preference. With this caveat in mind,

[22]Matching the universe of firms certified as B Corp and Great Place to Work (GPTW) prior to 2023 in Brazil to LinkedIn data, we find that only 25.43% of GPTW-certified firms explicitly mention their certification, but 72.34% of B Corp firms do mention it.

our experimental design mitigates this concern by explicitly instructing respondents with the following: "When no information related to ESG practices is shown, it means that the company does not have an ESG system in place."

Second, posted wages may differ from final negotiated wages. If respondents, especially highly educated respondents, treat posted wages as lower bounds, posted pay may be a noisy measure of expected compensation, which would overstate the wage-equivalent value of ESG. In Brazil, jobs offered under the CLT regime are subject to more standardized compensation practices compared to PJ contractor arrangements. Hence, a negotiation channel predicts stronger responses to posted wages in CLT jobs. Column (3) of Appendix Table A18 does not support this prediction: the interaction between log wage and the CLT indicator is small and statistically insignificant. We also find little evidence of stronger wage responses in sectors with high collective bargaining coverage (de Holanda Barbosa Filho and Peruchetti, 2025), as shown in Column (5). We collect additional supportive evidence that wage negotiation is unlikely to be a first-order driver of the magnitudes of our effects in our May 2026 survey of job-seekers. As shown in Appendix Figure A12, when asked "How close was the pay eventually discussed during the hiring process to the amount shown in the job posting?", the majority of respondents reported that the pay was about the same, and those who mentioned it changed were broadly split between higher and lower pay.

A third concern is that we may overstate ESG preferences if workers under-react to high wages due to a perceived lower likelihood of being hired, perhaps due to perceived higher competition. First, as noted in Section 3.3, our instructions directed participants to evaluate the job without reference to the likelihood of being hired. Moreover, several tests argue against this. In our supplemental May 2026 survey, respondents were asked how they interpret a posting offering a salary above the average for a similar role (Appendix Figure A13). 33% of the respondents interpret above-average pay as a sign of greater selectivity and lower hiring chances, but a larger 52% share of the participants interpret above-average pay as a sign that the employer is having difficulty hiring. The remaining 14% report no clear interpretation. More directly, Appendix Table A6 from our primary experiment shows that higher wages increase the perceived likelihood of being hired.

A final concern is that ESG information may have been unusually salient because certification logos were visually prominent or because ESG information occupied a prominent place in the posting. Several pieces of evidence argue against this interpretation. First, Appendix Table A5 shows that respondents react not only to certification logos, but also to textual ESG sentences. Second, respondents distinguish sharply across ESG content: the effects are concentrated in environmental sentences, while social and governance sentences generate coefficients close to zero. To probe salience more directly, in the May 2026 survey we showed respondents job postings constructed from the same templates used in the main experiment

and then asked which posting elements they recalled. ESG sentences and certifications were recalled at rates comparable to other prominent posting elements, such as hiring stages, work arrangements, and company information. Taken together, this pattern is difficult to reconcile with a pure salience explanation.[23]

4.5.3. *External Validity.* Our experiment is conducted in Brazil, so the estimates should be interpreted as specific to that labor market. Nevertheless, the sociodemographic heterogeneity in ESG preferences is consistent with broader cross-country patterns. Using existing survey data, Appendix F shows that college-educated individuals tend to report stronger prosocial and pro-environmental preferences, and that the relationship between education and these outcomes in Brazil is similar to that observed in other countries.

5. Quantitative Impact of ESG on Equilibrium Labor Outcomes

Motivated by our firm survey and experimental reduced-form results, we now use an equilibrium model to quantify how workers' ESG preferences affect labor market sorting, wages, aggregate labor value added, and worker welfare. The model features workers who are heterogeneous in preferences and productivity, firms that are heterogeneous in productivity and nonwage amenities, monopsonistic competition, and endogenous ESG adoption. The model allows us to distinguish the direct utility workers derive from ESG adoption from the indirect effects of equilibrium adjustments in employment and wages.

We discipline the model using our randomized job-rating experiment, matched employer-employee administrative data, and firm survey. We then use the estimated structural model to perform counterfactual simulations that quantify the consequences of shifts in workers' preferences for firm ESG activities.

5.1. **Model of Labor Market Equilibrium and Firm ESG Adoption.** We consider a labor market with monopsonistic competition, heterogeneous firms, and heterogeneous workers. Workers differ in both productivity and preferences. Their productivity depends not only on their demographic group, but also on the firm at which they are employed. Their preferences depend on wages, firm ESG certifications, other nonwage amenities, and idiosyncratic tastes over employers. Firms differ in productivity, in the productivity of different worker groups, and in the nonwage amenities they offer. They also decide endogenously whether to obtain a costly ESG certification, while we take other nonwage firm characteristics as given.

Time is divided into two subperiods. In subperiod t_0, firms observe their productivities and nonwage amenities and decide whether to adopt ESG. In subperiod t_1, the labor market

[23]In various unreported tests, we also show that our estimates remain largely consistent across various splits of the data where ESG salience may vary (e.g., depending on the number of ESG signals displayed, or the number of job postings with ESG signals).

opens: firms post wages, workers choose employers, firms hire workers, and production occurs.

5.1.1. *Workers.* The labor market consists of a mass $\bar{L}$ of workers indexed by i and a set of firms indexed by $j = 1, \ldots, J$. Each worker belongs to a demographic group $g = 0, \ldots, G$, which captures observable characteristics such as education, race, and gender. We denote the mass of workers in group g by $\bar{L}_g$. Workers supply one unit of labor inelastically and choose among firms based on wages, firm ESG activities, other nonwage amenities, and idiosyncratic preferences over employers.

The indirect utility of worker i, belonging to group $g(i)$, from working at firm j in period t is

$$u_{ijt} = \log W_{jg(i)t} + \beta_{g(i)} E_{jt} + \log X_{jg(i)t} + \tau_{g(i)} \varepsilon_{ijt}. \tag{5.1}$$

Here, $W_{jg(i)t}$ is the wage paid by firm j to workers in group $g(i)$, and E_{jt} is an indicator equal to one if firm j has an ESG certification. The parameter β_g captures the log-wage-equivalent utility value that workers in group g place on firm ESG activities. This allows ESG preferences to differ across demographic groups, which is central to our empirical findings. The term X_{jgt} is a wage-equivalent index of firm j's other nonwage amenities for workers in group g, excluding ESG. Finally, ε_{ijt} is an iid Type-1 Extreme Value shock capturing worker-specific horizontal preferences over firms, and τ_g governs the dispersion of these idiosyncratic preferences.

Workers observe posted wages and choose the firm that maximizes utility:

$$j_t(i) = \arg\max_j u_{ijt}. \tag{5.2}$$

Firms commit to hiring any worker willing to work at the posted wage. Wages may vary by worker group, but firms cannot condition wages on the idiosyncratic taste shock ε_{ijt}, which is private information to the worker.

The Type-1 Extreme Value assumption implies the following logit choice probability:

$$\mathrm{P}(j_t(i) = j) = \frac{\left[W_{jg(i)t} \exp\left(\beta_{g(i)} E_{jt}\right) X_{jg(i)t}\right]^{1/\tau_{g(i)}}}{\sum_{j'} \left[W_{j'g(i)t} \exp\left(\beta_{g(i)} E_{j't}\right) X_{j'g(i)t}\right]^{1/\tau_{g(i)}}}. \tag{5.3}$$

Thus, higher wages, valued ESG activities, and better nonwage amenities all increase the probability that a worker chooses a given firm. The idiosyncratic preference term generates firm monopsony power. A firm can lower its wage relative to an otherwise identical competitor without losing all of its workers. The parameter τ_g controls the strength of this force. Lower τ_g corresponds to less dispersed idiosyncratic preferences and a more elastic residual labor supply curve.[24]

[24]Differences in ESG certifications and nonwage amenities lead to equilibrium wage differentials, but do not themselves create firm monopsony power.

5.1.2. *Firms.* Firms differ in their ESG certification decisions E_{jt}, non-ESG amenities X_{jgt}, total factor productivity Ξ_{jt}, and group-specific productivity multipliers A_{jg}. Let L_{jgt} denote employment of group g at firm j in period t. Firm j's revenue value added, defined as revenue net of intermediate input expenditures, is

$$Y_{jt} = \Xi_{jt} L_{jt}^{1-\eta}, \tag{5.4}$$

where

$$L_{jt} = \sum_g A_{jg} L_{jgt} \tag{5.5}$$

is the firm's total efficiency units of labor. The parameter A_{jg} captures the productivity of workers in group g at firm j. The productivity of a worker thus depends on both her demographic group and the specific firm at which she works. The parameter η governs decreasing returns to labor, so that $1 - \eta$ is the elasticity of value added with respect to efficiency units of labor.

Firms compete monopsonistically for workers. When setting wages, each firm takes the market wage index as given. Combining worker choice probabilities from equation (5.3) above with the mass of workers in each group yields the residual labor supply curve facing firm j for group g:

$$L_{jgt}(W_{jgt}) = \frac{\bar{L}_g \left[W_{jgt} \exp(\beta_g E_{jt}) X_{jgt}\right]^{1/\tau_g}}{\Delta_{gt}}, \tag{5.6}$$

where

$$\Delta_{gt} = \sum_{j'} \left[W_{j'gt} \exp(\beta_g E_{j't}) X_{j'gt}\right]^{1/\tau_g} \tag{5.7}$$

is the market wage index for group g, taken as given by individual firms. The residual labor supply elasticity for group g is therefore $\sigma_g \equiv 1/\tau_g$.

Given these labor supply curves, firm j chooses wages to maximize profits:

$$\Pi_{jt} = \Xi_{jt} \left[\sum_g A_{jg} L_{jgt}\right]^{1-\eta} - \sum_g W_{jgt} L_{jgt}. \tag{5.8}$$

The first-order condition for the wage paid to group g is

$$(1 + \sigma_g) W_{jgt} = \sigma_g (1 - \eta) \Xi_{jt} A_{jg} L_{jt}^{-\eta}. \tag{5.9}$$

Thus, equilibrium wages are a fraction $\sigma_g/(1 + \sigma_g)$ of the marginal product of group-g labor, with smaller markdowns when labor supply is more elastic. Because this markdown is common across firms within each worker group, wage differences across firms also reflect differences in marginal products for that group.

5.1.3. *Endogenous ESG Adoption.* We model ESG certification as a nonwage amenity that workers may value but that does not directly affect firms' production technology. In subperiod t_0, firm j observes its total factor productivity Ξ_{jt}, group-specific productivity multipliers A_{jg}, and non-ESG amenities X_{jgt}. It then decides whether to obtain an ESG certification, anticipating the effects of adoption on equilibrium wages, employment, and profits when the labor market opens in subperiod t_1.

Let $Y_{jt,1}$ and $Y_{jt,0}$ denote the output of firm j with and without ESG adoption, respectively, and let $\Pi_{jt,1}$ and $\Pi_{jt,0}$ denote the corresponding operating profits before ESG adoption costs.[25] We assume that the cost of ESG adoption is

$$C_{jt} = F + cY_{jt,1}^{\nu} + \rho Y_{jt,1}\zeta_{jt}. \tag{5.10}$$

This specification allows ESG adoption costs to have three components. The parameter F captures a fixed cost of adoption. The term $cY_{jt,1}^{\nu}$ captures a scale-dependent variable cost, allowing the cost of adoption to rise flexibly with firm output. Finally, $\rho Y_{jt,1}\zeta_{jt}$ is an idiosyncratic adoption cost shock, where ζ_{jt} is iid and follows a logistic distribution. The term $\rho Y_{jt,1}$ allows the dispersion of idiosyncratic adoption costs to vary with firm scale.

Firm j adopts ESG if the profit gain from adoption $\Pi_{jt,1} - \Pi_{jt,0}$ exceeds the adoption cost C_{jt}. Given the logistic distribution of ζ_{jt}, the probability that firm j adopts ESG in period t is

$$\mathrm{P}(E_{jt} = 1) = \frac{1}{1 + \exp\left(\frac{\Pi_{jt,0} - \left(\Pi_{jt,1} - F - cY_{jt,1}^{\nu}\right)}{\rho Y_{jt,1}}\right)}. \tag{5.11}$$

This expression shows that firms are more likely to adopt ESG when adoption raises operating profits by more or when adoption costs are lower. In our counterfactual analysis, we use the firm survey to discipline the cost parameters, matching both the average level of ESG adoption and its heterogeneity across firms.

5.1.4. *Equilibrium and Worker Utility.* An equilibrium in period t consists of worker choices $j_t(i)$, wages W_{jgt}, employment allocations L_{jgt}, market wage indexes Δ_{gt}, and ESG adoption decisions E_{jt} such that the following conditions hold:

(1) Workers choose firms to maximize utility, generating the choice probabilities in equation (5.3).
(2) Firms set wages optimally, taking the residual labor supply curves and market wage indexes as given. Wages satisfy the first-order condition in equation (5.9), and employment satisfies $L_{jgt} = L_{jgt}(W_{jgt})$.

[25] Strictly speaking, since the number of firms is finite, subperiod t_1 outputs and profits are random at the time of the period-t_0 adoption decision. Each firm cluster will contain many firms, however, so realized adoption shares within a cluster are well approximated by their model-implied probabilities. We therefore treat the equilibrium outputs and profits entering the adoption decision as deterministic, invoking a law-of-large-numbers approximation.

(3) Market wage indexes Δ_{gt}, as defined by equation (5.7), are internally consistent and generated from workers' optimal decisions.
(4) Firms have rational expectations and choose ESG adoption optimally according to equation (5.11).

We use the model to evaluate changes in worker welfare across counterfactual equilibria. Given the Type-1 Extreme Value distribution of idiosyncratic preferences, ex-ante worker welfare is given by the standard log-sum expression:

$$U_t = \sum_g \bar{L}_g \tau_g \log \left[\sum_j \exp \left(\frac{\log W_{jgt} + \beta_g E_{jt} + \log X_{jgt}}{\tau_g} \right) \right]. \tag{5.12}$$

This object is the expected utility of workers before the realization of their idiosyncratic taste shocks. It summarizes the value workers receive from the equilibrium distribution of wages, ESG activities, and other nonwage amenities across firms.

5.1.5. *Impact of Shifts in ESG Preferences.* Motivated by our experimental findings, we compare a baseline economy in which $\beta_g = 0$ for every worker group with an economy in which ESG preferences take their experimentally estimated values. The shift can be interpreted as a change in workers' valuations of ESG, potentially reflecting greater awareness of firms' environmental and societal practices. It affects worker welfare through two channels. Directly, workers employed by ESG-adopting firms receive an amenity benefit, whose magnitude depends on their preferences and the extent of adoption. Indirectly, heterogeneous adoption changes the relative attractiveness of employers, inducing adjustments in worker sorting and equilibrium wages.

Because ESG does not enter the production function, any effect on aggregate labor value added must operate through changes in labor allocation. In the baseline economy, the allocation of workers need not maximize labor value added. Firms face upward-sloping labor supply curves and mark wages down below marginal products, which need not be equalized across firms. To illustrate, suppose that the group-specific productivity multiplier A_{jg} does not vary across firms. When high-TFP firms do not offer sufficiently greater non-ESG amenities X_{jgt}, they have higher equilibrium marginal products than low-TFP firms.[26] Reallocating workers toward these firms would increase aggregate labor value added.

ESG preferences can induce such a reallocation when adoption is concentrated among firms with higher marginal products of labor. High-TFP firms are more likely to adopt when fixed adoption costs are important and variable costs are not too convex, since these firms can spread fixed costs over a larger scale of production. Because skilled workers place a

[26]Suppose this were not the case. Since wages, by equation (5.9), are a constant markdown of marginal products within each worker group, low-TFP firms would offer (weakly) higher wages. Under the assumed restrictions on X_{jgt}, they would then attract (weakly) more workers, implying higher marginal products at high-TFP firms, a contradiction.

higher value on ESG, they respond more strongly to adoption. If the resulting sorting draws skilled workers toward firms with higher marginal products of skilled labor, aggregate labor value added rises. Equilibrium wages and the skilled–unskilled wage differential may also change.[27]

These equilibrium effects are not necessarily positive. Adoption may instead be concentrated among less productive firms if scale-dependent costs are sufficiently convex; firms with high TFP may already employ relatively large workforces because of their non-ESG amenities; and firm-specific skill productivity can reinforce or offset the effects of sorting by TFP. In contrast, workers can obtain direct ESG amenity benefits even when labor allocation and wages do not change, as when all firms adopt. The magnitudes of the direct welfare benefits and indirect equilibrium effects are therefore quantitative questions that depend on worker preferences, firms' adoption decisions, and the initial allocation of labor.

5.2. **Identification.** Having described the model, we now explain how we recover its key structural parameters. Identification combines the randomized job-rating experiment with observed wages and employment in the matched employer–employee data. Under the model's utility specification, the experiment identifies workers' valuation of ESG, β_g, and the residual labor supply elasticity, σ_g. Given these parameters, RAIS wages and employment shares allow us to recover other nonwage amenities. Conditional on the wage-setting equation and calibrated returns to scale, the same data then discipline firm TFP and the group-specific productivity multipliers. The firm survey separately disciplines the ESG adoption costs in Section 5.3.1. Thus, the direct ESG valuations are anchored in experimental variation, while the recovered productivity objects and output counterfactuals additionally rely on the model's production and wage-setting assumptions.

The Brazilian Ministry of Labor's RAIS database has nearly universal coverage of the Brazilian formal labor market and is widely considered a high-quality census of formal employment (Dix-Carneiro, 2014; Helpman et al., 2017). Following standard practice in using RAIS (Colonnelli and Prem, 2022; Bernstein et al., 2022), we keep the highest-paying job of a worker in cases where the worker is employed by more than one firm in a given year. We estimate the model with two demographic groups: skilled workers, defined as those with a college degree, and unskilled workers, defined as those without a college degree. We let $g = 0$ denote unskilled workers.

[27]When the labor supply elasticity is common across worker groups, $\sigma_g \equiv \sigma$, the aggregate wage bill equals the constant share $\frac{\sigma}{1+\sigma}(1-\eta)$ of labor value added. The estimated elasticities are approximately equal across groups.

On the production side, we calibrate the decreasing-returns parameter $\eta = 0.21$ following Lamadon et al. (2022).[28] Panels A and B of Figure 3 summarize the key RAIS moments underlying this step. Panel A plots the empirical CDF of firm employment and shows substantial heterogeneity in firm size. Panel B shows that the college share slightly declines with total employment, while both unskilled and skilled wages rise with employment. The wage-size gradient is steeper for skilled workers, so the skilled–unskilled wage differential increases with firm size.

To implement this estimation and the counterfactual analysis, we approximate the firm distribution using k-means clustering. Let z_j denote the vector of firm-level moments used in the structural estimation: skilled and unskilled employment and skilled and unskilled wages, measured in 2019. The k-means procedure partitions firms into clusters $\{\mathcal{C}_m\}_{m=1}^{M}$ with centroids $\bar{z}_m$. In the estimation and counterfactual analysis, each cluster m is represented by a firm type with wages and employment equal to the corresponding cluster averages and mass equal to the number of firms in the cluster. We then recover the structural objects $(X_{mgt}, A_{mg}, \Xi_{mt})$ at the cluster level and solve the equilibrium over these M representative firm types.

This discretization assumes that, for the purposes of the counterfactual equilibrium, within-cluster heterogeneity is well summarized by the cluster averages of wages and employment. Equivalently, the clustering variables are treated as sufficient statistics for the economically relevant dimensions of firm heterogeneity captured by the model: non-ESG amenities, group-specific productivity multipliers, firm TFP, and ESG adoption incentives. In our baseline specification, we use $M = 10$ clusters; in robustness exercises, we repeat the analysis with larger numbers of clusters and find similar quantitative results.

5.2.1. *Labor Supply.* We first use the ranking of synthetic job postings implied by each respondent's experimental ratings to estimate the dispersion of idiosyncratic preferences, τ_g, and the value of ESG certification, β_g. The former determines the residual labor supply elasticity faced by firms, $\sigma_g = 1/\tau_g$. For a given respondent in demographic group g, let $\mathcal{J}$ denote the set of job postings evaluated and let

$$\widehat{j} = (j_1, \ldots, j_{|\mathcal{J}|})$$

denote the resulting ranking, ordered from highest to lowest reported interest. Given the utility specification in equation (5.1) and the choice probabilities in equation (5.3), the

[28]Directly estimating η is challenging because effective labor is endogenous to firm productivity: more productive firms hire more workers, which would bias an OLS estimate. We therefore rely on Lamadon et al. (2022), who estimate this parameter using external and internal instruments.

probability of observing this ranking is

$$\begin{aligned} \mathrm{P}(\widehat{j} \mid g) &= \prod_{k=1}^{|\mathcal{J}|} \mathrm{P}\left(j_k = \underset{j' \in \mathcal{J} \setminus \{j_1, \ldots, j_{k-1}\}}{\operatorname{argmax}} u_{j'g} \right) \\ &= \prod_{k=1}^{|\mathcal{J}|} \frac{[W_{j_k g} \exp(\beta_g E_{j_k}) X_{j_k g}]^{1/\tau_g}}{\sum_{j' \in \mathcal{J} \setminus \{j_1, \ldots, j_{k-1}\}} [W_{j'g} \exp(\beta_g E_{j'}) X_{j'g}]^{1/\tau_g}}. \end{aligned} \tag{5.13}$$

This expression applies the logit choice probability in equation (5.3) sequentially to the ranked list of job postings. We estimate the model by maximum likelihood using the rank-ordered conditional logit.[29] The coefficient on log wages identifies $\sigma_g = 1/\tau_g$, while the coefficient on the ESG indicator identifies β_g/τ_g.

Appendix Table A12 reports the estimates. The ESG valuations are highly consistent with the reduced-form evidence in Section 4.3. Skilled workers place substantially greater value on ESG certifications than unskilled workers do. Quantitatively, skilled workers value firm ESG activities as equivalent to a 0.152-point increase in log wages, statistically significant at the 1% level. By contrast, unskilled workers value ESG as equivalent to a 0.019-point increase in log wages, which is statistically indistinguishable from zero. Both groups value other nonwage amenities, although the point estimate is larger for skilled workers.

The same estimation also recovers τ_g, and hence the firm-level labor supply elasticity $\sigma_g = 1/\tau_g$. The estimated elasticities are broadly similar across groups, with unskilled workers appearing slightly more elastic. The implied structural parameters are $\tau_g = 0.938$ for unskilled workers and $\tau_g = 1.058$ for skilled workers.

Finally, we use RAIS to recover the non-ESG amenity index X_{mgt} for firms in cluster m. In the historical RAIS data, we set $E_{mt} = 0$ for all clusters. We view this as a reasonable approximation because ESG practices were less widely documented in earlier years and were likely not salient to most workers. Normalizing $X_{m^*gt} = 1$ for a baseline cluster m^*, equation (5.3) implies

$$X_{mgt} = \left(\frac{L_{mgt}}{L_{m^*gt}} \right)^{\tau_g} \frac{W_{m^*gt}}{W_{mgt}}. \tag{5.14}$$

Thus, X_{mgt} is the structural residual that rationalizes the average per-firm employment L_{mgt} of group g workers within cluster m, given wages and the estimated labor supply elasticity.

5.2.2. *Labor Demand.* We next recover the labor demand objects at the representative firm-type level. Given the labor supply estimates from Section 5.2.1 and the calibrated decreasing-returns parameter $\eta = 0.21$, the remaining objects are the group-specific productivity multipliers A_{mg} and firm-type TFP Ξ_{mt}. We normalize the productivity of unskilled workers to

[29] See Beggs et al. (1981), Hausman and Ruud (1987), and Allison and Christakis (1994) for additional details on the rank-ordered logit. We implement the estimation using the `cmrologit` command in Stata, which is extended to handle ties.

one, $A_{m0} = 1$, and recover the productivity multiplier for skilled workers relative to unskilled workers.

Let lowercase variables denote logs, so that $a_{mg} = \log A_{mg}$ and $\xi_{mt} = \log \Xi_{mt}$. Taking logs of the firm first-order condition in equation (5.9) gives

$$w_{mgt} = c_g + \xi_{mt} + a_{mg} - \eta \ell_{mt}, \tag{5.15}$$

where

$$c_g \equiv \log \left[(1-\eta) \frac{\sigma_g}{1+\sigma_g} \right], \qquad \ell_{mt} \equiv \log \left(\sum_g A_{mg} L_{mgt} \right). \tag{5.16}$$

The constant c_g is pinned down by the calibrated production parameter η and the labor supply elasticity $\sigma_g = 1/\tau_g$ estimated from the experiment. The term ℓ_{mt} is the log of effective labor for representative firm type m.

Equation (5.15) identifies the skilled productivity multiplier from within-firm-type wage differences. Since $A_{m0} = 1$, we have

$$a_{m1} = w_{m1t} - w_{m0t} - (c_1 - c_0). \tag{5.17}$$

Thus, skilled workers are inferred to be more productive at firm types that pay a larger skilled–unskilled wage premium, after accounting for differences in labor supply elasticities across groups.

Given A_{m1}, we compute log effective labor ℓ_{mt} and recover firm-type TFP from the unskilled wage equation:

$$\xi_{mt} = w_{m0t} - c_0 + \eta \ell_{mt}. \tag{5.18}$$

Equivalently, TFP is the residual that rationalizes the observed wage and employment levels of each representative firm type, given the calibrated production technology and estimated labor supply elasticities.

Panels C and D of Figure 3 summarize the recovered labor demand objects. Panel C shows substantial dispersion in TFP across the 10 representative firm types. Panel D shows that the skilled productivity multiplier is increasing in TFP: skilled workers are relatively more productive at high-TFP firms. This positive relationship is important for the counterfactual analysis because it implies that when ESG preferences induce skilled workers to sort toward high-TFP adopters, labor is reallocated toward firms where skilled labor is especially productive.

5.3. **Counterfactual Simulations.** We now turn to our counterfactual analysis. Our experiment documents substantial preferences for ESG, especially among skilled workers, raising the question of how these preferences affect equilibrium labor market outcomes. To investigate this question, we compare a baseline economy calibrated to 2019 data in which workers do not value ESG, so that $\beta_g = 0$ for all g, to an economy in which ESG preferences

are set equal to our experimental estimates. This shift can be interpreted as reflecting evolving values or greater awareness of firms' environmental and societal practices. Firms respond endogenously through their ESG adoption decisions, while workers adjust their employment choices. The exercise allows us to quantify the effects on talent allocation and worker welfare, distinguishing direct amenity benefits from indirect equilibrium effects.

5.3.1. *Endogenous ESG Adoption by Firms.* To quantify the effects of this shift in worker ESG preferences, we need to discipline the parameters governing firms' endogenous ESG adoption decisions. Direct estimates of ESG adoption costs are difficult to obtain, so we use the firm survey described in Section 2. In the survey, firms report the likelihood that they will pursue ESG activities in the future, along with wages and employment for skilled and unskilled workers. These variables are also observed in RAIS, allowing us to map survey firms into the state variables of the structural model.

We estimate separate logistic regressions predicting firms' intentions to adopt ESG, using three alternative measures: environmental practices, average ESG practices, and B Corp certification. The results are reported in Panel A of Table 5. We then use these estimates to construct predicted adoption probabilities for each firm cluster from the k-means procedure applied to the RAIS data. Finally, for each adoption measure, we choose the structural cost parameters F, c, ν, ρ to minimize the squared deviations between the model-implied ESG adoption probabilities and these survey-predicted adoption probabilities across firm clusters.

Figure 4 illustrates the calibration for B Corp certification. The model matches the survey-predicted adoption probabilities well across firm clusters and reproduces the key pattern that higher-TFP firms are more likely to adopt ESG. The figure also illustrates how the adoption-cost parameters shape this pattern. Panel A shows that increasing the fixed cost F primarily reduces adoption among low-TFP firms, since high-TFP firms operate at larger scale and can spread the fixed cost over greater output. Conversely, Panel B shows that increasing the variable cost parameter c lowers adoption probabilities more uniformly across firms.[30]

5.3.2. *Impact of ESG Preferences on Labor Market Equilibrium.* The counterfactual results are reported in Panel B of Table 5. Depending on the definition of ESG adoption, average worker welfare is 0.61–0.83% higher in wage-equivalent terms in the economy with ESG preferences. These differences are unevenly distributed: welfare is 2.15–2.93% higher for skilled workers, compared with 0.20–0.28% for unskilled workers. These welfare differences reflect two channels. First, workers derive direct amenity benefits from ESG adoption, with the magnitude depending on their valuations of ESG and the extent of adoption across firms. Second, ESG preferences induce equilibrium adjustments in worker sorting and wages, which can generate additional welfare effects.

[30]Increasing the convexity parameter ν flattens the relationship between TFP and adoption and can eventually make high-TFP firms less likely to adopt than low-TFP firms.

To quantify the relative importance of these channels, we decompose the total welfare effect into direct amenity benefits and indirect equilibrium effects. We calculate the direct effect by introducing the ESG adoption pattern from the counterfactual equilibrium while holding workers at their baseline employers and wages. Across the three definitions of ESG adoption, the probability that a worker's baseline employer adopts ESG in the counterfactual equilibrium is approximately 13–18% for skilled workers and 10–14% for unskilled workers. These baseline employment-weighted adoption probabilities, together with workers' group-specific ESG valuations, determine the magnitude of the direct amenity effect. The indirect effect is then the difference between the total welfare change and this direct effect, capturing the additional consequences of equilibrium adjustments in worker sorting and wages. Across the three definitions of ESG adoption, the direct amenity channel accounts for 95–96% of the total welfare difference (96% for skilled workers and 93–94% for unskilled workers), with equilibrium adjustments in sorting and wages accounting for the remaining 4–5%.

Beyond these welfare effects, ESG preferences also affect aggregate production and wages through the reallocation of workers across firms. Since higher-TFP firms are more likely to adopt ESG, skilled workers, who value ESG more strongly, sort toward these firms. In the estimated economy, this reallocation reduces existing differences in marginal products of labor across firms, improving allocative efficiency and raising annual labor value added by 5–6 basis points. The resulting expansion of the total wage bill accrues disproportionately to skilled workers. Average skilled wages rise by 12–15 basis points, compared with 1–2 basis points for unskilled workers, increasing the skilled–unskilled wage differential by 29–36 basis points.[31]

To further understand the quantitative implications of ESG preferences and endogenous adoption, Figure 5 shows how equilibrium outcomes vary with the fixed cost F and variable cost coefficient c of ESG adoption. For each point on the (F, c) grid, we solve for the equilibrium, allowing firms to endogenously decide whether to adopt ESG, and report outcomes relative to the baseline economy without ESG preferences. The red dot indicates the calibrated parameter values.

Panels A and B show that the shift in ESG preferences increases the skilled–unskilled wage differential by roughly 0.3–0.9% and labor value added by 0.05–0.11% across the parameter values considered. Both effects increase with the fixed cost F, even though higher fixed costs reduce overall ESG adoption. The reason is that fixed costs disproportionately discourage adoption among low-TFP firms, which have less scope to spread these costs over a large scale of production. As F rises, adoption becomes more concentrated among high-TFP firms,

[31]Partial-equilibrium logic might suggest the opposite. If skilled workers value ESG, ESG-adopting firms could lower skilled wages while continuing to attract skilled labor. This logic does not apply in general equilibrium. For example, if all firms adopted ESG, the amenity would no longer differentiate firms, leaving equilibrium wages and labor allocations unchanged.

strengthening the reallocation of skilled workers toward more productive employers. This increases labor value added and, through the resulting expansion of the wage bill, widens the skilled–unskilled wage differential. Conversely, when fixed costs are low, adoption is more uniform across firms and these equilibrium effects are muted.

Increasing the variable cost coefficient c has the opposite effect. Higher variable costs reduce adoption more broadly across firms, weakening the sorting response and the associated improvements in allocative efficiency. As a result, the effects on labor value added and the skilled–unskilled wage differential decline as c increases.

Panel C shows the corresponding effects on worker welfare. Across the parameter grid, average welfare is 0.5–1.1% higher in wage-equivalent terms in the economy with ESG preferences. Welfare is highest when adoption costs are low, because more workers receive the direct amenity benefits of ESG adoption. Increasing fixed costs can strengthen the indirect equilibrium effects by concentrating adoption among high-TFP firms, but this is outweighed by the reduction in direct amenity benefits associated with lower overall adoption. Increasing variable costs reduces welfare by limiting ESG adoption and weakening the associated equilibrium responses.

6. Conclusion

Corporate social responsibility has become a central topic in public discourse, with many large firms increasingly divided between reaffirming and scaling back their ESG commitments. These divisions raise important questions about the impacts of shifting societal expectations and corporate priorities. We examine how corporate ESG practices shape job-seeker preferences, workforce composition, and broader labor market dynamics amid growing polarization in corporate approaches to social responsibility. Our study sheds new light on the importance of organizational values in influencing job-seekers' choices and shaping the talent landscape.

Using Brazil as our setting, we make two primary contributions. First, in partnership with a leading Brazilian job platform, we design a nondeceptive incentivized field experiment to estimate job-seekers' preferences to work for socially responsible firms. We find that, on average, job-seekers place a value on ESG signals equivalent to 8.04% (12.59%) of the mean (median) monthly wage in our experimental sample. These preferences are highly heterogeneous, with skilled workers valuing firm ESG activities substantially more than unskilled workers.

Second, we combine our experimental estimates with matched employer-employee administrative data and structurally estimate an equilibrium model of the labor market with heterogeneous workers, heterogeneous firms, and endogenous firm ESG adoption. We use the model to quantify the consequences of a structural shift in (skilled) worker preferences

for ESG, relative to a baseline economy in which no workers have ESG preferences. In the estimated model, high-TFP firms are more likely to adopt ESG. This pattern reflects the structure of adoption costs: fixed costs allow high-TFP firms to take advantage of economies of scale, while the convexity of variable costs determines how quickly adoption costs rise with firm output.

Our counterfactual results indicate that ESG preferences can reshape equilibrium labor market outcomes. Because skilled workers value ESG, they sort toward high-TFP, high-wage firms that adopt ESG. This reallocation modestly improves allocative efficiency and raises labor value added. Worker welfare is also higher in the economy with ESG preferences, primarily reflecting the direct amenity benefits of ESG adoption, with smaller contributions from equilibrium adjustments in worker sorting and wages. These welfare differences accrue disproportionately to skilled workers, who place substantially greater value on ESG amenities.

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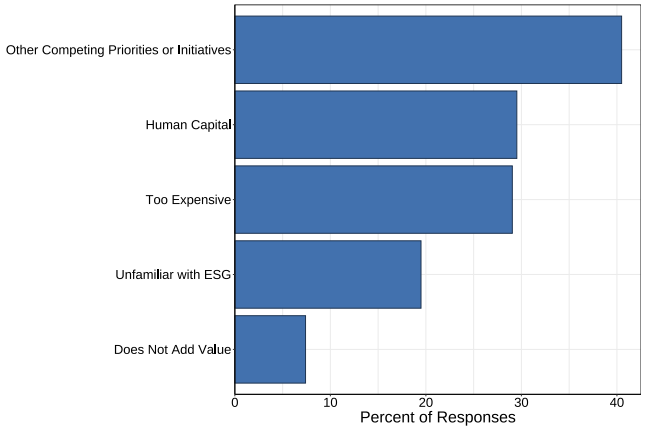
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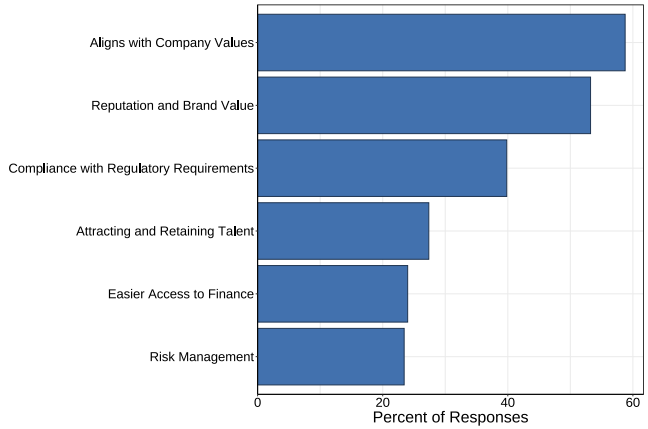
FIGURE 1. Firm Survey of ESG Practices

A. Firms' Current and Future ESG Practices

	Mean (1)	Standard Deviation (2)	10th Percentile (3)	25th Percentile (4)	Median (5)	75th Percentile (6)	90th Percentile (7)
Current ESG Adoption:							
Implemented ESG Practices (0/1=Yes)	0.81	0.39	0.00	1.00	1.00	1.00	1.00
Extensively Implemented ESG Practices (0/1=Yes)	0.41	0.49	0.00	0.00	0.00	1.00	1.00
Prior Knowledge of ESG Practices (1-Low; 5-High)	4.04	0.93	3.00	3.00	4.00	5.00	5.00
Prior Knowledge of B Corp (0-None; 1-Some; 2-Extensive)	0.99	0.67	0.00	1.00	1.00	1.00	2.00
Future ESG Implementation Likelihood:							
Likelihood of Implementing Environmental Practices (%)	70.13	25.05	32.60	54.00	75.00	90.00	100.00
Likelihood of Implementing Social Practices (%)	74.24	24.54	40.00	60.00	80.00	93.00	100.00
Likelihood of Implementing Governance Practices (%)	75.23	24.30	40.00	61.50	81.00	96.00	100.00
Likelihood of Achieving B Corp Certification (%)	73.06	25.55	31.60	59.00	80.00	94.00	100.00

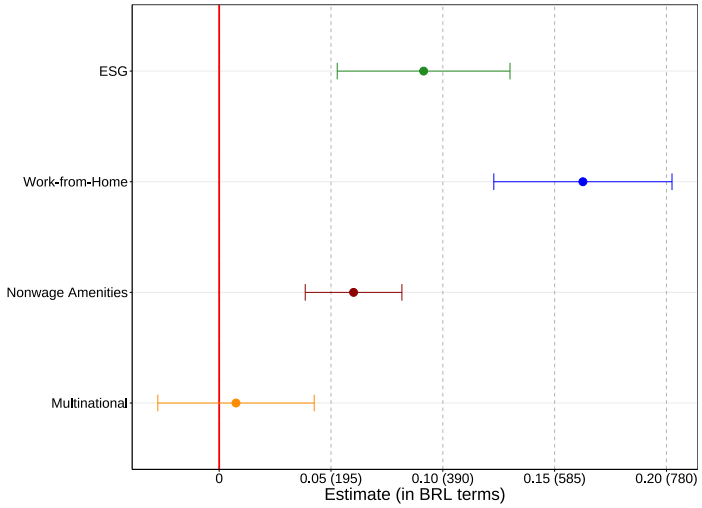


B. Perceived ESG Constraints

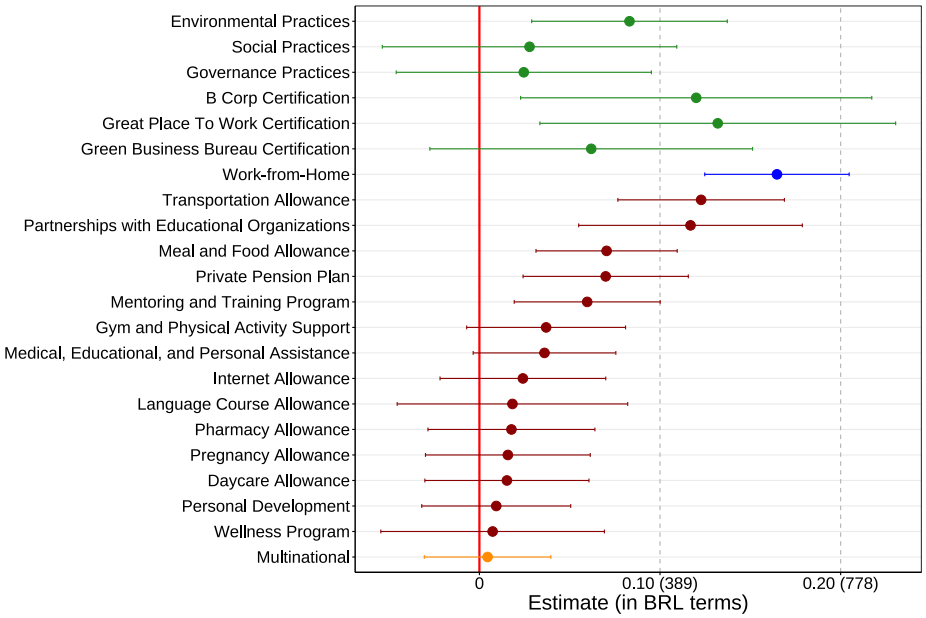


C. Perceived ESG Benefits

Note.—Panel A presents the summary statistics for firms' current and future ESG practices in our firm survey sample. For each measure of ESG practices, we provide the mean, standard deviation, tenth percentile (10th Percentile), twenty-fifth percentile (25th Percentile), median, seventy-fifth percentile (75th Percentile), and ninetieth percentile (90th Percentile). The survey sampled a total of 1,067 firms. For additional details, see Section 2. Panel B presents the responses to the question: "What are the main factors preventing your company from fully adopting or increasing your investment in ESG practices? Select up to three choices." Panel C presents the responses to the question: "What do you think are the main benefits of adopting ESG practices in your company? Select up to three choices."



A. Aggregate ESG Signals and Nonwage Amenities

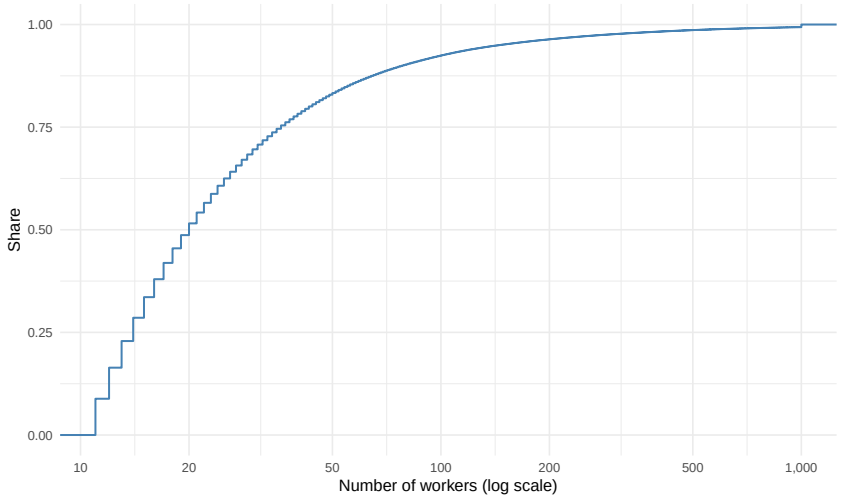


B. Granular ESG Signals and Nonwage Amenities

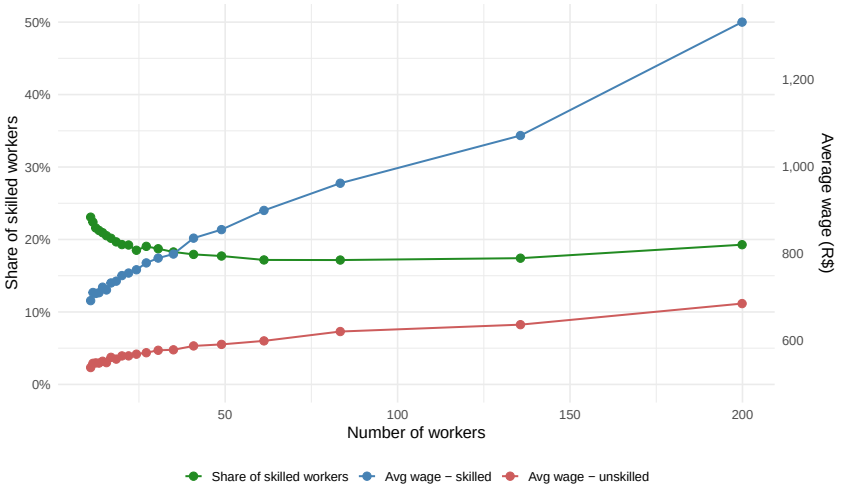
FIGURE 2. Job-Seeker Preferences for ESG Practices and Nonwage Amenities

Note.—Panel A shows the estimates and 95% confidence intervals for the coefficients β_2 to β_5 of the regression: $Interest_{ij} = \alpha + \beta_1 Wage_{ij} + \beta_2 ESG_{ij} + \beta_3 WFH_{ij} + \beta_4 NWA_{ij} + \beta_5 Multinational_{ij} + Strata\ FE + Individual\ FE + \epsilon_{ij}$. Panel B shows the estimates and 95% confidence intervals for the coefficients β_2 to β_9 and α_1 to α_K of the regression: $Interest_{ij} = \beta_0 + \beta_1 Wage_{ij} + \beta_2 Environmental_{ij} + \beta_3 Social_{ij} + \beta_4 Governance_{ij} + \beta_5 BCorp_{ij} + \beta_6 GPTW_{ij} + \beta_7 GBB_{ij} + \beta_8 WFH_{ij} + \sum_{k=1}^{K} \alpha_k NWA_{ijk} + \beta_9 Multinational_{ij} + Strata\ FE + Individual\ FE + \epsilon_{ij}$. i is the ith individual and j is the jth job posting rated by individual i. ESG is an indicator variable equal to one if the job posting displays at least one ESG sentence (see Appendix Table H12) or ESG certification (see Appendix Table H13). $Environmental$ is an indicator variable equal to one if the job posting displays at least one ESG sentence related to environmental practices. $Social$ is an indicator variable equal to one if the job posting displays at least one ESG sentence related to social practices. $Governance$ is an indicator variable equal to one if the job posting displays at least one ESG sentence related to governance practices. $BCorp$ is an indicator variable equal to one if the job displays a B Corporation certification. $GPTW$ is an indicator variable equal to one if the job displays a Great Place to Work certification. GBB is an indicator variable equal to one if the job displays a Green Business Bureau certification. WFH is an indicator variable equal to one if the job posting includes a work-from-home arrangement. NWA is the number of nonwage amenities. $Multinational$ is an indicator variable equal to one if the employer in the job posting is a multinational firm. NWA_k is an indicator variable equal to one if the kth out of K nonwage amenities is displayed in the job posting. We include strata fixed effects, which are binary indicators for each combination of respondent education level and preferred professional area, as well as individual fixed effects. In both panels, we report the equivalent value of the estimates in BRL in parentheses, representing how each regression's point estimate translates into an increase in monthly salary.

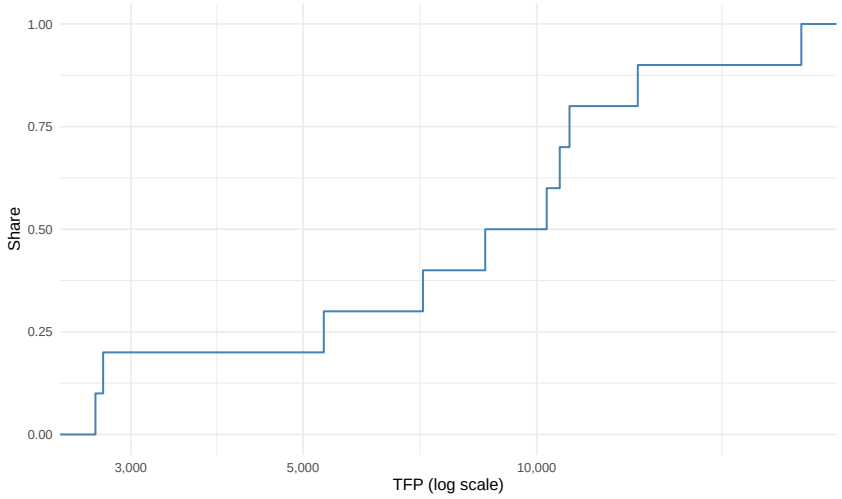
FIGURE 3. Firm Heterogeneity and Productivity



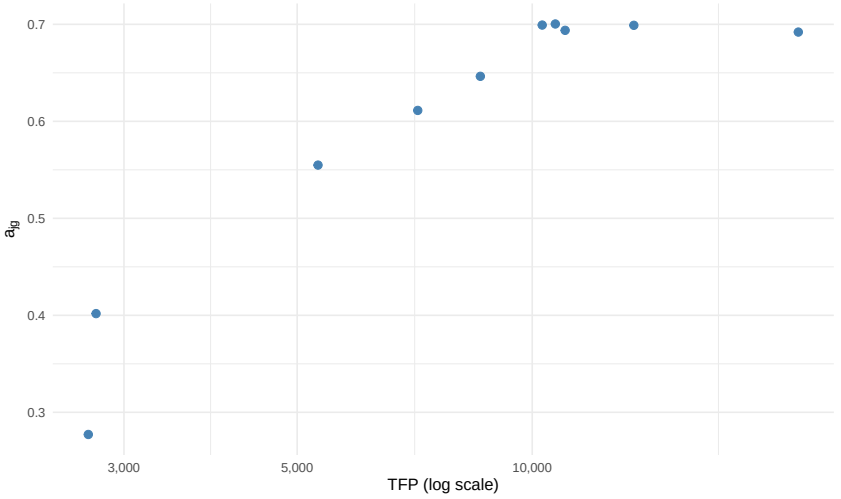
A. Firm Size Distribution



B. Skilled Share and Wages



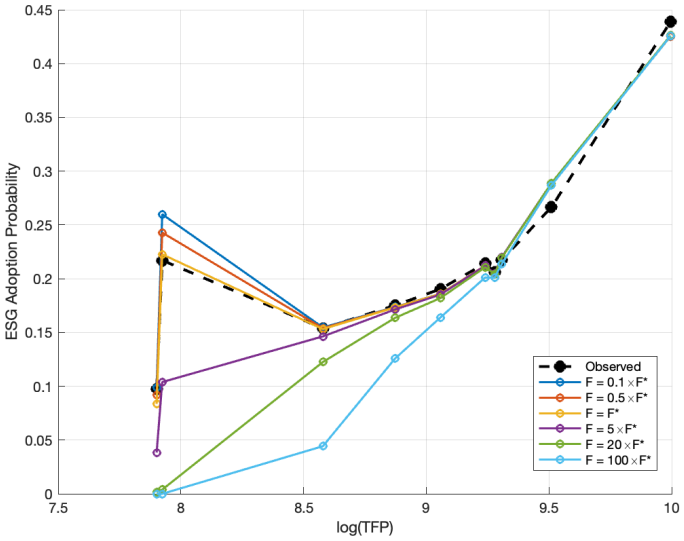
C. TFP Distribution



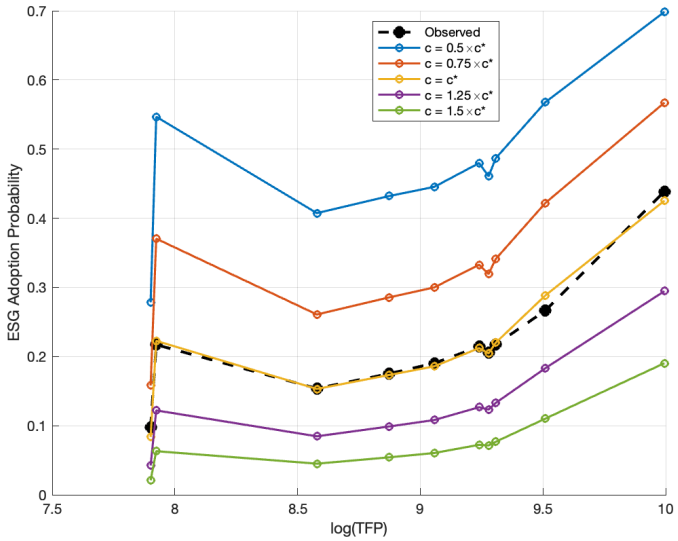
D. TFP vs. Productivity

Note.—Panel A shows the empirical cumulative distribution of firm size (number of workers, capped at 1,000) using the RAIS 2019 data. Panel B plots binned averages of the share of skilled workers (left axis) and mean wages for skilled and unskilled workers (right axis) against firm size (number of workers), using the RAIS 2019 data. Firms with 200 or more employees are grouped into a single bin. Panel C displays the empirical cumulative distribution of TFP across the 10 clusters estimated from the RAIS 2019 data. Panel D shows the relationship between the log skilled productivity multiplier $\hat{a}_{jg}$ and cluster-level TFP estimated from the RAIS 2019 data.

FIGURE 4. ESG Adoption Under Alternative Fixed and Variable Costs



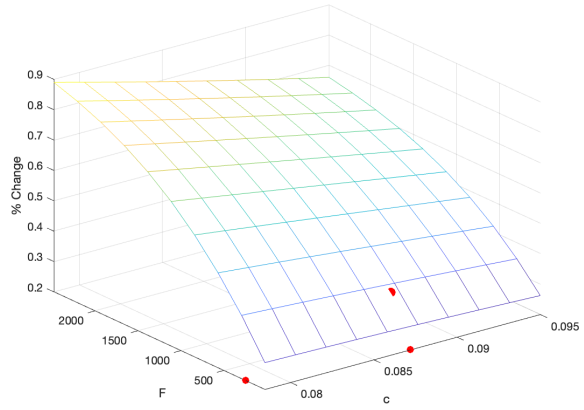
A. Fixed Cost



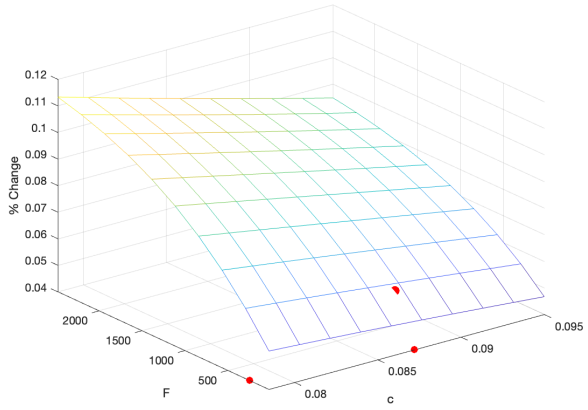
B. Variable Cost

Note.—This figure shows how the predicted probability of ESG adoption varies with cluster-level TFP under alternative values of the fixed cost (F) and the variable cost coefficient (c). In each panel, all other parameters are held fixed at their calibrated values. The dashed black line shows the observed B Corp adoption probability across clusters. Panel A varies the fixed cost F by applying different multiplicative factors to its calibrated value F^*. Panel B varies the variable cost coefficient c by applying different multiplicative factors to its calibrated value c^*. The solid line at the calibrated value ($F = F^*$ or $c = c^*$) corresponds to the model's baseline prediction. For details on the cost structure and calibration, see Section 5.3.

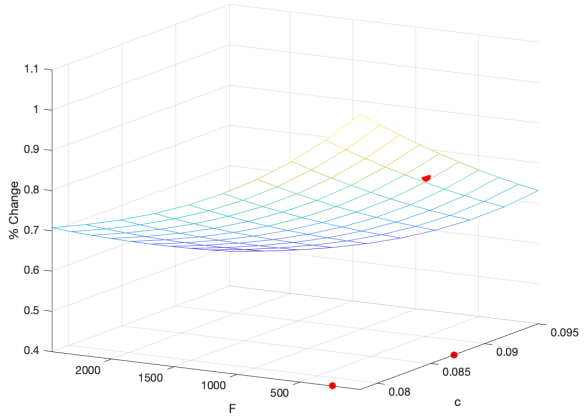
FIGURE 5. Model Counterfactuals



A. Skilled Wage Differential



B. Labor Value Added



C. Worker Utility

Note.—This figure presents the results of our counterfactual simulations, which evaluate how the cost of ESG adoption impacts the labor market equilibrium. We vary the fixed cost (F) and the variable cost coefficient (c) of ESG adoption and repeatedly solve for the labor market equilibrium across a grid of (F, c) values. Panel A shows the impact on the equilibrium wage differential between skilled and unskilled workers. Panel B shows the impact on labor value added. Panel C shows the impact on worker utility. The red dot indicates the calibrated parameter values. For additional details on the counterfactual simulations, see Section 5.3.

TABLE 1. Heterogeneous ESG Adoption Across Firms

	Implemented ESG (0/1=Yes) (1)	Extensively Implemented ESG (0/1=Yes) (2)	Environmental Practices (0% to 100%) (3)	Social Practices (0% to 100%) (4)	Governance Practices (0% to 100%) (5)	B Corp Certification (0% to 100%) (6)
Ln(Number of Employees)	0.047***	0.060***	1.632***	0.912*	0.961**	1.075**
	(0.007)	(0.009)	(0.476)	(0.467)	(0.462)	(0.473)
Ln(Average Wage)	0.044**	0.089***	4.289***	3.174**	4.017***	5.847***
	(0.022)	(0.029)	(1.526)	(1.481)	(1.478)	(1.503)
Employees with College Degree (%)	0.001**	0.003***	0.204***	0.227***	0.212***	0.246***
	(0.001)	(0.001)	(0.036)	(0.035)	(0.035)	(0.036)
Observations	1,067	1,067	1,067	1,067	1,067	1,067
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.141	0.195	0.176	0.158	0.153	0.193

Note.—This table reports the regression coefficients for the following specification: $\textit{ESG Adoption}_j = \alpha + \beta_1 ln(\textit{Number of Employees}_j) + \beta_2 ln(\textit{Average Wage}_j) + \beta_3 \textit{Employees with College Degree (\%)}_j + \textit{Industry FE} + \textit{State FE} + \epsilon_j$, where j is the jth firm. Columns (1) and (2) correspond to firms' self-reported current ESG adoption status, where $\textit{ESG Adoption}_j$ is an indicator variable equal to one if the firm has already implemented ESG practices (Column (1)) or has extensively implemented ESG practices (Column (2)). Columns (3) to (6) correspond to firms' expectations of future ESG adoption, where $\textit{ESG Adoption}_j$ represents the firm's stated likelihood (from 0% to 100%) of making a financial investment to meet ESG standards within the next 1–3 years. Specifically, Column (3) pertains to environmental practices, Column (4) to social practices, Column (5) to governance practices, and Column (6) to adopting a B Corp certification. *ln(Number of Employees)* is the natural logarithm of the firm's total number of employees. *ln(Average Wage)* is the natural logarithm of the average wage at the firm. *Employees with College Degree (%)* is the percentage of employees who have a college degree. All regressions include industry and state fixed effects. For additional details on our firm survey of ESG practices, see Section 2. Robust standard errors are reported in parentheses. *p<0.1; **p<0.05; ***p<0.01.

TABLE 2. Summary Statistics

	Catho Survey (1)	RAIS (2)	PNAD (3)
Observations	1,206	31,761,221	91,456,031
Female (%)	42.95	38.34	42.25
Race:			
White (%)	51.91	56.16	45.12
Mixed (%)	34.99	37.52	42.58
Black (%)	10.86	5.67	11.15
Asian (%)	1.08	0.50	0.77
Native (%)	0.66	0.14	0.35
Other (%)	0.50	0.00	0.03
Age:			
1st Qu.	32.00	31.00	29.00
Median	42.00	38.00	39.00
Mean	40.83	39.63	39.76
3rd Qu.	47.00	47.00	49.00
Wage:			
1st Qu. (BRL)	1,750.00	1,413.32	1,212.00
Median (BRL)	2,750.00	1,854.76	1,600.00
Mean (BRL)	4,308.41	2,871.21	2,636.67
3rd Qu. (BRL)	7,500.00	2,805.28	2,800.00
Education:			
Completed PhD (%)	0.50	0.21	0.50
Completed Masters (%)	4.39	0.64	1.03
Completed College (%)	51.41	19.38	27.98
Incomplete College (%)	16.67	6.62	4.87
Completed High School (%)	22.72	61.15	49.37
Incomplete High School (%)	2.32	5.01	8.03
Completed Middle School or Less (%)	1.99	6.99	8.22

Note.—This table provides summary statistics on the Catho survey sample, the 2020 Relação Anual de Informações Sociais (RAIS), and the Q2/2022 *Pesquisa Nacional por Amostra de Domicílios* (PNAD). We report the percentage of female individuals in RAIS using RAIS 2019. We only focus on the subset of individuals in PNAD that are active in the workforce. We provide additional details on RAIS and PNAD in Sections 3.1.1 and 5.2.

TABLE 3. Job-Seekers' Preferences for Corporate ESG

	Interest (1)	Interest (2)	Interest (3)	Interest (4)	BRL Terms (5)
ESG	0.083***	0.099***	0.084***	0.070**	346.244***
	(0.026)	(0.026)	(0.020)	(0.027)	(77.880)
Ln(Wage)	-	1.115***	1.203***	1.050***	-
	-	(0.031)	(0.026)	(0.034)	-
Nonwage Amenities	-	0.058***	0.062***	0.061***	239.815***
	-	(0.014)	(0.011)	(0.015)	(43.658)
Financial Strength	-	−0.005	0.012	−0.057	28.247
	-	(0.041)	(0.032)	(0.045)	(127.219)
Observations	24,120	24,120	24,120	24,120	24,120
Individual FE	No	No	Yes	Yes	Yes
Strata FE	Yes	Yes	Yes	Yes	Yes
Job Posting Controls	No	Yes	Yes	Yes	Yes
PNAD Re-Weight	No	No	No	Yes	No

Note.—This table reports regression coefficients for the following specifications. Column (1) specification: $Interest_{ij} = \alpha + \beta_1 ESG_{ij} + \text{Strata FE} + \varepsilon_{ij}$. Column (2) specification: $Interest_{ij} = \alpha + \beta_1 ESG_{ij} + \beta_2 \ln(Wage_{ij}) + \beta_3 NWA_{ij} + \beta_4 FS_{ij} + \mathbf{X}_{ij}\mathbf{\Gamma} + \text{Strata FE} + \varepsilon_{ij}$. Column (3) specification: $Interest_{ij} = \alpha + \beta_1 ESG_{ij} + \beta_2 \ln(Wage_{ij}) + \beta_3 NWA_{ij} + \beta_4 FS_{ij} + \mathbf{X}_{ij}\mathbf{\Gamma} + \text{Strata FE} + \text{Individual FE} + \varepsilon_{ij}$. Column (4) reports the same specification as Column (3), re-weighted using inverse-probability weights constructed from the PNAD household survey. Column (5) reports coefficients from the Column (3) specification estimated in wage levels, with coefficients expressed in BRL terms. i is the ith individual and j is the jth job posting rated by individual i. *ESG* is an indicator variable equal to one if the job posting displays at least one ESG sentence (see Appendix Table H12) or ESG certification (see Appendix Table H13). ln(*Wage*) is the natural logarithm of the monthly wage displayed in the job posting. NWA is equal to the number of nonwage amenities displayed in the job posting. FS is an indicator variable equal to one if the job posting displays a signal of financial strength (see Appendix Table H10). $\mathbf{X}_{ij}$ includes additional job-posting controls, accounting for the number of on-the-job activities, number of on-the-job opportunities, employer industry, employer establishment year, number of job prerequisites, an indicator for whether the job is not located in the respondent's primary chosen city, an indicator for work-from-home arrangements, and an indicator for whether the employer is a multinational. *Strata FE* are binary indicators for each combination of respondent education level and preferred professional area. Robust standard errors are reported in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

TABLE 4. Heterogeneous Preferences for ESG Across Sociodemographic Groups

	High Education (1)	White (2)	Liberal or Moderate (3)	Young (4)	Female (5)
ESG Interaction	0.113***	0.079**	0.090**	−0.047	−0.009
	(0.040)	(0.039)	(0.039)	(0.040)	(0.040)
ESG	0.019	0.043	0.041	0.112***	0.088***
	(0.030)	(0.028)	(0.027)	(0.031)	(0.025)
Ln(Wage)	1.204***	1.203***	1.202***	1.203***	1.203***
	(0.026)	(0.026)	(0.026)	(0.026)	(0.026)
Nonwage Amenities	0.061***	0.061***	0.062***	0.062***	0.062***
	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)
Financial Strength	0.012	0.012	0.012	0.012	0.012
	(0.032)	(0.032)	(0.032)	(0.032)	(0.032)
Observations	24,120	24,120	24,120	24,120	24,120
Individual FE	Yes	Yes	Yes	Yes	Yes
Strata FE	Yes	Yes	Yes	Yes	Yes
Job Posting Controls	Yes	Yes	Yes	Yes	Yes

Note.—This table reports regression coefficients for the following specification: $Interest_{ij} = \alpha + \beta_1 ESG\ Interaction_{ij} + \beta_2 ESG_{ij} + \beta_3 \ln(Wage_{ij}) + \beta_4 NWA_{ij} + \beta_5 FS_{ij} + \mathbf{X}_{ij}\mathbf{\Gamma}$+Strata FE+Individual FE+ ε_{ij}. i is the ith individual and j is the jth job posting rated by individual i. *ESG Interaction* is an interaction term between *ESG* and an indicator variable equal to one if the individual belongs to the sociodemographic group listed in the corresponding column heading. *ESG* is an indicator variable equal to one if the job posting displays at least one ESG sentence (see Appendix Table H12) or ESG certification (see Appendix Table H13). ln(*Wage*) is the natural logarithm of the monthly wage displayed in the job posting. NWA is equal to the number of nonwage amenities displayed in the job posting. FS is an indicator variable equal to one if the job posting displays a signal of financial strength (see Appendix Table H10). $\mathbf{X}_{ij}$ includes additional job-posting controls, accounting for the number of on-the-job activities, number of on-the-job opportunities, employer industry, employer establishment year, number of job prerequisites, an indicator for whether the job is not located in the respondent's primary chosen city, an indicator for work-from-home arrangements, and an indicator for whether the employer is a multinational. *Strata FE* are binary indicators for each combination of respondent education level and preferred professional area. Robust standard errors are reported in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

TABLE 5. ESG Implementation, Productivity, and Counterfactuals on Wages, Labor Value Added, and Welfare

	Environmental Practices (1)	Average ESG Practices (2)	B Corp Certification (3)
	A. Correlation Between Future ESG Implementation and Productivity		
Average Ln(Wage)	0.251*	0.357**	0.475***
	(0.144)	(0.144)	(0.136)
Skill Wage Gap	0.284*	0.410**	0.269*
	(0.159)	(0.161)	(0.150)
Ln(Number of Employees)	0.095**	0.083**	0.085**
	(0.041)	(0.041)	(0.039)
Employees with College Degree	0.018***	0.021***	0.019***
	(0.003)	(0.003)	(0.003)
Observations	1,067	1,067	1,067
	B. Counterfactual Effects		
Wage Differentials (bps)	31	29	36
Skilled Wage (bps)	13	12	15
Unskilled Wage (bps)	1	1	2
Value Added (bps)	5	5	6
Worker Welfare (%)	0.69	0.61	0.83
Skilled (%)	2.42	2.15	2.93
Unskilled (%)	0.23	0.20	0.28
Direct amenity share of welfare (%)			
All workers	95.6	95.4	95.8
Skilled	96.1	96.3	96.4
Unskilled	94.3	92.9	93.9

Note.—This table reports logistic regression coefficients for the following specification: $ESG\ Adoption_j = \alpha + \beta_1 \text{Average Ln}(Wage)_j + \beta_2 Skill\ Wage\ Gap_j + \beta_3 \text{Ln}(Number\ of\ Employees)_j + \beta_4\ Employees\ with\ College\ Degree_j + \epsilon_j$, where j is the jth firm in our firm survey of ESG practices (see Section 2 for additional details). In Column (1), the outcome variable is an indicator equal to one if the respondent reported at least a 90% likelihood that their firm will make a financial investment in environmental practices to meet ESG standards for strong performance within the next 1–3 years. In Column (2), the outcome variable is an indicator equal to one if the average of the respondent's stated likelihoods that their firm will make a financial investment in each of the three ESG categories—environmental, social, and governance—to meet ESG standards for strong performance within the next 1–3 years is at least 90%. We first compute the average of the firm's likelihoods for investing in each of the three categories and then apply the 90% threshold. In Column (3), the outcome variable is an indicator equal to one if the respondent stated at least a 90% likelihood that their firm will make a financial investment in obtaining a B Corp certification within the next 1–3 years. Average ln(*Wage*) is the average of the natural logarithm of the monthly wage across all employees at the firm. *Skill Wage Gap* is the difference between the natural logarithm of the average monthly wage of college-educated employees and the natural logarithm of the average monthly wage of non-college-educated employees. ln(*Number of Employees*) is the natural logarithm of the firm's total number of full-time employees. *Employees with College Degree* is the percentage of employees who have a college degree. Panel B reports the counterfactual analysis based on the estimated logistic regression results. Specifically, it reports the percentage changes in equilibrium wage differentials between skilled and unskilled workers (in basis points), average skilled wages (in basis points), average unskilled wages (in basis points), value added (in basis points), and worker welfare (expressed as a percentage in wage-equivalent terms) relative to the baseline economy without ESG preferences. Worker welfare is further decomposed into skilled and unskilled workers, reflecting the differential impact of ESG adoption on each group due to heterogeneous preferences for ESG amenities. Direct amenity shares divide the welfare change obtained with counterfactual ESG adoption probabilities and baseline employers and wages by the corresponding total welfare change, using unrounded values. These results are visualized in Figure 5. *p<0.1; **p<0.05; ***p<0.01.

Online Appendix

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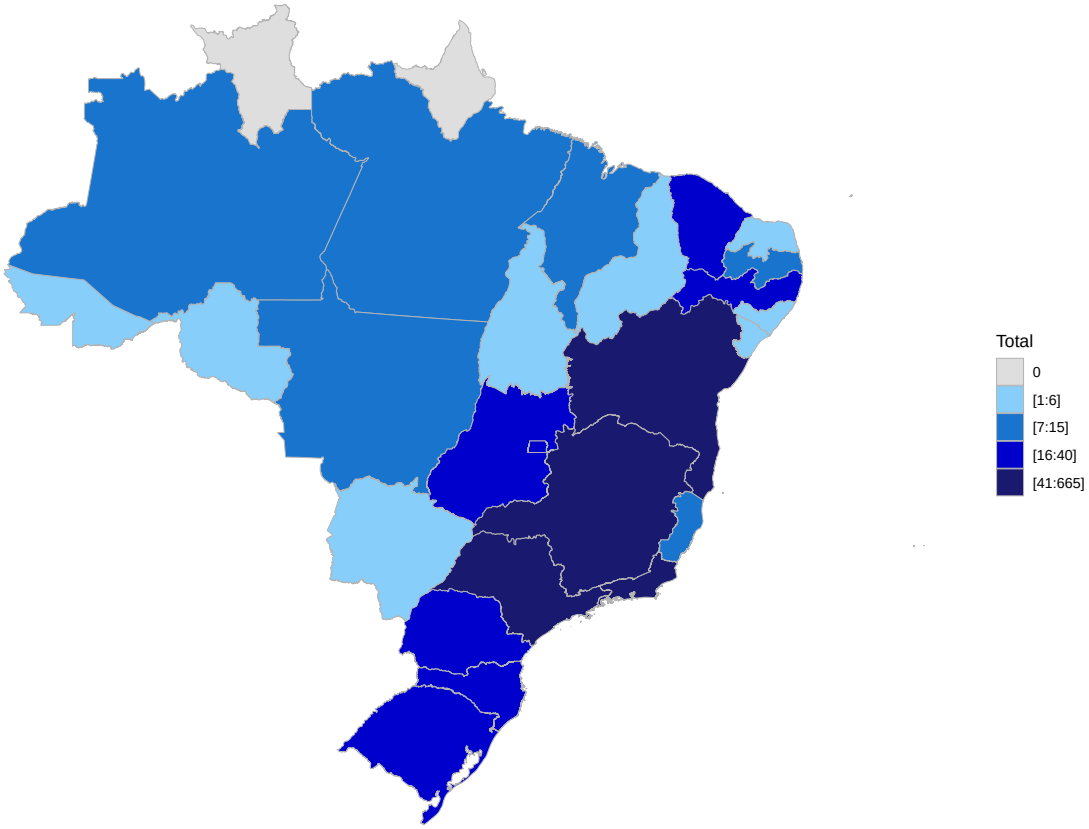
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Appendix A. Additional Figures and Tables

Figure A1. Location of Survey Respondents



Note.—This map shows the geographic distribution of survey respondents across different states in Brazil. Darker shades indicate a higher number of respondents from that state.

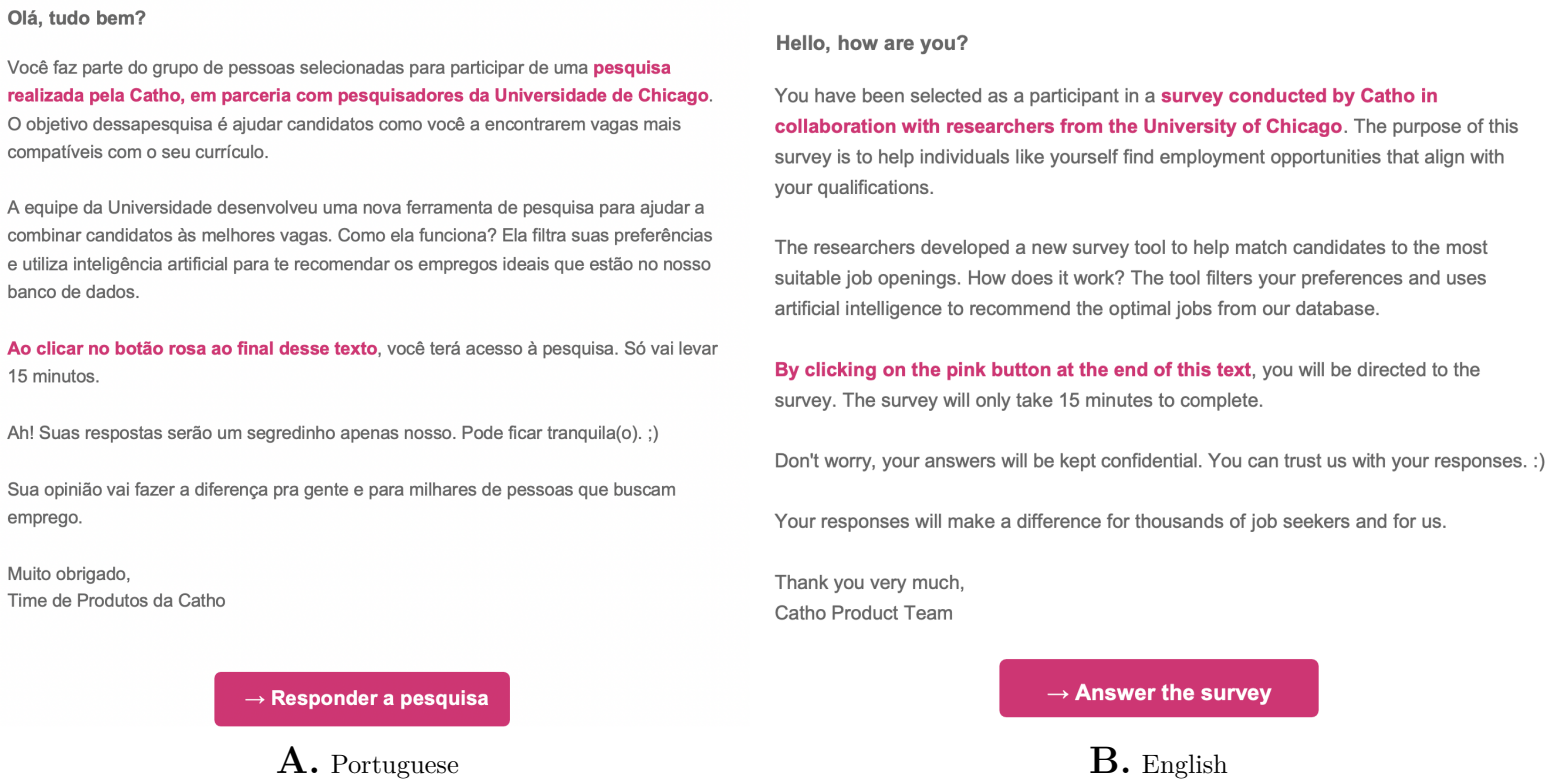


FIGURE A2. Email Sent to Survey Respondents

Note.—This figure presents the recruitment email sent to survey respondents. Panel A shows the actual email sent to respondents in the survey, in Portuguese. Panel B shows the English translation. The email subject line was as follows: *Queremos te ajudar a encontrar melhores vagas!* (*We want to help you find better jobs!*).

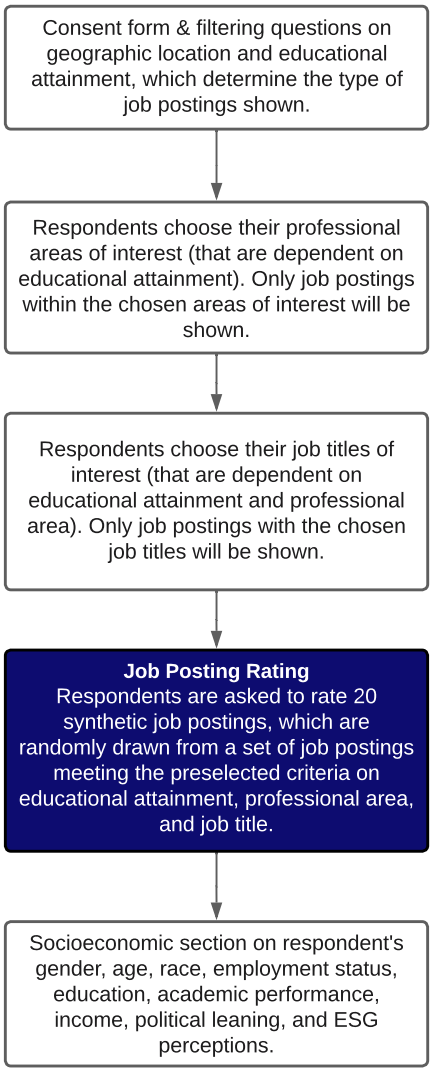
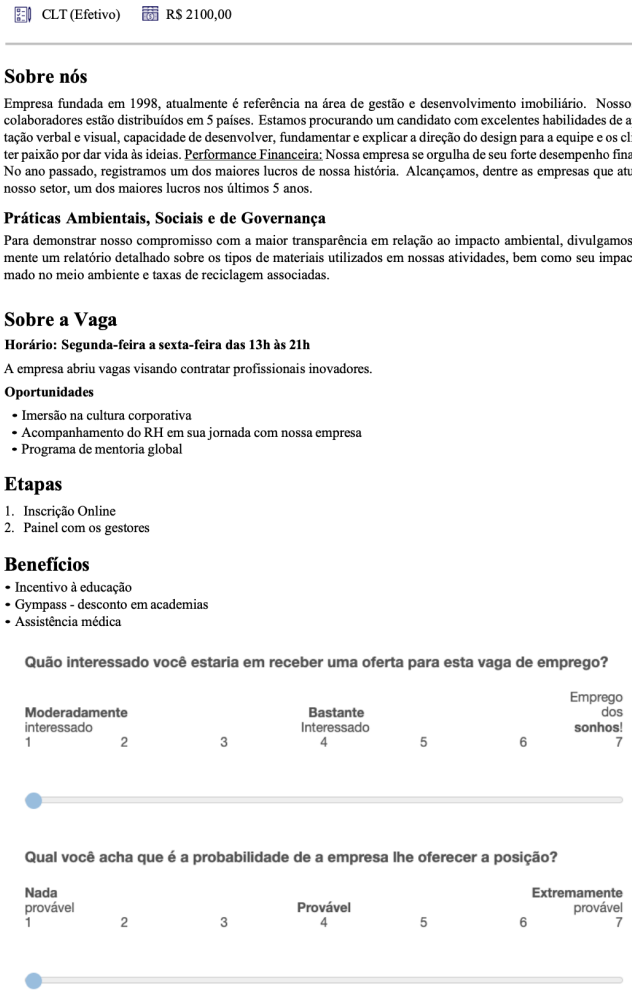
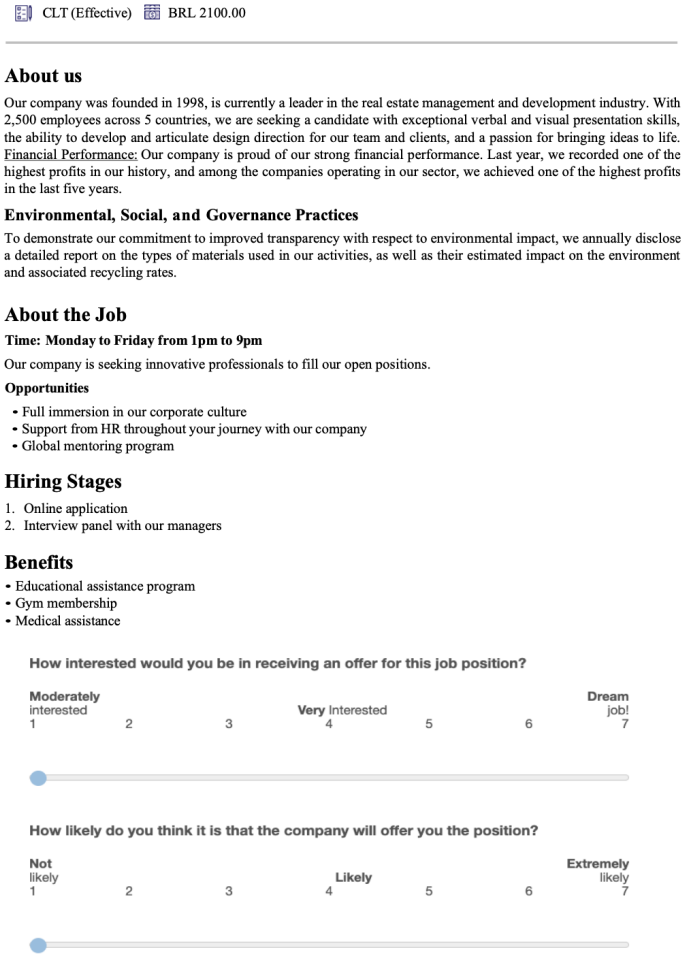


FIGURE A3. Experimental Survey Flow

Note.—This figure illustrates our experimental survey flow. We discuss additional details of the experimental design in Section 3.1.



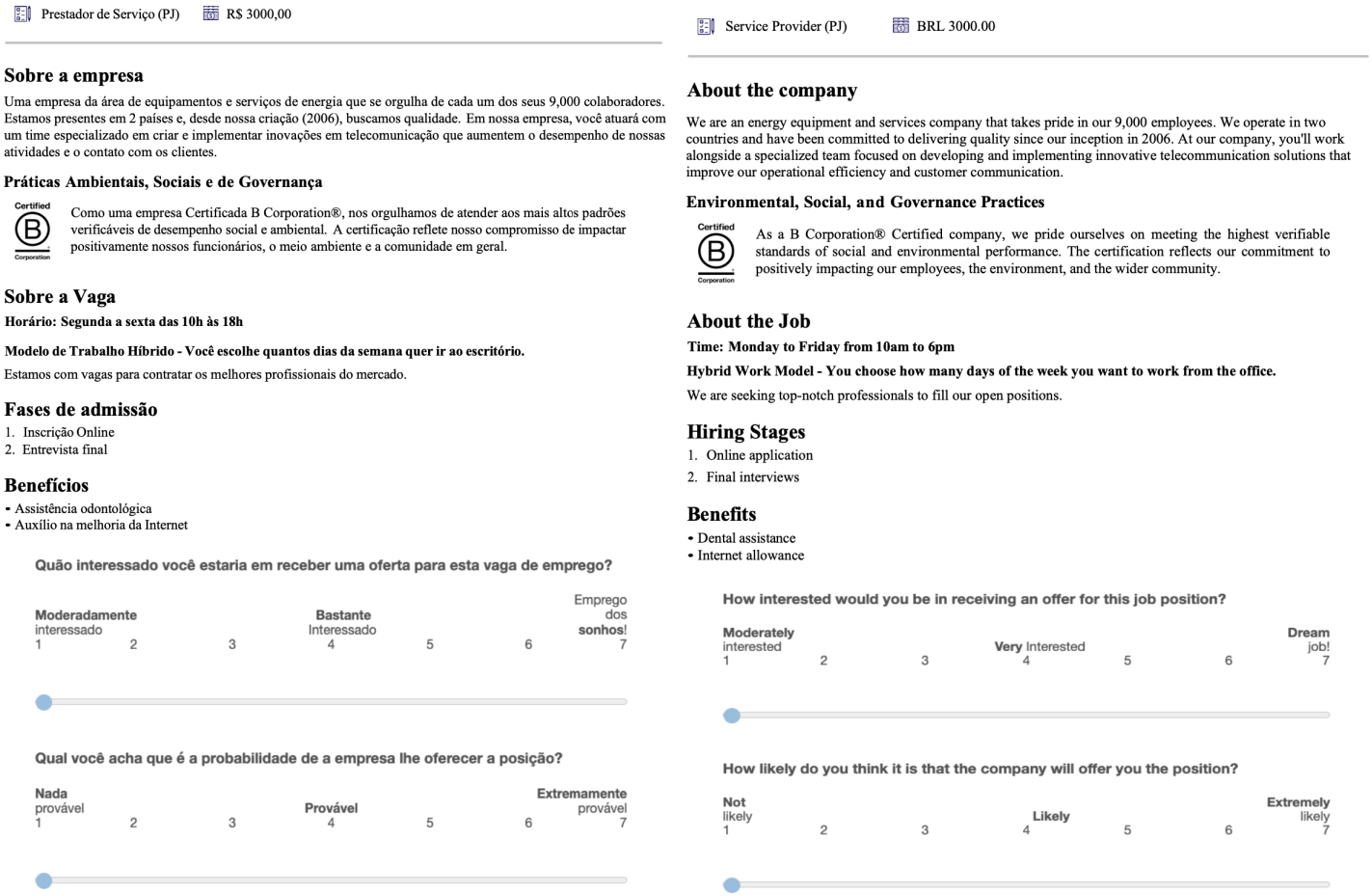
A. Portuguese



B. English

FIGURE A4. Example of Synthetic Job Posting E(SG) and Financial Performance

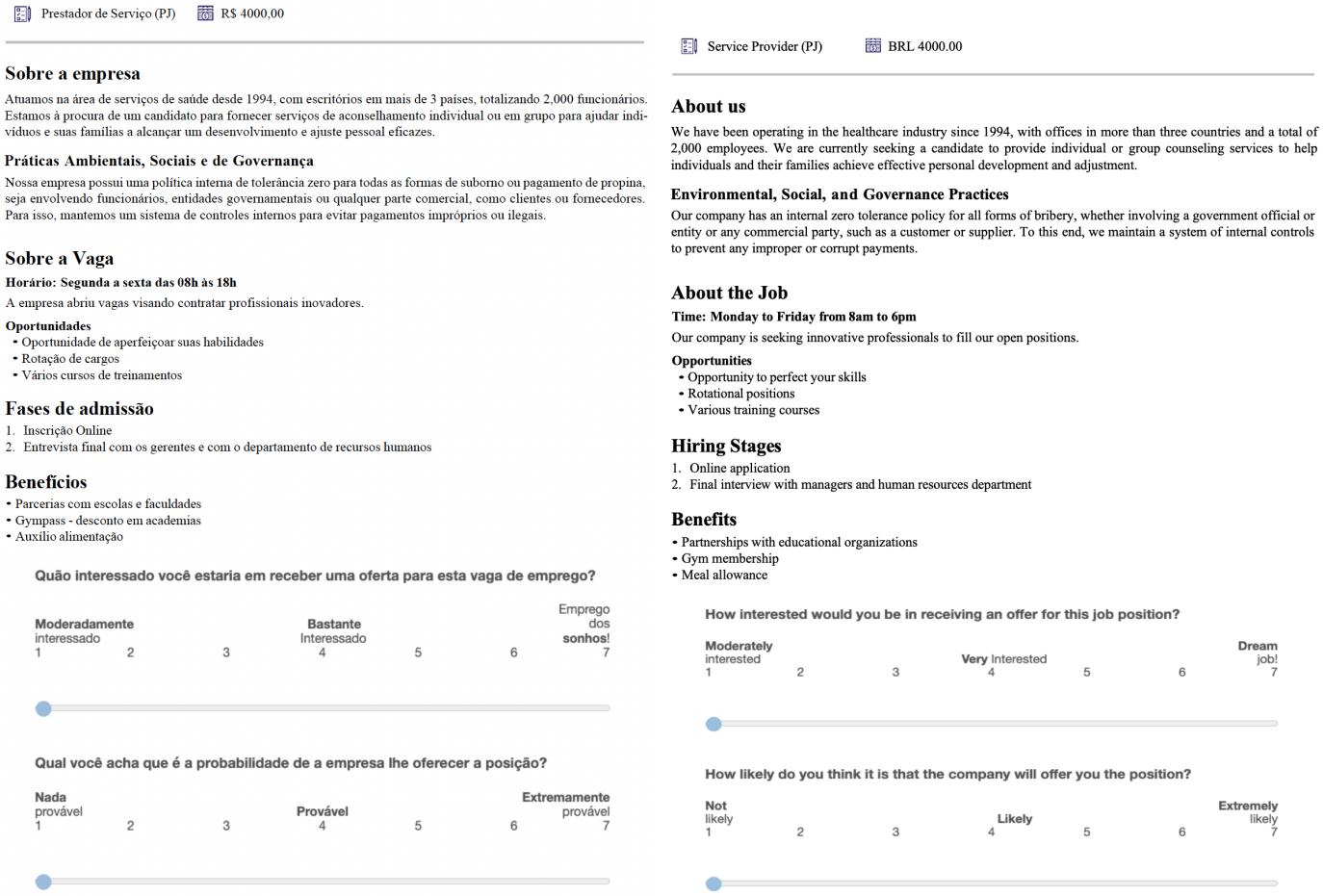
Note.—This figure presents an example of a synthetic job posting with environmental and financial performance signaling, as well as the two rating questions shown to respondents in the survey. For a description of each job posting component, see Section 3.2.



A. Portuguese **B.** English

FIGURE A5. Example of Synthetic Job Posting
ESG Certification (B Corporation)

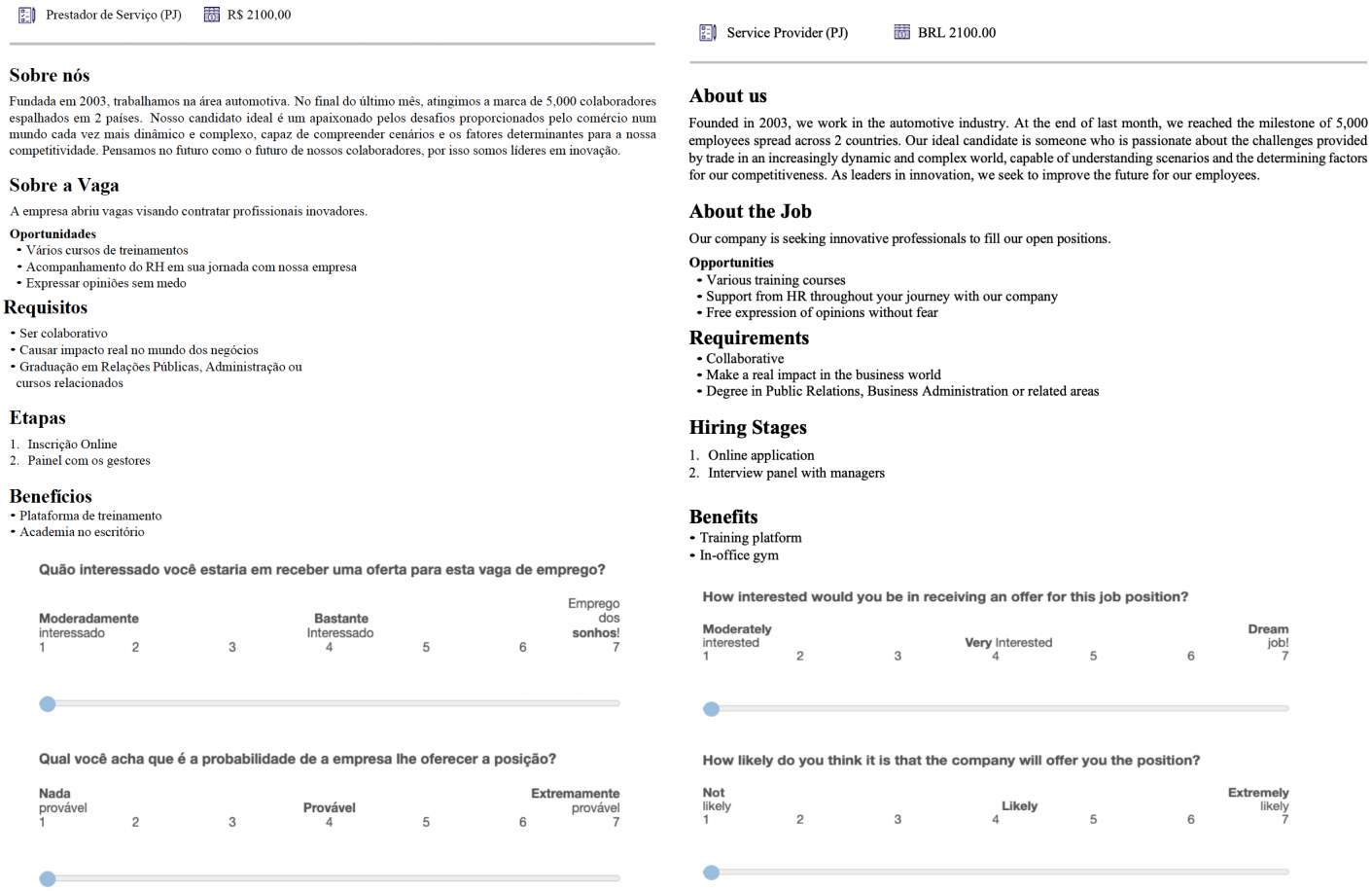
Note.—This figure presents an example of a synthetic job posting with an ESG certification, as well as the two rating questions shown to respondents in the survey. For a description of each job posting component, see Section 3.2.



A. Portuguese **B.** English

FIGURE A6. Example of Synthetic Job Posting (ES)G

Note.—This figure presents an example of a synthetic job posting with governance signaling, as well as the two rating questions shown to respondents in the survey. For a description of each job posting component, see Section 3.2.



A. Portuguese **B.** English

FIGURE A7. Example of Synthetic Job Posting Control

Note.—This figure presents an example of a synthetic job posting without any ESG signaling, as well as the two rating questions shown to respondents in the survey. For a description of each job posting component, see Section 3.2.

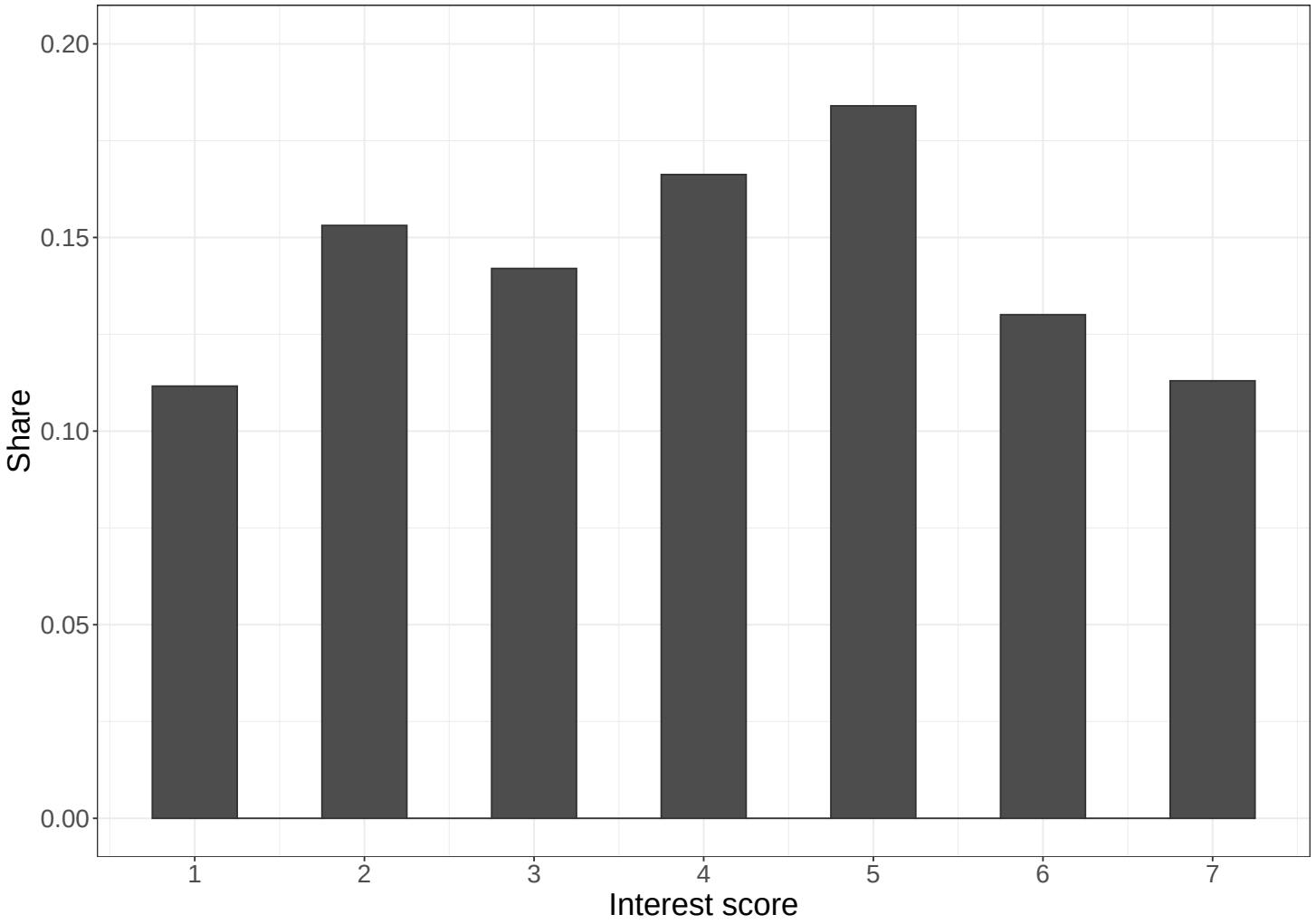


FIGURE A8. Distribution of Job Posting Ratings

Note.—This figure shows the distribution of our main outcome variable, *Interest*. *Interest* is measured on a scale of 1 to 7 and indicates the level of interest that respondents have in a job posting.



A. Words



B. Bigrams

FIGURE A9. ESG Word Clouds

Note.—This figure shows the responses visualized as word clouds to the open-ended question, "When you think of working for companies with Environmental, Social and Governance (ESG) practices in place, what are the main considerations that come to mind?" Panel A shows the word cloud generated for individual words, with the most frequently used words in the responses appearing larger. Panel B shows the word cloud generated for bigrams, with the most frequently used bigrams in the responses appearing larger. We remove all words present in the question from the word clouds.

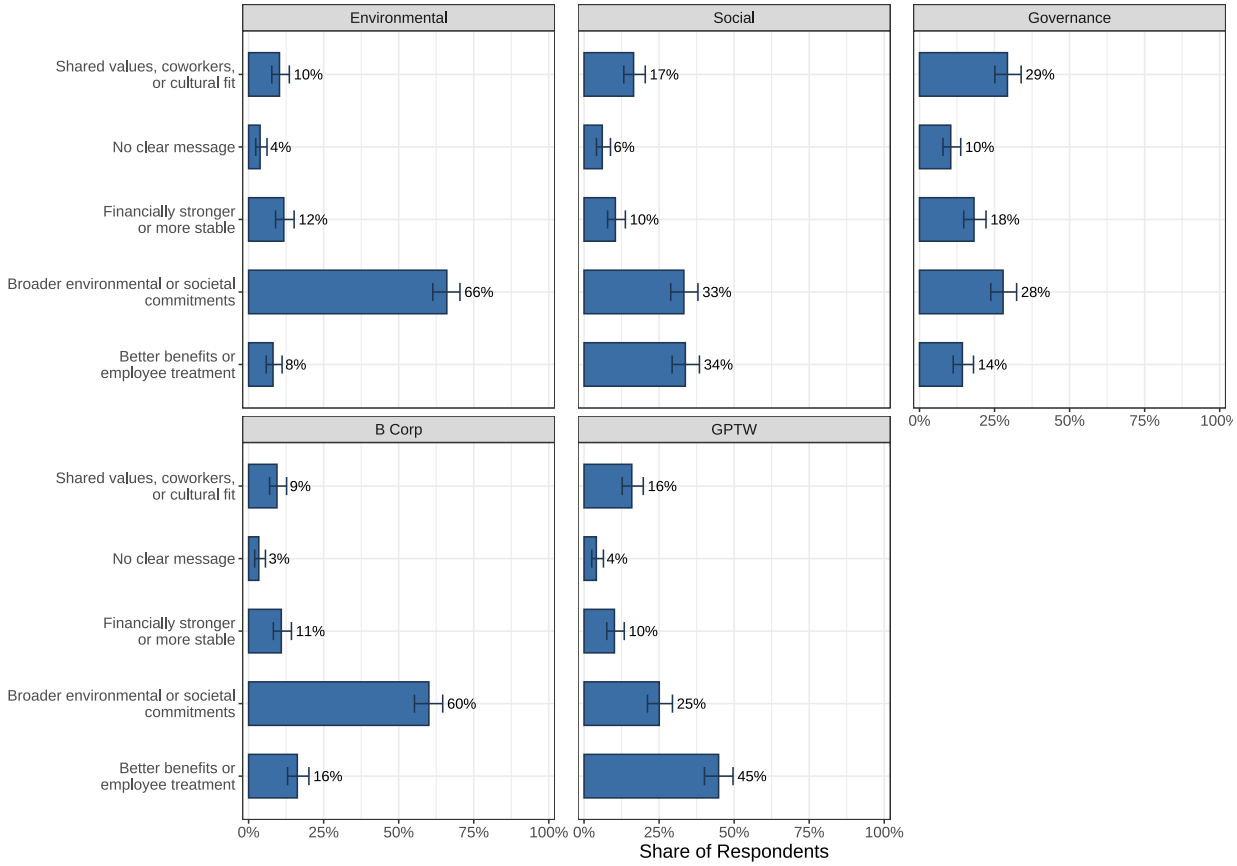


FIGURE A10. Interpretation of ESG Signals in Job Postings

Note.—This figure shows the distribution of responses to the closed-ended question: "If you had to choose one, what does this mainly tell you about the company?" for the five ESG signals used in the experiment: an environmental sentence, a social sentence, a governance sentence, a B Corp certification, and a Great Place to Work (GPTW) certification. Respondents selected one of five interpretations: (i) broader environmental or societal commitments; (ii) shared values, coworkers, or cultural fit; (iii) better benefits or employee treatment; (iv) greater financial strength or long-run stability; or (v) no clear message about the company. Bars show the share of respondents selecting each interpretation, separately by signal type. Horizontal error bars show 95% Wilson confidence intervals.

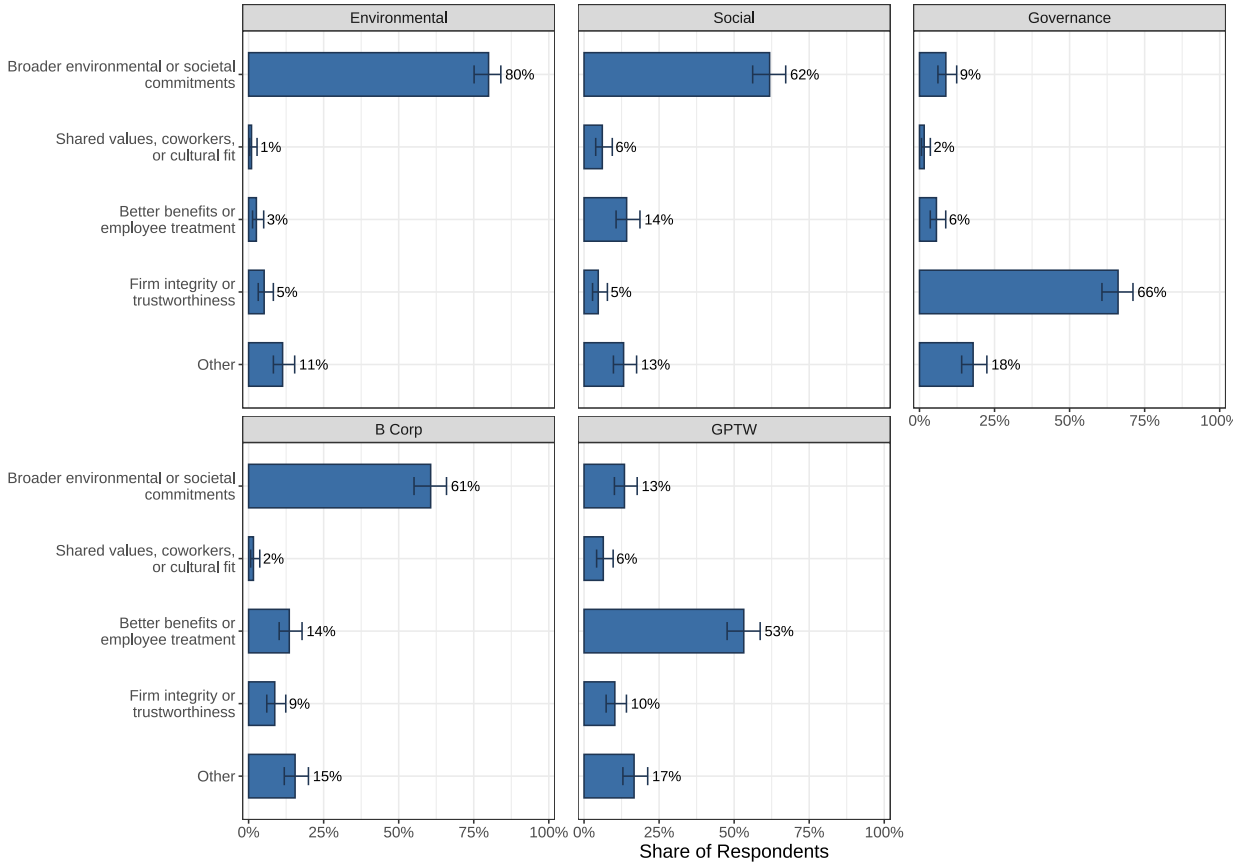


FIGURE A11. LLM Classification of Open-Ended Interpretations of ESG Signals in Job Postings

Note.—This figure shows the distribution of LLM-assigned categories for responses to the open-ended question: "In one or two sentences, what does this information mainly tell you about the company, if anything?" for the five ESG signals used in the experiment: an environmental sentence, a social sentence, a governance sentence, a B Corp certification, and a Great Place to Work (GPTW) certification. Each free-text response is classified by OpenAI `gpt-4o-mini` (temperature 0) into one of six mutually exclusive categories: (i) broader environmental or societal commitments; (ii) shared values, coworkers, or cultural fit; (iii) better benefits or employee treatment; (iv) firm integrity or trustworthiness; (v) other (coherent answers that do not fit (i)–(iv), including financial-strength and marketing-only readings); and (vi) non-substantive responses, including random characters or other responses that are not meaningful attempts to answer. Non-substantive responses are excluded from the figure. Bars show the share of respondents in each category, separately by signal type. Horizontal error bars show 95% Wilson confidence intervals.

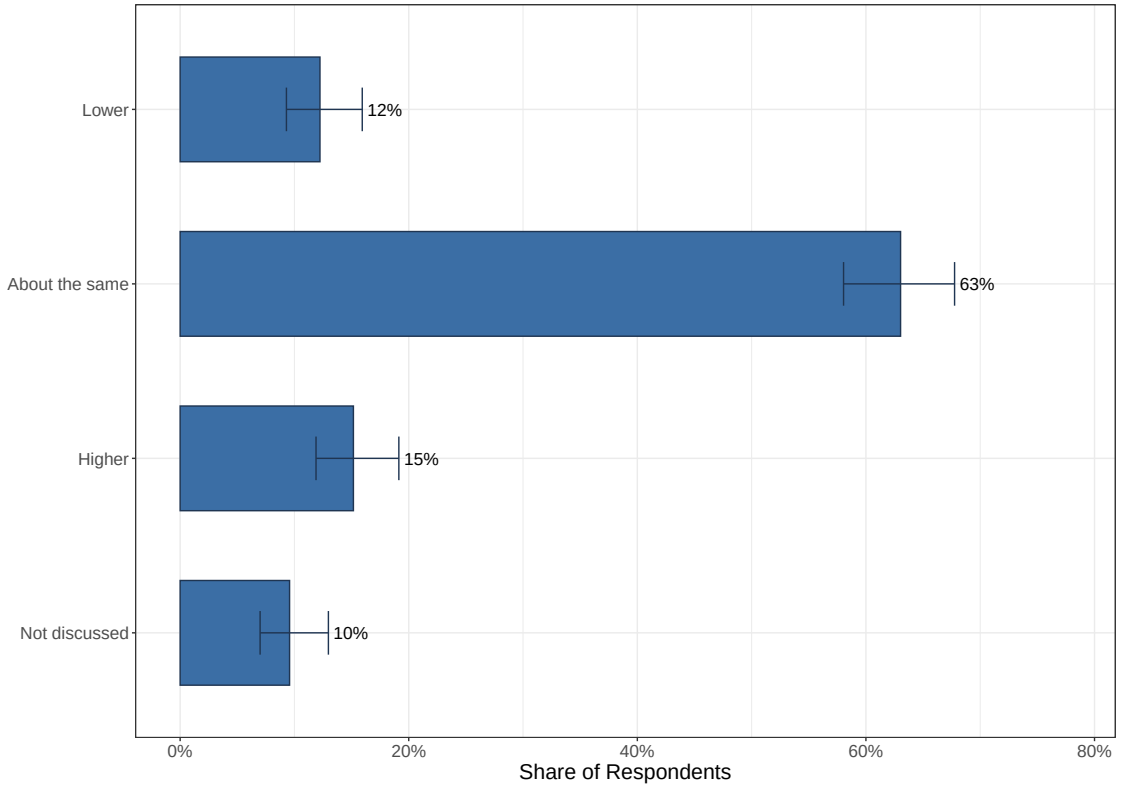


FIGURE A12. Pay Discussed During Hiring Relative to Posted Pay

Note.—This figure shows the distribution of responses from the 2026 qualitative survey to the question: "How close was the pay eventually discussed during the hiring process to the amount shown in the job posting?" This question is asked only of respondents whose most recent job posting listed a wage. Respondents selected one of five options: (i) "The pay discussed was lower than the amount shown in the job posting"; (ii) "The pay discussed was about the same as the amount shown in the job posting"; (iii) "The pay discussed was somewhat higher than the amount shown in the job posting"; (iv) "The pay discussed was much higher than the amount shown in the job posting"; and (v) "Pay was not discussed in detail during the hiring process." For presentation, responses are grouped into four categories: "lower" corresponds to the response indicating lower discussed pay; "about the same" corresponds to the response indicating about the same discussed pay; "higher" combines the two responses indicating somewhat or much higher discussed pay; and "not discussed" corresponds to the response indicating that pay was not discussed in detail. Bars report the share of respondents in each category. If respondents interpret the posted wage primarily as a floor, responses indicating somewhat or much higher discussed pay should be more common than responses indicating about the same or lower discussed pay.

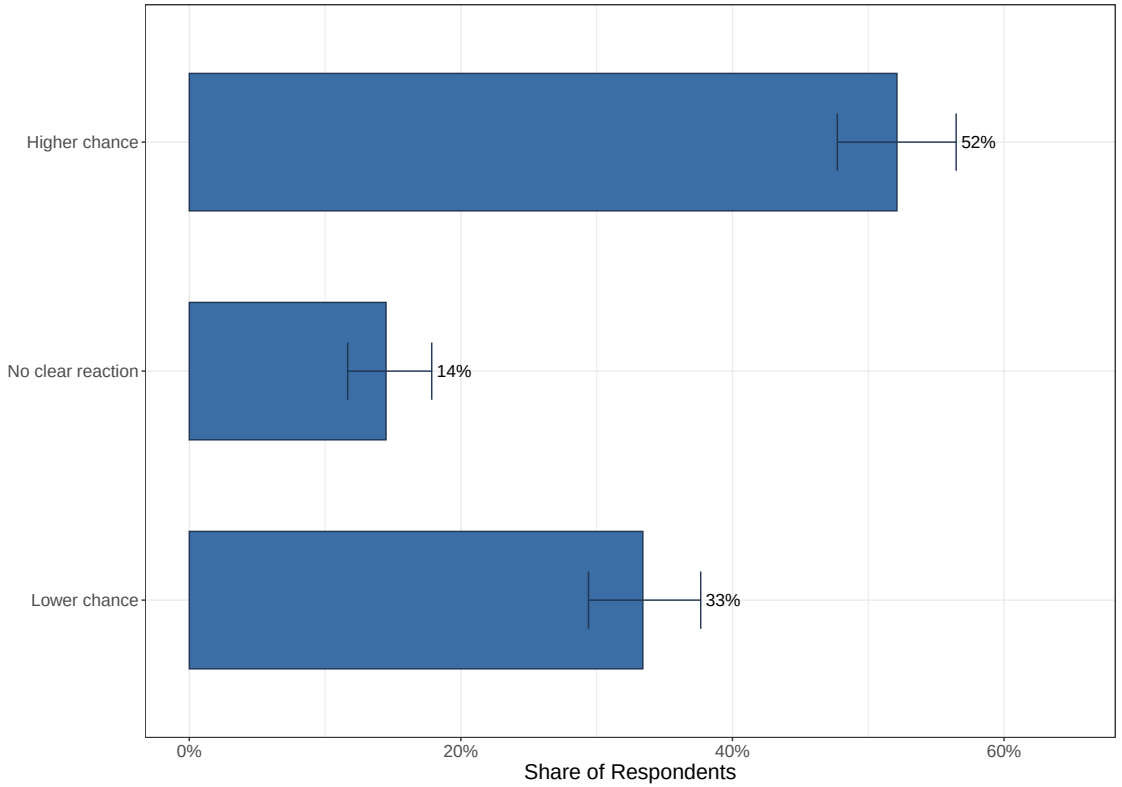


FIGURE A13. Above-Average Pay and Perceived Offer Chance

Note.— This figure reports responses from the 2026 qualitative survey to the question: "When a job posting offers a salary clearly above average for a similar role, what is your first reaction?" Respondents selected one of five options: (i) "The employer is having a lot of difficulty hiring, and my chances of being hired are much higher"; (ii) "The employer is having some difficulty hiring, and my chances of being hired are somewhat higher"; (iii) "I don't have a clear reaction"; (iv) "The employer is probably somewhat more selective, and my chances of being hired are somewhat lower"; or (v) "The employer is probably much more selective, and my chances of being hired are much lower." For presentation, responses are grouped into three categories: "higher chance" combines the two responses indicating higher chances of being hired; "no clear reaction" corresponds to the neutral response; and "lower chance" combines the two responses indicating lower chances of being hired. Bars report the share of respondents in each category.

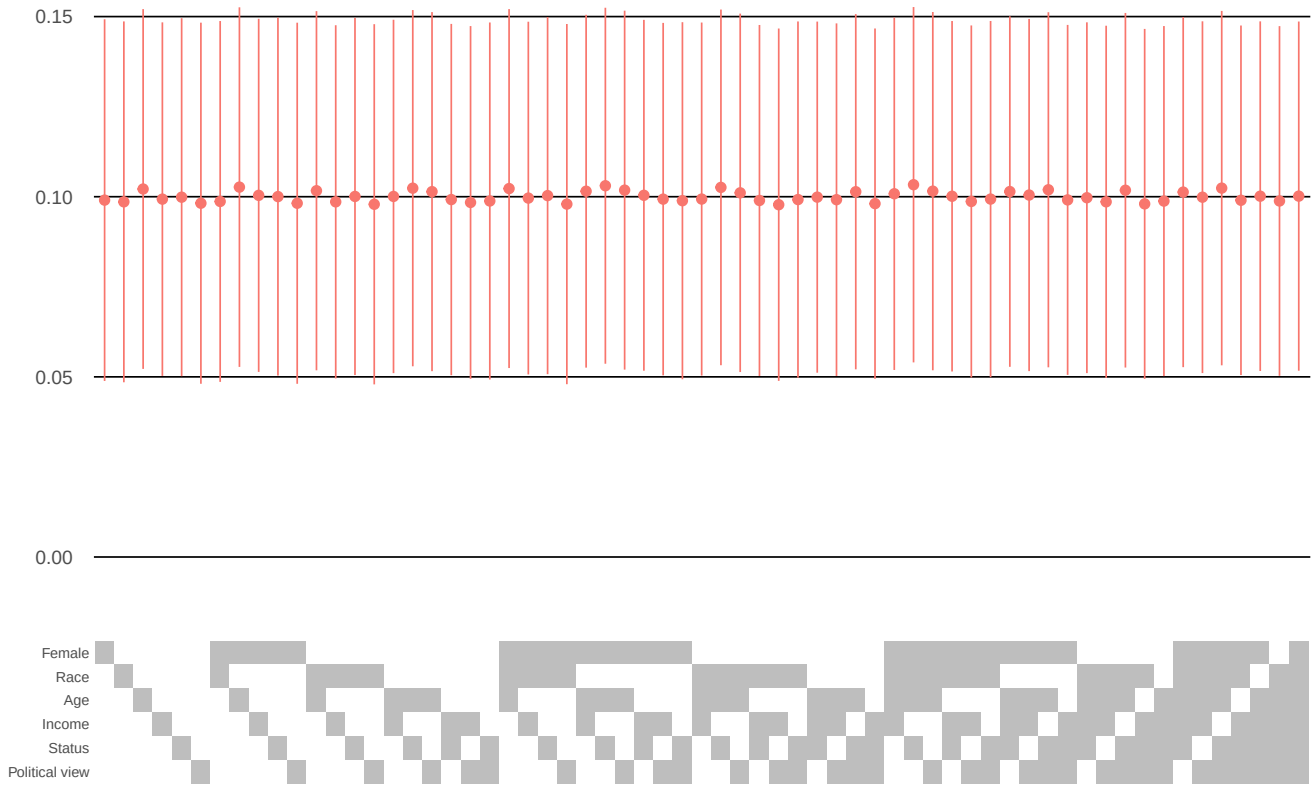


FIGURE A14. Coefficient Stability Plot

Note.—This figure shows the coefficient stability plot for the coefficient on the ESG indicator (β_1) in the following specification: $Interest_{ij} = \alpha + \beta_1 ESG_{ij} + \beta_2 \ln(Wage_{ij}) + \beta_3 NWA_{ij} + \beta_4 FS_{ij} + \mathbf{X}_{ij}\mathbf{\Gamma} + \mathbf{S}_i\mathbf{\Pi} + \text{Strata FE} + \varepsilon_{ij}$. i is the ith individual and j is the jth job posting rated by individual i. *ESG* is an indicator variable equal to one if the job posting displays at least one ESG sentence (see Appendix Table H12) or ESG certification (see Appendix Table H13). ln(*Wage*) is the natural logarithm of the monthly wage displayed in the job posting. *NWA* is equal to the number of nonwage amenities displayed in the job posting. *FS* is an indicator variable equal to one if the job posting displays a signal of financial strength (see Appendix Table H10). $\mathbf{X}_{ij}$ includes job-posting controls, accounting for the number of on-the-job activities, number of on-the-job opportunities, employer industry, employer establishment year, number of job prerequisites, an indicator for whether the job is not located in the respondent's primary chosen city, an indicator for work-from-home arrangements, and an indicator for whether the employer is a multinational. $\mathbf{S}_i$ includes socioeconomic controls for respondent gender, race, age, income, employment status, and political views. *Strata FE* are binary indicators for each combination of respondent education level and preferred professional area. Each point corresponds to the estimated ESG coefficient obtained from a separate regression that includes a different combination of socioeconomic control variables in $\mathbf{S}_i$. Vertical bars denote 95% confidence intervals based on heteroskedasticity-robust standard errors. The lower panel indicates which socioeconomic controls are included in each specification.

TABLE A1. Firm Survey of ESG Practices – Industry Distribution

Industry	Count	Share (%)
Information and Communication	184	17.24
Manufacturing	119	11.15
Retail	112	10.50
Scientific and Technical Activities	111	10.40
Other Service Activities	103	9.65
Construction	91	8.53
Finance, Insurance and Real Estate	85	7.97
Healthcare and Social Services	67	6.28
Agriculture, Livestock and Fishing	55	5.15
Transportation, Storage, and Mail	45	4.22
Electricity and Gas	24	2.25
Extractive Industries	14	1.31
Water, Sewage, and Waste Management	7	0.66
Other	50	4.69
Total	1,067	100

Note.—This table shows the industry distribution of the firms in our firm survey of ESG practices. The survey sampled a total of 1,067 firms.

TABLE A2. Correlation between ESG Implementation and Firm Characteristics

	B Corp Certification (0/1=Yes) (1)	GPTW Certification (0/1=Yes) (2)
Ln(Number of Employees)	0.055***	8.547***
	(0.016)	(0.262)
Ln(Average Wage)	0.148***	15.459***
	(0.035)	(0.517)
Employees with College Degree	0.006***	0.320***
	(0.001)	(0.009)
Observations	1,345,822	506,023
Industry FE	Yes	Yes
State FE	Yes	Yes
Year FE	Yes	Yes

Note.—This table reports the regression coefficients for the following specification: $Certification_{jt} \times 1000 = \alpha + \beta_1 \ln(Number\ of\ Employees_{jt}) + \beta_2 \ln(Average\ Wage_{jt}) + \beta_3 Employees\ with\ College\ Degree\ (\%)_{jt} + Industry\ FE + State\ FE + Year\ FE + e_{jt}$. In this specification, j is the jth firm. $Certification_{jt}$ is an indicator variable equal to one if the firm is certified. Specifically, in Column (1), $Certification_{jt}$ corresponds to the B Corp certification, while in Column (2), it refers to the Great Place to Work (GPTW) certification. In Columns (1) and (2), the dependent variable $Certification_{jt}$ is multiplied by 1000 to scale the coefficients due to the low number of certified firms in Brazil. ln(*Number of Employees*) is the natural logarithm of the firm's total number of employees. ln(*Average Wage*) is the natural logarithm of the average wage at the firm. *Employees with College Degree (%)* is the percentage of employees who have a college degree. In Columns (1) and (2), the sample includes all firms with more than 10 employees operating in the same state and sector as the certified firms, based on administrative data from RAIS. All regressions include industry, state, and year fixed effects. For additional details on RAIS, see Section 5.2. Robust standard errors are reported in parentheses. *p<0.1; **p<0.05; ***p<0.01.

TABLE A3. Job-Seekers' Preferences for Corporate ESG – Complete Raw Sample

	Interest (1)	Interest (2)	Interest (3)	Interest (4)	BRL Terms (5)
ESG	0.078***	0.094***	0.082***	0.067**	334.677***
	(0.026)	(0.025)	(0.019)	(0.027)	(75.940)
Ln(Wage)	-	1.129***	1.212***	1.064***	-
	-	(0.030)	(0.026)	(0.034)	-
Nonwage Amenities	-	0.056***	0.058***	0.055***	224.030***
	-	(0.014)	(0.011)	(0.015)	(42.494)
Financial Strength	-	−0.006	0.015	−0.048	39.916
	-	(0.040)	(0.031)	(0.045)	(123.729)
Observations	25,040	25,040	25,040	25,040	25,040
Individual FE	No	No	Yes	Yes	Yes
Strata FE	Yes	Yes	Yes	Yes	Yes
Job Posting Controls	No	Yes	Yes	Yes	Yes
PNAD Re-Weight	No	No	No	Yes	No

Note.—This table reports regression coefficients for the following specifications. Column (1) specification: $Interest_{ij} = \alpha + \beta_1 ESG_{ij} + \text{Strata FE} + \varepsilon_{ij}$. Column (2) specification: $Interest_{ij} = \alpha + \beta_1 ESG_{ij} + \beta_2 \ln(Wage_{ij}) + \beta_3 NWA_{ij} + \beta_4 FS_{ij} + \mathbf{X}_{ij}\mathbf{\Gamma} + \text{Strata FE} + \varepsilon_{ij}$. Column (3) specification: $Interest_{ij} = \alpha + \beta_1 ESG_{ij} + \beta_2 \ln(Wage_{ij}) + \beta_3 NWA_{ij} + \beta_4 FS_{ij} + \mathbf{X}_{ij}\mathbf{\Gamma} + \text{Strata FE} + \text{Individual FE} + \varepsilon_{ij}$. Column (4) reports the same specification as Column (3), re-weighted using inverse-probability weights constructed from the PNAD household survey. Column (5) reports coefficients from the Column (3) specification estimated in wage levels, with coefficients expressed in BRL terms. i is the ith individual and j is the jth job posting rated by individual i. *ESG* is an indicator variable equal to one if the job posting displays at least one ESG sentence (see Appendix Table H12) or ESG certification (see Appendix Table H13). ln(*Wage*) is the natural logarithm of the monthly wage displayed in the job posting. NWA is equal to the number of nonwage amenities displayed in the job posting. FS is an indicator variable equal to one if the job posting displays a signal of financial strength (see Appendix Table H10). $\mathbf{X}_{ij}$ includes additional job-posting controls, accounting for the number of on-the-job activities, number of on-the-job opportunities, employer industry, employer establishment year, number of job prerequisites, an indicator for whether the job is not located in the respondent's primary chosen city, an indicator for work-from-home arrangements, and an indicator for whether the employer is a multinational. *Strata FE* are binary indicators for each combination of respondent education level and preferred professional area. Robust standard errors are reported in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table A4. Randomization of Job Components

Job Posting Component	Number of Options Chosen	Inclusion Probability	Categories of Analysis Variable
Primary Job Characteristics:			
Job Title	1	1	Specified by survey respondent. See Appendix Tables H1, H2, and H3 for a curated list of job titles by level of education.
Location	1	1	Specified by survey respondent. If multiple cities are selected, *Primary Preference* (80%), *Secondary Preference(s)* (20%). See Appendix Table H4 for a list of cities by state.
Wage	1	1	See Appendix Table H5 for details on wage distribution.
Work Regime	1	0.5	*CLT* (50%), *Service Provider (PJ)* (50%).
General Firm Characteristics:			
Industry	1	1	See Appendix Table H6 for details on firm industry.
Establishment Year	1	1	See Appendix Table H7 for details on establishment year.
Number of Employees	1	1	See Appendix Table H7 for details on number of employees.
Number of Countries	1	1	One country (50%), and two to five countries (12.5% each). See Appendix Table H7 for details on number of countries.
Introductory Sentence	1	1	See Appendix Table H8 (H9) for introductory sentences included in job postings from domestic (multinational) firms.
Financial Strength	1	0.1	*Profit* (25%), *Credit Rating* (25%), *Growth Outlook* (25%), *Bankruptcy Probability* (25%). See Appendix Table H10 for details on financial strength.
Auxiliary Sentence - Firm Description	1	Conditional on no ESG Information – 1	See Appendix Table H14 for details on auxiliary sentences included.
Firm ESG Characteristics:			
ESG Sentences	1–2	80% – 0.2 20% – 0.5	*Environmental* (53%), *Social* (20%), *Governance* (27%). See Appendix Table H12 for details on ESG characteristics.
ESG Certification	1	0.1	*B Corporation* (33.3%), *Great Place to Work* (33.3%) and *Green Business Bureau* (33.3%). See Appendix Table H13 for details on ESG certifications.
General Job Characteristics:			
On-the-Job Opportunities	2–5	0.5	*Mentoring and Training* (33.3%), *Personal Development* (33.3%), *Company Culture* (33.3%). See Appendix Table H15 for opportunities in each category.
On-the-Job Activities	3–5	0.6	See Appendix Tables H16, H17 and H18 for a list of on-the-job activities split by respondent's level of education.
On-the-Job Activities (Sentences)	1	0.5	See Appendix Table H19 for sentences describing on-the-job activities split by professional area.
Workload Requirement	1	0.7	See Appendix Table H20 for details on estimated workload requirements.
Work-from-Home	1	Completed High School – 0.3 Technical School – 0 Completed College – 0.3	See Appendix Table H21 for details on work-from-home options.
Auxiliary Sentence - Job Opening	1	1	See Appendix Table H22 for details on auxiliary sentences included.
Job Prerequisites:			
Job Prerequisites	2–3	0.5	See Appendix Table H23 for details on job prerequisites.
Required Majors	1	Completed High School – 0 Technical School – 0 Completed College – 1	Specified by survey respondent. See Appendix Table H24 for the major requirements split by professional area presented to respondents that completed college.
Hiring Stages:			
Stage 1 - Application	1	1	See Appendix Table H25 for details on hiring stages.
Stage 2 - Online Assessments	1	Completed High School – 0 Technical School – 0 Completed College – 1	See Appendix Table H25 for details on hiring stages.
Stage 3 - Other Assessments	1	Completed High School – 0 Technical School – 0 Completed College – 1	See Appendix Table H25 for details on hiring stages.
Stage 4 - Final Interview	1	1	See Appendix Table H25 for details on hiring stages.
Nonwage Amenities:			
Nonwage Amenities	2–4	1	*Benefits* (70%), and *Amenities* (30%). See Appendix Table H26 for nonwage amenities in each category.

Note.—This table presents the components of each synthetic job posting and each associated number of options chosen, inclusion probability, and categories of the analysis variable. *Job Posting Component* specifies the component that is randomized and contained in the job posting, presented in descending order of appearance. *Number of Options Chosen* is the number of analysis variables that are randomized and included in the job posting. *Inclusion Probability* is the probability that the *Job Posting Component* appears in the hypothetical job posting. *Categories of Analysis Variable* is the category of the randomized characteristic included in each hypothetical job posting and the associated weight as a percentage in parentheses representing their selection probability (e.g., each job posting presented has a 50% chance of appearing from a domestic firm).

TABLE A5. Job-Seekers' Preferences for Corporate ESG Sentences & Certifications

	Interest (1)	BRL Terms (2)	Interest (3)	BRL Terms (4)
ESG Sentence	0.057**	232.422***	-	-
	(0.022)	(89.064)	-	-
Environmental	-	-	0.076***	301.305***
	-	-	(0.027)	(108.248)
Social	-	-	0.017	61.059
	-	-	(0.041)	(162.923)
Governance	-	-	0.025	137.305
	-	-	(0.036)	(140.726)
Certification	0.092***	395.004***	0.093***	396.756***
	(0.029)	(113.766)	(0.029)	(113.634)
Ln(Wage)	1.202***	-	1.203***	-
	(0.026)	-	(0.026)	-
Nonwage Amenities	0.062***	239.472***	0.062***	239.822***
	(0.011)	(43.685)	(0.011)	(43.647)
Financial Strength	0.012	26.146	0.013	30.678
	(0.032)	(127.254)	(0.032)	(127.185)
Observations	24,120	24,120	24,120	24,120
Individual FE	Yes	Yes	Yes	Yes
Strata FE	Yes	Yes	Yes	Yes
Job Posting Controls	Yes	Yes	Yes	Yes

Note.—This table reports regression coefficients for specifications that estimate job-seeker preferences for corporate ESG signals and certification. Columns (1) and (2) use an indicator variable equal to one if the job posting displays at least one environmental, social, or governance sentence. Columns (3) and (4) use separate indicators for environmental, social, and governance sentences. Column (1) specification: $Interest_{ij} = \alpha + \beta_1 ESGSentence_{ij} + \beta_2 Certification_{ij} + \beta_3 \ln(Wage_{ij}) + \beta_4 NWA_{ij} + \beta_5 FS_{ij} + \mathbf{X}_{ij}\mathbf{\Gamma} +$ Strata FE + Individual FE + ε_{ij}. Column (3) specification: $Interest_{ij} = \alpha + \beta_1 Environmental_{ij} + \beta_2 Social_{ij} + \beta_3 Governance_{ij} + \beta_4 Certification_{ij} + \beta_5 \ln(Wage_{ij}) + \beta_6 NWA_{ij} + \beta_7 FS_{ij} + \mathbf{X}_{ij}\mathbf{\Gamma} +$ Strata FE+Individual FE+ε_{ij}. Columns (2) and (4) report the corresponding effects in BRL terms, computed from the same specifications estimated in wage levels and expressed as the ratio of each coefficient to the wage coefficient using the delta method. i is the ith individual and j is the jth job posting rated by individual i. *ESGSentence* is an indicator variable equal to one if the job posting displays at least one environmental, social, or governance sentence. *Environmental*, *Social*, and *Governance* are indicators equal to one if the job posting displays a sentence in the corresponding ESG category. *Certification* is an indicator variable equal to one if the job posting displays any certification, including B Corp, Green Business Bureau, or Great Place to Work (GPTW). ln(*Wage*) is the natural logarithm of the monthly wage displayed in the job posting. *NWA* is equal to the number of nonwage amenities displayed in the job posting. *FS* is an indicator variable equal to one if the job posting displays a signal of financial strength. $\mathbf{X}_{ij}$ includes additional job-posting controls, which account for the number of on-the-job activities, number of on-the-job opportunities, employer industry, employer establishment year, number of job prerequisites, an indicator for whether the job is not located in the respondent's primary chosen city, an indicator for work-from-home arrangements, and an indicator for whether the employer is a multinational firm. *Strata FE* are binary indicators for each combination of respondent education level and preferred professional area. Robust standard errors are reported in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

TABLE A6. Job-Seekers' Perceived Reciprocal Interest in Job Postings

	Reciprocal Interest (1)	Reciprocal Interest (2)	Reciprocal Interest (3)	Reciprocal Interest (4)
ESG	0.013	0.022	0.029	0.034
	(0.024)	(0.024)	(0.018)	(0.024)
Ln(Wage)	-	0.491***	0.524***	0.466***
	-	(0.029)	(0.023)	(0.030)
Nonwage Amenities	-	0.035***	0.039***	0.049***
	-	(0.013)	(0.010)	(0.013)
Financial Strength	-	−0.010	0.003	−0.029
	-	(0.039)	(0.028)	(0.040)
Observations	24,120	24,120	24,120	24,120
Individual FE	No	No	Yes	Yes
Strata FE	Yes	Yes	Yes	Yes
Job Posting Controls	No	Yes	Yes	Yes
PNAD Re-Weight	No	No	No	Yes

Note.—This table reports regression coefficients for the following specifications. Column (1) specification: $ReciprocalInterest_{ij} = \alpha + \beta_1 ESG_{ij} +$ Strata FE$+\varepsilon_{ij}$. Column (2) specification: $ReciprocalInterest_{ij} = \alpha + \beta_1 ESG_{ij} + \beta_2 \ln(Wage_{ij}) + \beta_3 NWA_{ij} + \beta_4 FS_{ij} + \mathbf{X}_{ij}\mathbf{\Gamma} +$ Strata FE$+\varepsilon_{ij}$. Column (3) specification: $ReciprocalInterest_{ij} = \alpha + \beta_1 ESG_{ij} + \beta_2 \ln(Wage_{ij}) + \beta_3 NWA_{ij} + \beta_4 FS_{ij} + \mathbf{X}_{ij}\mathbf{\Gamma} +$ Strata FE + Individual FE$+\varepsilon_{ij}$. Column (4) reports the same specification as Column (3), re-weighted using inverse-probability weights constructed from the PNAD household survey. i is the ith individual and j is the jth job posting rated by individual i. $ReciprocalInterest_{ij}$ is the respondent's perceived likelihood of being hired for job posting j. ESG is an indicator variable equal to one if the job posting displays at least one ESG sentence (see Appendix Table H12) or ESG certification (see Appendix Table H13). $\ln(Wage)$ is the natural logarithm of the monthly wage displayed in the job posting. NWA is equal to the number of nonwage amenities displayed in the job posting. FS is an indicator variable equal to one if the job posting displays a signal of financial strength (see Appendix Table H10). $\mathbf{X}_{ij}$ includes additional job-posting controls, which account for the number of on-the-job activities, number of on-the-job opportunities, employer industry, employer establishment year, number of job prerequisites, an indicator for whether the job is not located in the respondent's primary chosen city, an indicator for work-from-home arrangements, and an indicator for whether the employer is a multinational. $Strata\ FE$ are binary indicators for each combination of respondent education level and preferred professional area. Robust standard errors are reported in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

TABLE A7. Job-Seekers' Preferences for Corporate ESG – Re-Weighted

	Interest (1)	Interest (2)	Interest (3)	BRL Terms (4)
	A. Re-Weighted Using PNAD			
ESG	0.077**	0.085**	0.070**	288.529***
	(0.036)	(0.036)	(0.027)	(104.971)
Ln(Wage)	-	0.948***	1.050***	-
	-	(0.043)	(0.034)	-
Nonwage Amenities	-	0.056***	0.061***	231.107***
	-	(0.019)	(0.015)	(57.871)
Financial Strength	-	−0.081	−0.057	−231.641
	-	(0.058)	(0.045)	(173.873)
	B. Re-Weighted Using RAIS			
ESG	0.096**	0.105**	0.079**	292.990**
	(0.047)	(0.047)	(0.036)	(125.320)
Ln(Wage)		0.915***	1.049***	-
		(0.060)	(0.049)	-
Nonwage Amenities		0.057**	0.069***	237.005***
		(0.025)	(0.020)	(70.867)
Financial Strength		−0.110	−0.064	−234.766
		(0.076)	(0.059)	(206.151)
Observations	24,120	24,120	24,120	24,120
Individual FE	No	No	Yes	Yes
Strata FE	Yes	Yes	Yes	Yes
Job Posting Controls	No	Yes	Yes	Yes

Note.—This table reports regression coefficients for re-weighted versions of the specifications reported in Table 3. Panel A reports estimates re-weighted using inverse-probability weights constructed from the PNAD household survey. Panel B reports estimates re-weighted using inverse-probability weights constructed from the RAIS administrative employment records. Column (1) specification: $Interest_{ij} = \alpha + \beta_1 ESG_{ij} + \text{Strata FE} + \varepsilon_{ij}$. Column (2) specification: $Interest_{ij} = \alpha + \beta_1 ESG_{ij} + \beta_2 \ln(Wage_{ij}) + \beta_3 NWA_{ij} + \beta_4 FS_{ij} + \mathbf{X}_{ij}\mathbf{\Gamma} + \text{Strata FE} + \varepsilon_{ij}$. Column (3) specification: $Interest_{ij} = \alpha + \beta_1 ESG_{ij} + \beta_2 \ln(Wage_{ij}) + \beta_3 NWA_{ij} + \beta_4 FS_{ij} + \mathbf{X}_{ij}\mathbf{\Gamma} + \text{Strata FE} + \text{Individual FE} + \varepsilon_{ij}$. Column (4) reports coefficients from the Column (3) specification estimated in wage levels, with coefficients expressed in BRL terms. i is the ith individual and j is the jth job posting rated by individual i. ESG is an indicator variable equal to one if the job posting displays at least one ESG sentence (see Appendix Table H12) or ESG certification (see Appendix Table H13). ln(*Wage*) is the natural logarithm of the monthly wage displayed in the job posting. NWA is equal to the number of nonwage amenities displayed in the job posting. FS is an indicator variable equal to one if the job posting displays a signal of financial strength (see Appendix Table H10). $\mathbf{X}_{ij}$ includes additional job-posting controls, accounting for the number of on-the-job activities, number of on-the-job opportunities, employer industry, employer establishment year, number of job prerequisites, an indicator for whether the job is not located in the respondent's primary chosen city, an indicator for work-from-home arrangements, and an indicator for whether the employer is a multinational. *Strata FE* are binary indicators for each combination of respondent education level and preferred professional area. Robust standard errors are reported in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

TABLE A8. Job-Seekers' Preferences for Corporate ESG

	Interest (1)	Interest (2)	Interest (3)	Interest (4)	BRL Terms (5)
	A. Excluding GPTW from ESG Indicator				
ESG	0.087***	0.095***	0.072***	0.057**	301.042***
	(0.027)	(0.026)	(0.020)	(0.028)	(80.467)
GPTW Certification	0.023	0.087	0.146***	0.122*	548.692***
	(0.068)	(0.066)	(0.050)	(0.068)	(197.102)
Ln(Wage)	-	1.115***	1.203***	1.051***	-
	-	(0.031)	(0.026)	(0.034)	-
Nonwage Amenities	-	0.058***	0.062***	0.061***	239.605***
	-	(0.014)	(0.011)	(0.015)	(43.654)
Financial Strength	-	−0.005	0.013	−0.057	29.032
	-	(0.041)	(0.032)	(0.045)	(127.205)
Observations	24,120	24,120	24,120	24,120	24,120
	B. Excluding Job Postings with GPTW Certification				
ESG	0.089***	0.099***	0.074***	0.061**	311.092***
	(0.027)	(0.027)	(0.020)	(0.029)	(81.371)
Ln(Wage)	-	1.114***	1.207***	1.056***	-
	-	(0.031)	(0.026)	(0.035)	-
Nonwage Amenities	-	0.056***	0.059***	0.056***	230.709***
	-	(0.014)	(0.011)	(0.015)	(44.308)
Financial Strength	-	−0.009	0.016	−0.064	39.530
	-	(0.042)	(0.033)	(0.046)	(128.861)
Observations	23,392	23,392	23,392	23,392	23,392
Individual FE	No	No	Yes	Yes	Yes
Strata FE	Yes	Yes	Yes	Yes	Yes
Job Posting Controls	No	Yes	Yes	Yes	Yes
PNAD Re-Weight	No	No	No	Yes	No

Note.—This table reports regression coefficients for the following specifications. Panel A reports estimates where the GPTW certification is excluded from the ESG indicator and enters separately as its own regressor. Panel A Column (1) specification: $Interest_{ij} = \alpha + \beta_1 ESG_{ij} + \beta_2 GPTW_{ij} + \text{Strata FE} + \varepsilon_{ij}$. Column (2) specification: $Interest_{ij} = \alpha + \beta_1 ESG_{ij} + \beta_2 GPTW_{ij} + \beta_3 \ln(Wage_{ij}) + \beta_4 NWA_{ij} + \beta_5 FS_{ij} + \mathbf{X}_{ij}\mathbf{\Gamma} + \text{Strata FE} + \varepsilon_{ij}$. Column (3) specification: $Interest_{ij} = \alpha + \beta_1 ESG_{ij} + \beta_2 GPTW_{ij} + \beta_3 \ln(Wage_{ij}) + \beta_4 NWA_{ij} + \beta_5 FS_{ij} + \mathbf{X}_{ij}\mathbf{\Gamma} + \text{Strata FE} + \text{Individual FE} + \varepsilon_{ij}$. Column (4) reports the same specification as Column (3), re-weighted using inverse-probability weights constructed from the PNAD household survey. Column (5) reports coefficients from the Column (3) specification estimated in wage levels, with coefficients expressed in BRL terms. Panel B reports estimates where all job postings displaying a GPTW certification are excluded from the sample; the same specifications are estimated, but the separate GPTW certification indicator is omitted. i is the ith individual and j is the jth job posting rated by individual i. ESG is an indicator variable equal to one if the job posting displays at least one ESG sentence (see Appendix Table H12) or ESG certification (see Appendix Table H13) other than GPTW. $GPTW$ is an indicator variable equal to one if the job posting displays a Great Place to Work certification. $\ln(Wage)$ is the natural logarithm of the monthly wage displayed in the job posting. NWA is equal to the number of nonwage amenities displayed in the job posting. FS is an indicator variable equal to one if the job posting displays a signal of financial strength (see Appendix Table H10). $\mathbf{X}_{ij}$ includes additional job-posting controls, accounting for the number of on-the-job activities, number of on-the-job opportunities, employer industry, employer establishment year, number of job prerequisites, an indicator for whether the job is not located in the respondent's primary chosen city, an indicator for work-from-home arrangements, and an indicator for whether the employer is a multinational. $Strata\ FE$ are binary indicators for each combination of respondent education level and preferred professional area. Robust standard errors are reported in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

TABLE A9. Preferences for Corporate ESG with Company Culture and Wellness Program Interactions

	Company Culture (1)	Wellness Program (2)
ESG	0.089***	0.077***
	(0.025)	(0.021)
ESG × Company Culture	−0.003	-
	(0.041)	-
ESG × Wellness Program	-	0.126*
	-	(0.071)
Company Culture	−0.011	-
	(0.031)	-
Wellness Program	-	−0.069*
	-	(0.036)
Wage	0.255***	0.255***
	(0.007)	(0.007)
Nonwage Amenities	0.061***	0.062***
	(0.011)	(0.011)
Financial Strength	0.007	0.007
	(0.032)	(0.032)
Observations	24,120	24,120
Individual FE	Yes	Yes
Strata FE	Yes	Yes
Job Posting Controls	Yes	Yes

Note.—This table reports coefficients from specifications that allow the effect of ESG to differ depending on whether the job posting also displays a company-culture feature or a wellness program. Column (1) estimates: $Interest_{ij} = \alpha + \beta_1 ESG_{ij} + \beta_2(ESG_{ij} \times Company\ Culture_{ij}) + \beta_3 Company\ Culture_{ij} + \beta_4 Wage_{ij} + \beta_5 NWA_{ij} + \beta_6 FS_{ij} + \mathbf{X}_{ij}\boldsymbol{\Gamma} + \text{Strata FE} + \text{Individual FE} + \varepsilon_{ij}$. Column (2) estimates an analogous specification in which ESG is interacted with a wellness program. ESG is the standard indicator for the presence of any ESG sentence or ESG certification. i is the ith individual and j is the jth job posting evaluated by individual i. *Company Culture* indicates an on-the-job opportunity in the company-culture category. $Wellness$ indicates a wellness program. $Wage$ is the monthly wage displayed in the job posting, measured in thousands of Brazilian reais. NWA is the number of nonwage amenities, and FS indicates financial strength. $\mathbf{X}_{ij}$ includes job-posting controls accounting for the number of on-the-job activities, number of on-the-job opportunities, employer industry, employer establishment year, number of job prerequisites, an indicator for whether the job is not located in the respondent's primary chosen city, an indicator for work-from-home arrangements, and an indicator for whether the employer is a multinational. *Strata FE* are binary indicators for each combination of respondent education level and preferred professional area. Robust standard errors are reported in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

TABLE A10. Job-Seekers' Preferences for Corporate ESG – Robustness Check Removing Uncommon Job Postings

	Baseline Sample As Reported in Table 3 (1)	Excluding Wage Outliers Bottom/Top 5% (2)	Excluding Wage Outliers Bottom/Top 10% (3)	Excluding Wage Outliers Bottom/Top 25% (4)	Excluding High-Paying Jobs Outside Capitals (5)
ESG	0.084***	0.085***	0.087***	0.105***	0.087***
	(0.020)	(0.020)	(0.021)	(0.027)	(0.022)
Ln(Wage)	1.203***	1.217***	1.247***	1.328***	1.188***
	(0.026)	(0.032)	(0.038)	(0.083)	(0.029)
Nonwage Amenities	0.062***	0.058***	0.055***	0.069***	0.066***
	(0.011)	(0.011)	(0.012)	(0.015)	(0.012)
Financial Strength	0.012	0.023	0.034	0.114***	0.014
	(0.032)	(0.033)	(0.034)	(0.044)	(0.036)
Observations	24,120	22,118	20,040	12,713	20,108
Individual FE	Yes	Yes	Yes	Yes	Yes
Strata FE	Yes	Yes	Yes	Yes	Yes
Job Posting Controls	Yes	Yes	Yes	Yes	Yes
	Excluding Outlier High-Paying Jobs at Non-Multinationals (6)	Excluding Posts with Excessive ESG Information (7)	Excluding Posts with More than 4 Nonwage Amenities (8)	Excluding Uncommon Amenities for Typical Employers (9)	Excluding Uncommon Amenities for Low-Skill Employers (10)
ESG	0.087***	0.077***	0.086***	0.082***	0.086***
	(0.020)	(0.022)	(0.027)	(0.020)	(0.020)
Ln(Wage)	1.222***	1.198***	1.195***	1.197***	1.204***
	(0.030)	(0.027)	(0.035)	(0.027)	(0.026)
Nonwage Amenities	0.061***	0.056***	0.058***	0.062***	0.062***
	(0.011)	(0.011)	(0.016)	(0.011)	(0.011)
Financial Strength	0.012	0.008	0.012	0.018	0.017
	(0.033)	(0.033)	(0.047)	(0.033)	(0.032)
Observations	22,829	22,062	13,315	23,078	23,673
Individual FE	Yes	Yes	Yes	Yes	Yes
Strata FE	Yes	Yes	Yes	Yes	Yes
Job Posting Controls	Yes	Yes	Yes	Yes	Yes

Note.—This table reports regression coefficients from the specification $Interest_{ij} = \alpha + \beta_1 ESG_{ij} + \beta_2 \ln(Wage_{ij}) + \beta_3 NWA_{ij} + \beta_4 FS_{ij} + \mathbf{X}_{ij}\mathbf{\Gamma} + \text{Strata FE} + \text{Individual FE} + \varepsilon_{ij}$. i is the ith individual and j is the jth job posting rated by individual i. ESG is an indicator variable equal to one if the job posting displays at least one ESG sentence (see Appendix Table H12) or ESG certification (see Appendix Table H13). $\ln(Wage)$ is the natural logarithm of the monthly wage displayed in the job posting. NWA is equal to the number of nonwage amenities shown in the job posting. FS is an indicator variable equal to one if the job posting displays a signal of financial strength (see Table H10). $\mathbf{X}_{ij}$ includes additional job-posting controls, which account for the number of on-the-job activities, number of on-the-job opportunities, employer industry, employer establishment year, number of job prerequisites, an indicator for whether the job posting is not located in the respondent's primary chosen city, an indicator for work-from-home arrangements, and an indicator for whether the employer is a multinational. $Strata\ FE$ are binary indicators for each combination of respondent education level and preferred professional area. Column (1) is the baseline with the full sample. In Columns (2) to (10), we exclude certain job postings based on specific criteria. Column (2) removes job postings in the bottom or top 5% of the wage distribution per educational level and professional area; Column (3) applies the same exclusion for the bottom or top 10%; Column (4) for the bottom or top 25%. Column (5) drops job postings outside state capitals that, within the set of 20 job postings presented to each respondent, offer wages higher than the lowest-paying job in capital cities. Column (6) excludes job postings where non-multinational companies, within the set of 20 job postings presented to each respondent, offer higher wages than multinational companies. Column (7) excludes job postings that contain both an ESG sentence and certification or include two ESG sentences. Column (8) removes job postings where the total number of nonwage amenities and on-the-job opportunities exceeds four. Column (9) excludes job postings that feature startup-specific amenities uncommon in Brazil, such as in-office gyms, training platforms, wellness programs, physical activity support, gym memberships, educational assistance programs, personal support programs, personal development, mentoring, and training. Column (10) applies the same exclusions as Column (9), but only for individuals whose highest educational attainment is high school or technical school. Robust standard errors are reported in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

TABLE A11. Preferences for Corporate ESG by Sociodemographic Subsamples

	High Education (1)	Low Education (2)	White (3)	Non-White (4)	Liberal or Moderate (5)	Conservative (6)	Young (7)	Older (8)	Female (9)	Male (10)
	A. Coefficients									
ESG	0.126***	0.031	0.122***	0.050*	0.131***	0.052*	0.072***	0.111***	0.088***	0.092***
	(0.026)	(0.030)	(0.027)	(0.029)	(0.028)	(0.027)	(0.025)	(0.031)	(0.031)	(0.026)
Wage	0.233***	0.431***	0.255***	0.256***	0.265***	0.244***	0.255***	0.255***	0.247***	0.260***
	(0.007)	(0.022)	(0.009)	(0.011)	(0.009)	(0.010)	(0.009)	(0.010)	(0.011)	(0.009)
Nonwage Amenities	0.058***	0.066***	0.060***	0.061***	0.051***	0.071***	0.065***	0.052***	0.069***	0.059***
	(0.015)	(0.016)	(0.015)	(0.016)	(0.016)	(0.015)	(0.014)	(0.018)	(0.017)	(0.014)
Financial Strength	0.055	−0.062	0.030	−0.018	−0.040	0.051	0.017	−0.019	0.031	−0.007
	(0.044)	(0.049)	(0.046)	(0.046)	(0.046)	(0.045)	(0.042)	(0.052)	(0.049)	(0.043)
	B. BRL Terms									
ESG	542.094***	71.890	480.576***	193.295*	493.908***	211.678*	281.896***	436.289***	354.636***	352.951***
	(112.208)	(70.872)	(108.696)	(111.784)	(108.944)	(112.425)	(100.138)	(124.120)	(126.085)	(99.160)
Nonwage Amenities	246.652***	152.646***	235.714***	239.191***	193.910***	289.480***	256.210***	202.645***	280.564***	226.010***
	(64.561)	(38.760)	(60.968)	(62.628)	(60.807)	(63.458)	(56.181)	(69.648)	(70.734)	(55.567)
Financial Strength	236.371	−144.886	117.536	−71.217	−152.677	210.667	66.352	−75.817	127.468	−26.661
	(187.714)	(113.076)	(182.524)	(177.976)	(175.007)	(186.544)	(162.806)	(204.075)	(198.669)	(166.294)
Observations	13,580	10,540	12,520	11,600	11,500	12,620	14,880	9,240	10,360	13,760
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Strata FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Job Posting Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note.—This table reports job-seeker preferences for corporate ESG by sociodemographic subgroup. Panel A reports regression coefficients from specifications estimated separately by subgroup. Panel B reports the corresponding wage-equivalent values in Brazilian reais. Each subgroup regression includes individual fixed effects, strata fixed effects, and job-posting controls. The BRL values in Panel B are obtained by dividing each coefficient by the estimated wage coefficient from the same subgroup-specific specification estimated with wage in levels; standard errors in Panel B are computed using the delta method. ESG is an indicator variable equal to one if the job posting displays at least one ESG sentence or ESG certification. $Wage$ is the monthly wage displayed in the job posting, measured in thousands of Brazilian reais in Panel A. NWA is equal to the number of nonwage amenities displayed in the job posting. FS is an indicator variable equal to one if the job posting displays a signal of financial strength. Job-posting controls account for the number of on-the-job activities, number of on-the-job opportunities, employer industry, employer establishment year, number of job prerequisites, an indicator for whether the job is not located in the respondent's primary chosen city, an indicator for work-from-home arrangements, and an indicator for whether the employer is a multinational firm. *Strata FE* are binary indicators for each combination of respondent education level and preferred professional area. Robust standard errors are reported in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

TABLE A12. Rank-Ordered Logit by Sociodemographic Subsample

	Not Highly Educated (1)	Highly Educated (2)
ESG	0.020	0.144***
	(0.028)	(0.025)
Ln(Wage)	1.066***	0.945***
	(0.041)	(0.027)
Nonwage amenities	0.046***	0.052***
	(0.015)	(0.014)
Financial Strength	-0.074*	0.077*
	(0.044)	(0.040)
Observations	10,540	13,580
Number of Groups	527	679
Job Posting Controls	Yes	Yes

Note.—This table reports the rank-ordered logit regression coefficients, with the underlying choice probabilities given by equation (5.13). The regressors are ESG, ln(*Wage*), NWA, FS, and additional job-posting controls. We estimate a rank-ordered logit regression for each demographic subsample based on the characteristic of interest, *Highly Educated*, which is an indicator for whether the respondent has obtained a college degree. In Column (1), we subsample for respondents who are not highly educated. In Column (2), we subsample for respondents who are highly educated. Each respondent's ranking of the 20 job postings is constructed from their *Interest* ratings (1–7), with tied ratings accounted for using the exact marginal likelihood. ESG is an indicator variable equal to one if the job posting displays at least one ESG sentence (see Appendix Table H12) or ESG certification (see Appendix Table H13). ln(*Wage*) is the natural logarithm of the monthly wage displayed in the job posting. NWA is equal to the number of nonwage amenities. FS is an indicator variable equal to one if the job posting displays a signal of financial strength (see Appendix Table H10). $^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$.

TABLE A13. Financial Strength and ESG Interaction

	Full Sample (1)	Low Education (2)	High Education (3)
ESG	0.085***	0.018	0.135***
	(0.020)	(0.032)	(0.027)
ESG x Financial Strength	−0.009	0.064	−0.070
	(0.072)	(0.112)	(0.096)
Ln(Wage)	1.202***	1.256***	1.182***
	(0.026)	(0.046)	(0.032)
Nonwage Amenities	0.062***	0.067***	0.057***
	(0.011)	(0.016)	(0.015)
Financial Strength	0.015	−0.072	0.083
	(0.038)	(0.056)	(0.052)
Observations	24,120	10,540	13,580
Individual FE	Yes	Yes	Yes
Strata FE	Yes	Yes	Yes
Job Posting Controls	Yes	Yes	Yes

Note.—This table reports the coefficients for the following specification: $Interest_{ij} = \alpha + \beta_1 ESG_{ij} + \beta_2 ESG_{ij} \times FS_{ij} + \beta_3 \ln(Wage_{ij}) + \beta_4 NWA_{ij} + \beta_5 FS_{ij} + \mathbf{X}_{ij}\mathbf{\Gamma} + \text{Strata FE} + \text{Individual FE} + \varepsilon_{ij}$. i is the ith individual and j is the jth job posting rated by individual i. ESG is an indicator variable equal to one if the job posting displays at least one ESG sentence or ESG certification. FS is an indicator variable equal to one if the job posting displays a signal of financial strength. ln(*Wage*) is the natural logarithm of the monthly wage. NWA is the number of nonwage amenities displayed. $\mathbf{X}_{ij}$ includes additional job-posting controls, accounting for the number of on-the-job activities, number of on-the-job opportunities, employer industry, employer establishment year, number of job prerequisites, an indicator for whether the job is not located in the respondent's primary chosen city, an indicator for work-from-home arrangements, and an indicator for whether the employer is a multinational. *Strata FE* are binary indicators for each combination of respondent education level and preferred professional area. Column (1) uses the full sample. Column (2) restricts the sample to low-education (not college educated) respondents. Column (3) restricts the sample to high-education (college educated) respondents. Robust standard errors are reported in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

TABLE A14. Education and Prosocial Policy Preferences

	Support for Equality-of-Opportunity Policies (1-5)	Share (%) Budget Allocation to Education	Support for Estate Tax (1-5)
College	0.088^{***}	0.566^{***}	0.111^{***}
	(0.027)	(0.207)	(0.034)
Observations	8,997	8,997	8,996
Demographic Controls	Yes	Yes	Yes
Country FE	Yes	Yes	Yes

Note.—This table reports estimates based on the survey data from Alesina, Stantcheva, and Teso (2018). The table reports coefficients from the following specification: $Y_{ic} = \alpha + \beta\, College_i + \mathbf{X}_i\mathbf{\Gamma} + \text{Country FE}_c + \varepsilon_{ic}$, where i indexes individuals and c indexes countries. The dependent variables are, respectively, support for equality-of-opportunity policies (measured on a 1–5 scale), the preferred share of the national budget allocated to education (in percentage points), and support for the estate tax (measured on a 1–5 scale). $College_i$ is an indicator equal to one if the respondent has completed a college degree. $\mathbf{X}_i$ includes demographic controls for gender, age, marital status, number of children, employment status, household income, place of birth, and experimental treatment assignment. Country fixed effects absorb all time-invariant cross-country differences. Robust standard errors are reported in parentheses. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

TABLE A15. Education and Environmental Engagement

	Social Activism: Donated Group or Campaign (1/0)	Prioritizes Environment Over Economic Growth (1/0)	Member of Any Environmental Organization (1/0)	Active Member of an Environmental Organization (1/0)	High Confidence in Environmental Protection Movement (1/0)
College	0.106***	0.082***	0.023***	0.011***	0.025***
	(0.004)	(0.004)	(0.003)	(0.002)	(0.004)
College × Brazil	0.022	−0.034	−0.012	−0.001	0.040
	(0.035)	(0.034)	(0.015)	(0.011)	(0.033)
Observations	85,196	81,088	86,172	86,172	81,850
Demographic Controls	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes

Note.—This table reports estimates using World Values Survey data from 2017–2022. This table reports coefficients from the following specification: $Y_{ict} = \alpha + \beta_1 College_i + \beta_2 (College_i \times Brazil_c) + \mathbf{X}_i\mathbf{\Gamma} + \text{Country FE}_c + \text{Year FE}_t + \varepsilon_{ict}$, where i indexes individuals, c indexes countries, and t indexes survey years. The dependent variables are indicators for: donating to a social group or campaign, prioritizing environmental protection over economic growth, being a member of an environmental organization, being an active member of an environmental organization, and expressing high confidence in the environmental protection movement. $College_i$ is an indicator equal to one if the respondent has completed a college degree. $Brazil_c$ is an indicator for respondents residing in Brazil. $\mathbf{X}_i$ includes controls for gender, age, marital status, number of children, employment status, household income, and whether the respondent was born in the country. Country and year fixed effects absorb cross-country and time-specific shocks. Robust standard errors are reported in parentheses. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

TABLE A16. AKM Individual FE Prediction

	Individual FE
White	0.125^{***}
	(0.000)
Completed College	0.500^{***}
	(0.000)
Female	-0.191^{***}
	(0.000)
Observations	78,912,958

Note.—This table reports estimates from a regression of AKM individual fixed effects on observable worker characteristics. The specification estimated is: $AKM\ Individual\ FE_i = \alpha + \beta_1 Female_i + \beta_2 White_i + \beta_3 CollegeMore_i + \varepsilon_i$. $AKM\ Individual\ FE_i$ denotes the individual fixed effect estimated from an AKM wage regression using RAIS administrative data from 2002 to 2019. $Female_i$ is an indicator equal to one if the individual is female, $White_i$ is an indicator for race, and $CollegeMore_i$ is an indicator equal to one if the individual has completed college or more. Robust standard errors are reported in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

TABLE A17. AKM Individual FE Quartile and Grouped Quartile Heterogeneity

	(1) Interest Continuous	(2) Interest Quartile Indicators	(3) Interest Grouped Quartiles
ESG	0.044*	−0.025	−0.025
	(0.024)	(0.035)	(0.035)
Ln(Wage)	1.204***	1.204***	1.205***
	(0.026)	(0.026)	(0.026)
Nonwage Amenities	0.061***	0.061***	0.061***
	(0.011)	(0.011)	(0.011)
Financial Strength	0.012	0.013	0.012
	(0.032)	(0.032)	(0.032)
ESG x AKM FE	0.203***	-	-
	(0.068)	-	-
ESG x Q2 AKM FE	-	0.208***	-
	-	(0.055)	-
ESG x Q3 AKM FE	-	0.127**	-
	-	(0.053)	-
ESG x Q4 AKM FE	-	0.148***	-
	-	(0.054)	-
ESG x Q2–Q4 AKM FE	-	-	0.160***
	-	-	(0.042)
Observations	24,120	24,120	24,120
Individual FE	Yes	Yes	Yes
Strata FE	Yes	Yes	Yes
Job Posting Controls	Yes	Yes	Yes

Note.—This table reports regression coefficients from specifications examining heterogeneous preferences for ESG across predicted AKM individual fixed effects, which proxy for workers' labor-market productivity component. Each column reports estimates from a separate regression of the following form: $Interest_{ij} = \alpha + \beta_1(ESG_{ij} \times Q_i) + \beta_2 ESG_{ij} + \beta_3 \ln(Wage_{ij}) + \beta_4 NWA_{ij} + \beta_5 FS_{ij} + \mathbf{X}_{ij}\mathbf{\Gamma} + \text{Strata FE} + \text{Individual FE} + \varepsilon_{ij}$. i is the ith individual and j is the jth job posting rated by individual i. In Column (1), Q_i is the continuous predicted AKM individual fixed effect. In Column (2), Q_i is replaced by indicator variables for AKM FE quartiles Q2, Q3, and Q4 (with Q1 omitted). In Column (3), Q_i is an indicator equal to one if the individual belongs to AKM FE quartiles Q2–Q4 (with Q1 omitted). ESG is an indicator variable equal to one if the job posting displays at least one ESG sentence (see Appendix Table H12) or ESG certification (see Appendix Table H13). ln(*Wage*) is the natural logarithm of the monthly wage displayed in the job posting. NWA is equal to the number of nonwage amenities displayed in the job posting. FS is an indicator variable equal to one if the job posting displays a signal of financial strength (see Appendix Table H10). $\mathbf{X}_{ij}$ includes additional job-posting controls, accounting for the number of on-the-job activities, number of on-the-job opportunities, employer industry, employer establishment year, number of job prerequisites, an indicator for whether the job is not located in the respondent's primary chosen city, an indicator for work-from-home arrangements, and an indicator for whether the employer is a multinational. $Strata\ FE$ are binary indicators for each combination of respondent education level and preferred professional area. Robust standard errors are reported in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

TABLE A18. Heterogeneity in Responsiveness to Posted Wages

	High Education (1)	Female (2)	CLT Regime (3)	Large Firm (4)	Collective Bargaining (5)
Ln(Wage) Interaction	−0.075	−0.126**	0.011	−0.022	−0.034
	(0.055)	(0.053)	(0.051)	(0.044)	(0.056)
Ln(Wage)	1.256***	1.255***	1.196***	1.214***	1.208***
	(0.045)	(0.034)	(0.047)	(0.034)	(0.028)
ESG	0.083***	0.084***	0.080***	0.084***	0.084***
	(0.020)	(0.020)	(0.020)	(0.020)	(0.020)
Nonwage Amenities	0.062***	0.062***	0.061***	0.062***	0.062***
	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)
Financial Strength	0.012	0.013	0.012	0.012	0.012
	(0.032)	(0.032)	(0.032)	(0.032)	(0.032)
Observations	24,120	24,120	24,120	24,120	24,120
Individual FE	Yes	Yes	Yes	Yes	Yes
Strata FE	Yes	Yes	Yes	Yes	Yes
Job Posting Controls	Yes	Yes	Yes	Yes	Yes

Note.—This table reports coefficients for the following specification: $Interest_{ij} = \alpha + \beta_1 \ln(Wage_{ij}) + \beta_2\big(\ln(Wage_{ij}) \times Z\big) + \beta_3 ESG_{ij} + \beta_4 NWA_{ij} + \beta_5 FS_{ij} + \mathbf{X}_{ij}\mathbf{\Gamma} + \text{Strata FE} + \text{Individual FE} + \varepsilon_{ij}$. i denotes the individual and j denotes the job posting evaluated by individual i. Each column reports estimates from a separate regression, where Z is the characteristic listed in the column heading: Z_i in Columns (1) and (2), and Z_{ij} in Columns (3)–(5). In Column (1), Z_i indicates whether the respondent has completed college; in Column (2), whether the respondent is female; in Column (3), Z_{ij} indicates whether the job is offered under the CLT hiring regime; in Column (4), whether the employer is a large firm, defined as one with at least 10,500 employees, the median of the firm size distribution in our experiment; and in Column (5), whether the job belongs to a sector with high collective bargaining coverage. The main effects of the respondent-level characteristics in Columns (1) and (2) are absorbed by individual fixed effects. The regressions in Columns (3) and (4) include the main effects of the CLT hiring-regime indicator and the large-firm indicator, respectively, though these coefficients are not reported. The main effect of the high-bargaining-sector indicator in Column (5) is absorbed by the included employer-industry controls, since the indicator is defined at the industry level. $\ln(Wage)$ is the natural logarithm of the monthly wage displayed in the job posting. ESG is an indicator equal to one if the job posting displays at least one ESG sentence or ESG certification. NWA is equal to the number of nonwage amenities displayed in the job posting. FS is an indicator equal to one if the job posting displays a signal of financial strength. $\mathbf{X}_{ij}$ includes additional job-posting controls accounting for the number of on-the-job activities, number of on-the-job opportunities, employer industry, employer establishment year, number of job prerequisites, an indicator for whether the job is not located in the respondent's primary chosen city, an indicator for work-from-home arrangements, and an indicator for whether the employer is a multinational firm. $Strata\ FE$ are binary indicators for each combination of respondent education level and preferred professional area. Robust standard errors are reported in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Appendix B. Theory of ESG Adoption and Labor Market Equilibrium

In this appendix, we lay out a simplified version of the model with ESG as a nonwage amenity and its effects on labor market equilibrium so as to develop a series of formal theoretical results that rationalize our quantitative findings.

B.1. **Workers.** The labor market is comprised of a large number of workers indexed by i. There are two representative firms, labeled A and B. The total mass of skilled workers is $\bar{L}_S$ and the total mass of unskilled workers is $\bar{L}_U$. Each worker inelastically supplies one unit of labor. Workers have preferences over the wage W_{jg} and any ESG rating $E_j \in [0,1]$ of the firm. In the baseline economy, we assume that firms do not engage in any ESG activities.

In particular, we assume that the utility of worker i of type $g \in \{S,U\}$ at firm $j \in \{A,B\}$ is given by:

$$u_{ij} = \log W_{jg} + \log \Upsilon_g(E_j) + \tau \varepsilon_{ij}, \tag{B1}$$

where ε_{ij} is a Type-1 Extreme Value idiosyncratic shock and τ controls the dispersion of idiosyncratic preferences. We normalize the ESG utility such that $\Upsilon_g(E_{jt}=0)=1$. The model therefore allows firms, fixing worker type g, to differ in their ESG activities and other nonwage amenities. It also allows workers of different demographic groups to value ESG and other nonwage amenities differently, and allows workers to have idiosyncratic preferences over firms, captured by the Type-1 Extreme Value shock. Motivated by our empirical results, we further assume that $\Upsilon_U(\cdot)=1$. That is, unskilled workers place no value on firm ESG activities.

Workers observe posted wages and firms agree to hire immediately any worker willing to work at that wage. Wages are allowed to be a function of the worker's skill type, but cannot be conditioned on the idiosyncratic taste shock ε_{ij}, which is private information to the worker. Workers thus choose the job that maximizes their utility given the posted wages. Standard logit math gives the probability

$$Pr(j(i)=j) = \frac{[W_{jg}\Upsilon_g(E_j)]^{1/\tau}}{\sum_{j'}[W_{j'g}\Upsilon_g(E_{j'})]^{1/\tau}} \tag{B2}$$

that worker i chooses firm j. Higher wages and more ESG increase the probability that a worker chooses firm j.

B.2. **Firms.** Firm $j \in \{A,B\}$ faces an isoelastic (value-added) production function:

$$Y_j = \Xi_j L_j^{1-\eta} \tag{B3}$$

where:

$$L_j = \sum_g A_g L_{jg} \tag{B4}$$

are the efficiency units of labor. Here, Ξ_j is firm TFP, A_g is the productivity multiplier of skilled/unskilled labor, and η is the return to scale in the production function. We normalize

$A_U = 1$. Without loss of generality, we further assume Firm B is more productive than Firm A, that is $\Xi_B > \Xi_A$. Firms engage in monopsonistic competition. When setting wages, firms ignore their impact on the overall market wage index. Under this assumption, each firm faces an upward-sloping labor supply curve for each skill group g:

$$L_{jg}(W) = \frac{\bar{L}_g \left[W_{jg} \Upsilon_g (E_j)\right]^{1/\tau}}{\Delta_g}, \tag{B5}$$

with overall market wage index:

$$\Delta_g = \sum_{j'} \left[W_{j'g} \Upsilon_g (E_{j'})\right]^{1/\tau} \tag{B6}$$

taken as given. Thus, the labor supply elasticity is given by $\sigma \equiv 1/\tau$. Under the assumption that firms set wages to maximize profits, the firm's first-order condition is:

$$(1 + \sigma_g) W_{jg} = \sigma_g (1 - \eta) \Xi_j A_g L_j^{-\eta}. \tag{B7}$$

That is, wages are marked down from marginal products of labor (MPL) according to the wedge $\sigma/(1+\sigma)$. Equilibrium wages are below MPL due to the idiosyncratic horizontal differentiation captured by the Type-1 Extreme Value preference shock. Due to this horizontal differentiation, a firm that is otherwise identical to another will not lose all of its workers if it reduces its wages. This creates individual firm market power in the labor market and results in markdowns from perfectly competitive wages.

Note that we do not model the ESG adoption decision of firm j. Instead, we take the ESG characteristics as given, allowing us to focus on their labor market equilibrium effects. In subsequent analyses, we will specifically examine how the labor market equilibrium is affected when higher-TFP firms engage in ESG activities, which is expected if there are fixed costs associated with ESG adoption. This is also the empirically relevant case, as our firm survey evidence and structural work demonstrate.

B.3. **Equilibrium and Worker Utility.** Equilibrium in the labor market constitutes worker decisions $j(i)$, wages W_{jg}, labor demand L_{jg}, and market wage indexes Δ_g such that:

(1) Workers optimize over firms according to their utility as reflected in equation (B2).
(2) Firms set wages W_{jg} optimally to maximize profits, taking the labor supply curve and market wage indices as given, as in equation (B7), and labor demand $L_{jg} = L_{jg}(W_{jg})$.
(3) Market wage indexes Δ_g as defined by equation (B6) are internally consistent and generated from worker optimal decisions.

Finally, total ex-ante worker welfare is given by the usual log-sum exponential:

$$U = \sum_g \bar{L}_g \tau \log \left[\sum_j \exp\left(\frac{\log W_{jg} + \log \Upsilon_g (E_j)}{\tau}\right)\right], \tag{B8}$$

reflecting the distribution of wages and ESG across firms.

B.4. **Theoretical Results.** We now develop three theoretical results to illustrate how ESG activities by firms can impact the labor market equilibrium. In particular, we examine how ESG activities might impact the allocative efficiency of worker sorting in terms of value added, the equilibrium wage differential between skilled and unskilled labor, and worker welfare. All formal proofs are in the appendix.

We first show that, relative to a baseline economy with no firm ESG activities, ESG by more productive firms can in fact correct for allocative distortions introduced by firm monopsony power, and thereby improve allocative efficiency and increase value added.

THEOREM **B.1.** *Suppose that Firm B is more productive than Firm A, that is $\Xi_B > \Xi_A$. Suppose also that Firm A has no ESG activities ($E_A = 0$). Then labor value added $Y = Y_A + Y_B$ is increasing in the ESG activities E_B of Firm B for sufficiently small values of E_B.*

Proof. See Appendix Section C.1. ■

Intuitively, due to the horizontal differentiation across firms, driven by workers' Type-1 Extreme Value idiosyncratic preferences, firms have monopsony power in the labor market. That is, workers are not perfectly elastic in their labor supply across firms. Consequently, there are MPL wedges in the baseline equilibrium without firm ESG activities. That is, the marginal product of labor for skilled and unskilled workers is not equalized across firms. Since efficiency in terms of value added requires the absence of MPL wedges, this implies that the baseline equilibrium is inefficient in terms of value added.

To understand this, suppose that there were no MPL wedges across firms. Since workers are paid a constant markdown of their marginal productivity of labor, wages for skilled and unskilled workers would be equalized across firms. However, workers would then be equally distributed across the two firms due to the Type-1 Extreme Value idiosyncratic shocks. Since Firm B is more productive than Firm A, this would imply a higher marginal product of labor at Firm B, which creates a contradiction. A similar argument shows that the MPLs cannot be higher at Firm A in the baseline equilibrium. If this were the case, wages would be higher at Firm A than at Firm B and more workers would work at Firm A. Yet, since Firm A is less productive than Firm B, the MPL would be lower at Firm A, again leading to a contradiction.

Thus, in the baseline equilibrium, MPLs are higher at Firm B. This implies that Firm B hires too few workers, both skilled and unskilled, relative to the labor allocation that would maximize value added. At the margin, relative to this baseline, an increase in ESG activities by Firm B corrects for this, as it reallocates skilled workers from Firm A to Firm B, reducing the equilibrium MPL wedge and increasing labor value added.

Another labor market equilibrium outcome of particular interest to both researchers and policymakers is wage inequality across demographics. Using our framework, we show that

an increase in ESG activities by the more productive firm not only generates higher value added, but also increases the equilibrium wage differential between skilled and unskilled workers. We have the following result:

THEOREM **B.2.** *Suppose that Firm B is more productive than Firm A, that is $\Xi_B > \Xi_A$. Suppose also that Firm A has no ESG activities ($E_A = 0$). Define the total wage bill of skill group g as $W_g = L_{Ag}W_{Ag} + L_{Bg}W_{Bg}$. Then the total wage differential $\Delta_{SU} = W_S - W_U$ between skilled and unskilled workers is increasing in the ESG activities E_B of Firm B for sufficiently small values of E_B.*

Proof. See Appendix Section C.2. ■

This result is subtle and may at first glance appear counterintuitive. Given the higher valuation of ESG by skilled workers relative to unskilled workers, one might suspect firms could offer lower wages to skilled workers while still attracting the same amount of skilled labor, thus compressing the wage differential. This, however, is partial equilibrium logic.

First, note from Theorem B.1 that labor value added increases in response to the ESG activities of Firm B due to greater allocative efficiency. It can further be shown that the total wage bill in the economy is a constant fraction $(1-\eta)\sigma/(1+\sigma)$ of value added. From this, we can conclude that the total wage bill of the economy must increase. It thus suffices to show that the total wage bill of the unskilled workers actually declines in response to a marginal increase in Firm B's ESG activities.

To this end, let us also observe from the previous theorem that the total effective labor rises at Firm B and declines at Firm A, which increases unskilled wages at Firm A and decreases them at Firm B. There are now two first-order effects on the total unskilled wage bill to consider in response to a marginal increase in Firm B's ESG activities. First, since unskilled workers do not value ESG, the marginal worker at Firm B switches to Firm A. Since the wages at Firm A are lower than those at Firm B in the baseline equilibrium due to the MPL wedge, this lowers the total wage bill at the margin.

Turning to the second first-order effect, unskilled wages decline at Firm B and rise at Firm A, which impacts the total wage bill of the inframarginal workers at the two firms. One can show that the increase (decrease) in the total wage bill of the unskilled workers at Firm A (Firm B) is proportional to the current wage W_j, with proportionality constant $\eta L_{jU}/L_j$ for $j \in \{A, B\}$, equal to the scale parameter multiplied by the ratio of unskilled labor to total effective labor. In the baseline economy, however, the ratio of wages is the same for skilled and unskilled workers, which implies that skilled workers work at Firm A with the same probability that unskilled workers work at Firm A, and similarly for Firm B. This further implies that the ratio of unskilled labor to total effective labor is the same at both firms, equal to $\bar{L}_U/(\bar{L}_U + A_S\bar{L}_S)$. Thus, the constants of proportionality are the same. Since wages are lower at Firm A in the baseline economy, the increase in the unskilled wage bill of the

inframarginal workers at Firm A is dominated by the decline in the unskilled wage bill at Firm B.

Thus, both first-order effects are negative, which implies that the total unskilled wage bill declines in response to a marginal increase in Firm B's ESG activities. Since the total wage bill increases, this implies that the increase in wages accrues to the skilled workers, increasing the wage differential between skilled and unskilled workers. Indeed, for the skilled workers, the marginal worker switches from Firm A to Firm B to take advantage of the nonwage amenities that ESG offers. In other words, the increase in the wage bill accrues to skilled workers since they are precisely the workers who respond to the introduction of ESG by migrating to the high-productivity, high-wage firm.

We finally turn to understanding the overall welfare impact of ESG activities on workers, who care both about the wages they receive and the nonwage amenities, along with the idiosyncratic match value to the firm. We show that at $E_A = E_B = 0$, the local gradient of worker welfare with respect to ESG reflects only the direct utility effect of ESG. Specifically, we have the following result:

THEOREM **B.3.** *Suppose that Firm B is more productive than Firm A. That is, $\Xi_B > \Xi_A$. Further assume that $E_A = E_B = 0$, so that neither Firm A nor Firm B is pursuing ESG activities. Then the local derivative of worker welfare with respect to increases in ESG is given by:*

$$\frac{dU}{dE_{j^*}} = L_{jS}\Upsilon'_S(0),$$

for $j^* \in \{A, B\}$.

Proof. See Appendix Section C.3. ■

Intuitively, in discrete choice settings, a version of the envelope theorem holds, such that the re-optimizing behavior of workers does not have a first-order effect on total worker welfare. From this, it follows that the local impact of an increase in firm ESG activities reflects only the direct utility effect $L_{jS}\Upsilon'_S(0)$ and the inframarginal effects of changes in the log wage. It is straightforward to show that the sum of these inframarginal effects is proportional to $\eta \sum_g (L_{Ag}/L_A - L_{Bg}/L_B)$. But this term is zero in the baseline economy since, as discussed above, the ratio of skilled/unskilled workers to total effective labor is constant across the two firms. Thus, the marginal increase in utility from increasing firm ESG activities, relative to the baseline economy, reflects only the direct effect.

Note that this result hinges crucially on the fact that the returns-to-scale parameter η is constant across firms. If this were not the case, then the effect would be proportional to $\sum_g (\eta_A L_{Ag}/L_A - \eta_B L_{Bg}/L_B)$, which would not be equal to zero if $\eta_A \neq \eta_B$. Intuitively, as labor reallocation occurs, wage gains at one firm come at the expense of the other. In the specific case where the returns to scale are constant across firms, the gains and losses exactly

offset.[32] Moreover, the sum of the inframarginal effects may not be zero in the presence of existing nonwage amenities, since then the distributions of skilled and unskilled labor across firms may differ.

[32]This is analogous to a result in urban economics showing that, absent spatial transfers, there are no welfare gains from reallocating workers across space relative to the free mobility equilibrium, even with agglomeration or congestion externalities, as long as the agglomeration/congestion spillover elasticity is constant across space. See, for example, Glaeser and Gottlieb (2008), Kline and Moretti (2014), and Fajgelbaum and Gaubert (2020).

Appendix C. Model Proofs

C.1. **Proof of Theorem B.1.** Note that labor value added is:

$$Y = \Xi_A L_A^{1-\eta} + \Xi_B L_B^{1-\eta},$$

so that:

$$\begin{aligned}\frac{dY}{dE_B} &= (1-\eta)\left[\Xi_A L_A^{-\eta}\frac{dL_A}{dE_B} + \Xi_B L_B^{-\eta}\frac{dL_B}{dE_B}\right] \\ &= (1-\eta)\left[-\Xi_A L_A^{-\eta}\frac{dL_B}{dE_B} + \Xi_B L_B^{-\eta}\frac{dL_B}{dE_B}\right] \\ &= \frac{1+\sigma}{\sigma}\left[W_{BU} - W_{AU}\right]\frac{dL_B}{dE_B}.\end{aligned}$$

At $E_B = 0$, this derivative will be positive if $W_{BU}/W_{AU} > 1$ and $dL_B/dE_B > 0$. To the first point, suppose that $W_{BU}/W_{AU} \leq 1$. Since the ratio of firm wages is the same for skilled and unskilled labor, we then have $W_{BS}/W_{AS} \leq 1$ as well. Then, $L_{Bg} \leq L_{Ag}$ for $g \in \{U, S\}$, which implies $L_B \leq L_A$. But then:

$$\frac{W_{BU}}{W_{AU}} = \frac{\Xi_B}{\Xi_A}\left(\frac{L_A}{L_B}\right)^{\eta} > 1,$$

which is a contradiction. Finally, suppose $dL_B/dE_B \leq 0$. Then, wages would increase at Firm B and decline at Firm A. Since workers (weakly) value ESG, this would increase the probability that workers choose Firm B, a contradiction.

C.2. **Proof of Theorem B.2.** We will show that for a sufficiently small increase in E_B, relative to $E_B = 0$, the total wage bill of the unskilled workers declines and the total wage bill of the skilled workers increases. First note that the total wage bill in the economy is:

$$\begin{aligned}W_S + W_U &= \sum_j (1-\eta)\frac{\sigma}{1+\sigma}\left(L_{jU} + A_S L_{jS}\right)\Xi_j L_j^{-\eta} \\ &= (1-\eta)\frac{\sigma}{1+\sigma}\left(Y_A + Y_B\right).\end{aligned}$$

That is, the total wage bill in the economy is a constant fraction of the labor value added. It suffices to show that the wage bill of the unskilled workers declines, since by Theorem B.1, value added in the economy increases and thus the total wage bill of the skilled workers must also increase. To this end, note that:

$$W_U = \sum_j (1-\eta)\frac{\sigma}{1+\sigma} L_{jU}\Xi_j L_j^{-\eta}.$$

We therefore have:

$$\begin{aligned}\frac{dW_U}{dE_B} &= \sum_j \frac{dL_{jU}}{dE_B}W_{jU} - \sum_j (1-\eta)\,\eta\frac{\sigma}{1+\sigma}L_{jU}\Xi_j L_j^{-\eta-1}\frac{dL_j}{dE_B} \\ &= \frac{dL_{AU}}{dE_B}\left(W_{AU} - W_{BU}\right) + (1-\eta)\,\eta\frac{\sigma}{1+\sigma}\frac{dL_A}{dE_B}\left(-L_{AU}\Xi_A L_A^{-\eta-1} + L_{BU}\Xi_B L_B^{-\eta-1}\right).\end{aligned}$$

From Theorem B.1, we know that $dL_B/dE_B > 0$, which lowers unskilled wages at Firm B. Since unskilled workers do not value ESG, this implies that $dL_{AU}/dE_B > 0$. Since $W_{AU} < W_{BU}$ at $E_B = 0$, also from the proof of Theorem B.1, it follows that the first term in the equation above is negative. The proof will therefore be complete if we can show:

$$-L_{AU}\Xi_A L_A^{-\eta-1} + L_{BU}\Xi_B L_B^{-\eta-1} > 0,$$

since $dL_A/dE_B < 0$. Now:

$$-L_{AU}\Xi_A L_A^{-\eta-1} + L_{BU}\Xi_B L_B^{-\eta-1} = L_{BU}\Xi_B L_B^{-\eta-1}\left[-\frac{L_{AU}}{L_{BU}}\frac{\Xi_A}{\Xi_B}\left(\frac{L_A}{L_B}\right)^{-\eta-1} + 1\right].$$

However, since the ratio of wages across firms is the same for skilled and unskilled workers, it is easy to see that $L_{AU}/L_{BU} = L_A/L_B$ when $E_B = 0$. Therefore:

$$\begin{aligned}-\frac{L_{AU}}{L_{BU}}\frac{\Xi_A}{\Xi_B}\left(\frac{L_B}{L_A}\right)^{\eta+1} + 1 &= -\frac{\Xi_A}{\Xi_B}\left(\frac{L_A}{L_B}\right)^{-\eta} + 1 \\ &= \frac{-W_A}{W_B} + 1 \\ &> 0,\end{aligned}$$

since $0 < W_A/W_B < 1$, which completes the proof.

C.3. **Proof of Theorem B.3.** We can write total worker welfare as:

$$U = \sum_g \bar{L}_g \tau \log\left[\sum_j \exp\left(\frac{\log W_{jg} + \log \Upsilon_g\left(E_j\right)}{\tau}\right)\right].$$

Taking the derivative at $E_A = E_B = 0$ gives:

$$\begin{aligned}\frac{dU}{dE_{j^*}} &= \sum_g \sum_j L_{jg}\left(\frac{d\log W_{jg}}{dE_{j^*}} + \frac{d\log \Upsilon_g\left(E_j\right)}{dE_{j^*}}\right) \\ &= L_{jS}\Upsilon_S'\left(0\right) + \sum_g \sum_j L_{jg}\frac{d\log W_{jg}}{dE_{j^*}},\end{aligned}$$

since $\Upsilon_U'\left(\cdot\right) = 0$ and $\Upsilon_S\left(0\right) = 1$. The second term is:

$$\begin{aligned}\sum_g \sum_j L_{jg}\frac{d\log W_{jg}}{dE_{j^*}} &= \sum_g \sum_j L_{jg}\left(-\eta\frac{d\log L_j}{dE_{j^*}}\right) \\ &= \sum_g \sum_j -\eta\frac{L_{jg}}{L_j}\frac{dL_j}{dE_{j^*}} \\ &= -\eta\frac{dL_A}{dE_{j^*}}\sum_g\left(\frac{L_{Ag}}{L_A} - \frac{L_{Bg}}{L_B}\right) \\ &= -\eta\frac{dL_A}{dE_{j^*}}\left(\frac{L_{AU} + L_{AS}}{L_{AU} + A_S L_{AS}} - \frac{L_{BU} + L_{BS}}{L_{BU} + A_S L_{BS}}\right) \\ &= -\eta\frac{dL_A}{dE_{j^*}}\left(\frac{L_U + L_S}{L_U + A_S L_S} - \frac{L_U + L_S}{L_U + A_S L_S}\right) \\ &= 0,\end{aligned}$$

since $L_{jU}/\bar{L}_U = L_{jS}/\bar{L}_S$ when $E_A = E_B = 0$, as discussed in the proof of Theorem B.2. This completes the proof.

Appendix D. Main Survey Questionnaire

Introduction Page.

Q1. Would you like to continue?
Yes; No.

Filtering Questions.

Q2. What is the highest level of education you have completed?
Completed high school and/or pursuing an undergraduate degree; Completed technical school (or professional training course); Completed college (or more).

Q3. In which state would you like to work?
List with 26 states and the Federal District.

Q4. In which city would you like to work?
List with the main cities of the state chosen in the previous question.

Q5. In which locations are you also open to working? (*optional*)
List of cities not chosen in the previous question.

Q6. In which of the following professional areas would you like to work?
List of professional areas, see Appendix Tables H1, H2, and H3 for the professional areas based on level of education.

Introduction Page 1.

Introduction Page 2.

Job Posting Rating.

Demographic Questions.

Prompt. **Thanks for rating the job openings! There are only a few questions left for us to finish!**

Remember that it is important that you complete the questionnaire, so we can offer you the best job openings available.

Q7. What gender do you identify as?
Male; Female; Prefer not to answer.

Q8. What age group are you in?
16 to 19 years old; 20 to 24 years old; 25 to 29 years old; 30 to 34 years old; 35 to 39 years old; 40 to 44 years old; 45 to 49 years old; 50 to 54 years old; 55 to 59 years old; 60 years old or older.

Q9. What ethnicity do you identify as?
Indigenous; White; Asian; Black; Mixed; Other.

Q10. At the moment are you:
Employed, but looking for a new job; Employed, just seeing the job market; Unemployed and looking for a new job; Unemployed, but just seeing the job market; Looking for my first job; Other (specify):

Q11. What is your level of education?
Completed middle school; Incomplete high school; Completed high school; Incomplete college; Completed college; Completed masters; Completed doctorate.

Q12. How was your academic performance in high school? You were among:
Top 1%; Top 10%; Top 20%; Top 50%; Bottom 50%; I'm uncertain.

Q13. What is your monthly salary range at your current job (or at your last job if you are unemployed)?
Less than R$1,000; Between R$1,001 and R$1,500; Between R$1,501 and R$2,000; Between R$2,001 and R$2,500; Between R$2,501 and R$3,000; Between R$3,001 and R$4,000; Between R$4,001 and R$5,000; Between R$5,001 and R$10,000; Between R$10,001 and R$15,000; R$15,001+; Other (specify):

Q14. What is your political affiliation?
Liberal; Moderate; Conservative; Prefer not to answer.

Q15. Do you agree that you will get a better job after taking part in this survey?
Completely disagree; Disagree; Neither disagree nor agree; Agree; Completely agree.

Q16. When you think about working for companies with Environmental, Social and Governance (ESG) practices in place, what are the main considerations that come to mind?
Open-ended response.

Conclusion.

Q. Conclusion. **Thank you very much for completing our survey!**
We are always looking to improve our surveys and would love to hear from you. Feel free to make any comments, criticisms, etc. about your experience in the box below.

Appendix E. 2026 Qualitative Survey

In May 2026, we conducted a qualitative survey of Brazilian job seekers to provide additional suggestive evidence on the mechanisms underlying our experimental results. In this appendix, we describe the survey sample and its structure.

We fielded the survey through the commercial survey company Dynata and obtained responses from 497 job seekers. Eligible respondents were residents of Brazil, at least 18 years old, and had either searched for jobs on online platforms or participated in a hiring process in the past 12 months. The survey was administered in Portuguese. The survey sample is broadly comparable to our experimental survey with Catho, but respondents are slightly more likely to be female (56% vs. 43%), are younger (37 vs. 41 years), are more likely to have a university degree (64% vs. 56%), and earn higher wages (BRL 5,686 vs. BRL 4,308).

The survey began with basic screening questions to determine eligibility. We organized the survey into four parts. Part A measures recall of sample job postings from our main experiment, broken down by job posting component. Part B elicits respondents' beliefs about how posted pay relates to the pay eventually negotiated during a hiring process. Part C asks how respondents interpret each of the ESG signals used in the main experiment and also asks them to interpret other nonwage amenities shown in the job postings. Part D measures respondents' priors about how often ESG content appears in real job postings and how they interpret postings that do not mention ESG. The survey ended with a set of sociodemographic questions. Below, we list the main questions from each of the four survey parts.

Survey Questionnaire (main questions).

Part A: Recall.

Respondents are shown three job postings, one at a time. After each posting, respondents answer the following questions:

- *Core posting elements.* "Thinking about the job posting you just saw, do you remember whether it included each of the following?" (A) a monthly wage; (B) information about benefits (e.g., health insurance, meal vouchers); (C) information about the work schedule or work arrangement (e.g., hours, remote work); (D) information about the company (e.g., industry, size); and (E) hiring or application stages (e.g., interview rounds, assessments). Options: (1) Yes; (2) No; (3) Not sure.
- *ESG-relevant elements.* "Do you remember whether the posting included either of the following?" (A) a company certification or seal, such as a quality or sustainability badge; and (B) a short sentence about company practices, such as environmental or social initiatives. Options: (1) Yes; (2) No; (3) Not sure.

Part B: Salary information in hiring.

- *Closeness of discussed wage to posted amount.* "How close was the pay eventually discussed during the hiring process to the amount shown in the posting?" Options: (1) The discussed wage was lower than the wage shown in the posting; (2) The discussed wage was about the same as the wage shown in the posting; (3) The discussed wage was somewhat higher than the wage shown in the posting; (4) The discussed wage was much higher than the wage shown in the posting; (5) Wage was not discussed during the hiring process.
- *Above-average wage.* "When a job posting offers a wage clearly above average for a similar role, what is your first reaction?" Options: (1) The employer is having a lot of difficulty hiring, and my chances of being hired are much higher; (2) The employer is having some difficulty hiring, and my chances of being hired are somewhat higher; (3) No clear reaction; (4) The employer is probably somewhat more selective, and my chances of being hired are somewhat lower; (5) The employer is probably much more selective, and my chances of being hired are much lower.

Part C: Interpretation of ESG and nonwage amenities.

C.1: ESG signals. We first show the five ESG signals used in the main experiment one at a time: an environmental sentence, a social sentence, a governance sentence, a B Corp certification, and a Great Place to Work certification. For each signal, we elicit both a closed-ended and an open-ended interpretation:

- *Closed-ended meaning.* For each ESG signal, participants were asked: "If you had to choose one, what does this mainly tell you about the company?" ("this" refers to the ESG signal shown immediately before the question). Options: (1) The company has broader commitments to the environment or society; (2) Employees at the company share similar values and ways of thinking; (3) The company offers better benefits or treatment for its employees; (4) The company is financially stronger or more stable in the long run; (5) This does not send me any clear message about the company.
- *Open-ended meaning.* In addition, for each ESG signal, participants were asked: "In one or two sentences, what does this information mainly tell you about the company, if anything?" We classify free-text responses using an LLM-based coding procedure. Each response is assigned to exactly one of six mutually exclusive categories: (i) broader environmental or societal commitments; (ii) shared values, coworkers, or cultural fit; (iii) better benefits or employee treatment; (iv) firm integrity or trustworthiness; (v) other (coherent answers that do not fit (i)–(iv), including financial-strength and marketing-only readings); and (vi) non-substantive responses.

C.2: Nonwage amenities. We also ask respondents to interpret common nonwage amenities using the same closed-ended categories as for the ESG signals:

- *Amenities.* Respondents were asked: "Now consider the following items that sometimes appear in job postings. For each one, if you had to choose one, what does it mainly tell you about the company?" Items: On-site daycare or childcare support; Extended maternity or paternity leave; Flexible working hours; Remote or hybrid work arrangement; Health insurance; Meal vouchers or food allowance; Career development or training programs; Wellness programs (e.g., gym membership, mental health support). Options: (1) Broader commitments to the environment or society; (2) Employees share similar values and ways of thinking; (3) Better benefits or treatment of employees; (4) Financially stronger or more stable; (5) No clear message.

Part D: ESG disclosure in job postings.

- *Frequency of ESG practices in postings.* "For job postings you have seen recently, how often do they mention a company's ESG practices, such as environmental or social initiatives?" Options: (1) Never: I have not seen this in any job posting; (2) Rarely: only in a few job postings; (3) Sometimes: in some job postings but not most; (4) Often: in many of the job postings I see; (5) Very often: in most or all of the job postings I see.
- *Closed-ended interpretation of silence.* "Suppose a job posting does not mention anything about ESG practices or certifications. In your opinion, what is the most likely reason?" Options: (1) The company does not have any ESG practices in place; (2) The company may have ESG practices, but does not consider them important enough to mention; (3) The job posting simply does not include that kind of information; it does not mean anything about the company; (4) It depends entirely on the company; there is no single explanation; (5) I am not sure.

Appendix F. External Validity

Although our experimental setting is the Brazilian labor market and we cannot carry out our experiment in multiple countries, in this appendix we explore the external validity of our results using existing cross-country survey data.

First, we use microdata from Alesina et al. (2018), a cross-country survey covering the United States and several European countries with comparable measures of economic beliefs and policy preferences. We regress prosocial policy preferences on an indicator for college completion, with detailed individual controls and country fixed effects. Table A14 finds that college-educated respondents express significantly stronger support for equality-of-opportunity policies, allocate a larger share of the budget to education, and express greater support for the estate tax.

Second, we use data from the World Values Survey, a large-scale international survey designed to be nationally representative for more than 60 countries. We focus on five concrete measures: donating to social causes, prioritizing environmental protection over economic growth, membership in environmental organizations, active participation in such organizations, and confidence in environmental protection movements. As reported in Table A15, college-educated respondents score higher on all five measures. We then interact the college indicator with a Brazil dummy to test whether the Brazilian relationship differs from that observed elsewhere; the interaction is statistically insignificant in every column. That is, the relationship between education and pro-social or pro-environmental preferences in Brazil is not systematically different from that observed in other countries.

Overall, this additional cross-country evidence suggests that the education-related differences in ESG preferences we document in Brazil are broadly consistent with patterns observed in other settings.

Appendix G. Match Delivery

To translate the experimental preferences into recommendations of real vacancies, we constructed a group-level scoring rule for active job postings on Catho's platform. Because data-protection restrictions prevented us from observing individual user identities, the rule was defined at the level of demographic groups rather than individual respondents. Groups were formed using gender, age, and completed college education, including observed combinations of these characteristics. For each demographic group (g), we estimated preferences over job-posting attributes that could be mapped into Catho vacancy tags, including ESG signals, work-from-home arrangements, contract type, and nonwage amenities. Specifically, we estimated:

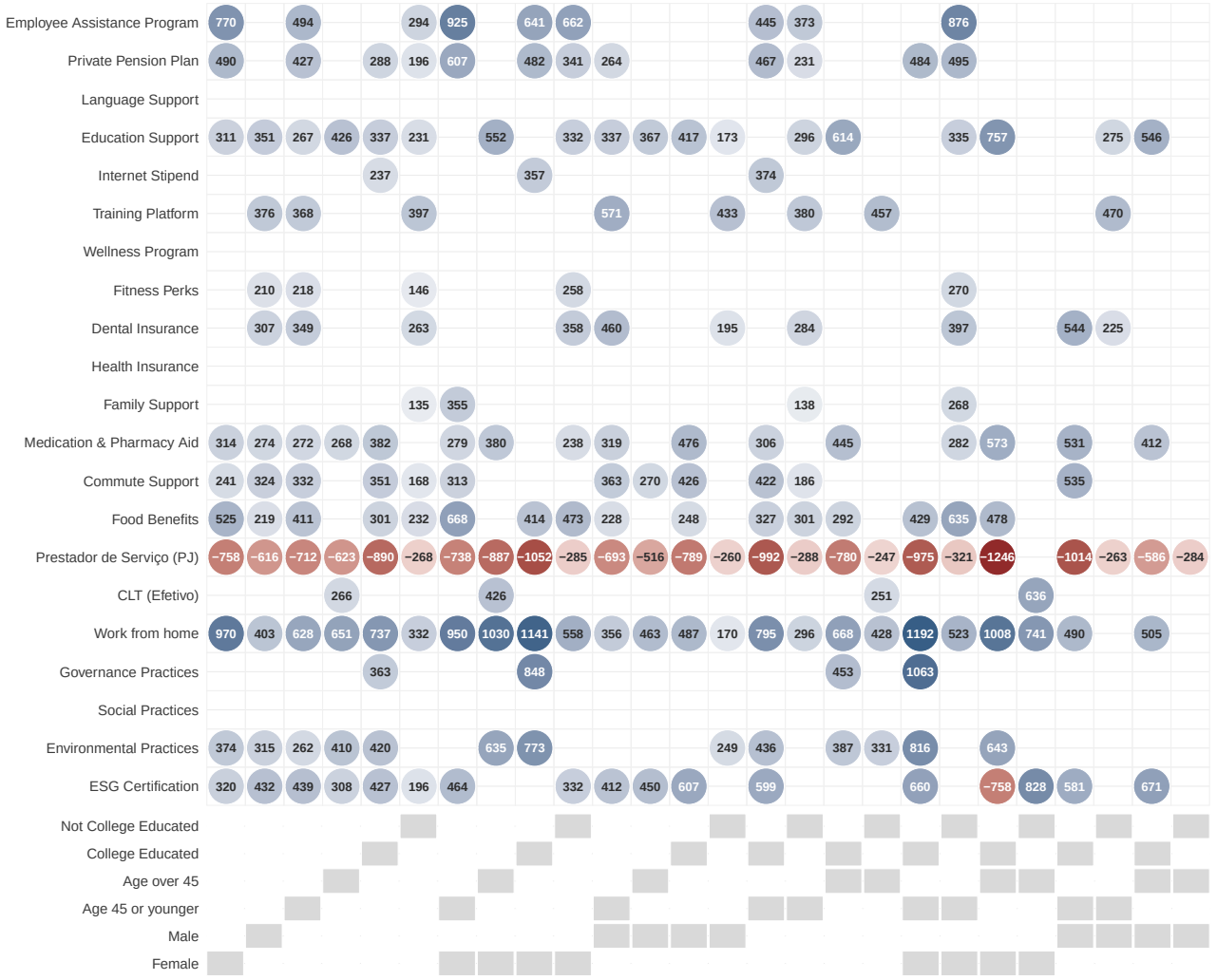
$$Interest_{ij} \;=\; x_j^{\top}\beta_g \;+\; \gamma_g\, Wage_{ij} \;+\; \text{Strata FE} \;+\; \text{Individual FE} \;+\; \varepsilon_{ij},$$

where x_j contains the relevant job attributes and $Wage_j$ is the posted wage. We converted statistically significant attribute coefficients into wage-equivalent values, $\left(\widehat{\beta}_{gk}/\widehat{\gamma}_g\right)$, setting insignificant coefficients to zero. For each posting j, we then constructed a group-specific Match Score:

$$\text{Match Score}_{jg} \;=\; Wage_j \;+\; \sum_{k \in x_j = 1} \left(\frac{\widehat{\beta}_{gk}}{\widehat{\gamma}_g}\right).$$

The resulting score was designed to rank vacancies within a user's city and professional area, thereby avoiding mechanical mismatches in location or occupation. We supplied Catho with the resulting preference matrix for use in ranking active postings. Appendix Figure G1 reports the wage-equivalent values by demographic group, and Appendix Table G1 lists the job-posting characteristics used in the scoring rule.

FIGURE G1. Preference Matrix



Note.—Circles mark job characteristics whose coefficients are statistically significant at the 10% level in group–specific OLS estimation with strata and individual fixed effects. Blue (red) dots indicate positive (negative) effects; blank cells indicate no statistically significant effect. Color intensity encodes magnitude: darker shades indicate larger values in absolute terms, while lighter shades indicate smaller values. Wage enters each regression in levels but is not reported in the figure. Integers inside dots report the wage–equivalent value (in BRL) computed as $\left(\widehat{\beta}_{gk}/\widehat{\gamma}_g\right)$. Columns are demographic groups observed in the data (positive–only combinations); the gray strip below the matrix shows which demographics define each column.

TABLE G1. Job Posting Characteristics Used for Recommendations

Category Used	Granular Characteristics Within Category
ESG Certification	See Appendix Table H13 for details on ESG certifications.
Environmental Sentence	Presence of sentences indicating environmental practices or commitments (Appendix Table H12 for details on ESG characteristics).
Social Practices Sentence	Presence of sentences indicating social practices or commitments (Appendix Table H12 for details on ESG characteristics).
Governance Practices Sentence	Presence of sentences indicating governance practices or commitments (Appendix Table H12 for details on ESG characteristics).
WFH Arrangements	See Appendix Table H21 for details on work-from-home (remote or hybrid) options.
Contract Type	CLT (Efetivo); Prestador de Serviço (PJ).
Food Benefits	Meal allowance; Food allowance.
Commute Support	Transportation allowance; Office bike storage and changing room.
Medication & Pharmacy Aid	Medication allowance; Pharmacy allowance.
Family Support	Pregnancy allowance; Daycare allowance.
Health Insurance	Medical assistance.
Dental Insurance	Dental assistance.
Fitness Perks	In-office gym; Physical activity support; Gym membership.
Wellness Program	Wellness program.
Training Platform	Training platform.
Internet Stipend	Internet allowance.
Education Support	Educational assistance program; Partnerships with educational organizations.
Language Support	Language course allowance.
Private Pension Plan	Private pension plan.
Employee Assistance Program	Personal support program.

Note.—This table lists the job-posting characteristics used in the recommendation algorithm and maps granular tags to the broader categories employed in the analysis. ESG certification and ESG sentences (Environmental, Social, and Governance) and WFH arrangements are summarized here (see Appendix Tables H13, H12 and H21 for further detail). A full list of nonwage amenities appears in Appendix Table H26. Wage is included in the regressions and used only to rank otherwise similar posts; see Appendix Table H5 for wage distribution details.

Appendix H. Job Posting Components

Table H1. Job Titles per Professional Area – Completed High School

Professional Area	Job Titles
Agriculture, Livestock and Veterinary	Livestock Worker; Junior Veterinary Assistant; Equipment Maintenance Assistant; Veterinary Consultant; Veterinary Assistant; Junior Agricultural Buyer; Store Advisor; Digital Agricultural Assistant; Lab Assistant; Livestock Handler; Seed Quality Tester; Sales Representative Junior Agronomist; Harvester; Agricultural Support Assistant; Agricultural Worker; Livestock Technician; Farm Manager; Scientific Advisor; Agronomist Assistant; Agricultural Foreman; Digital Agricultural Consultant; Junior Technical Sales Representative; Seed Genetics Specialist; Agricultural Extension Officer; Technical Manager; Agricultural Production Supervisor; Agricultural Manager; CLP Automation Supervisor
Architecture and Design	Designer; CAD Draftsman; Junior Designer; Architectural Designer Draftsman; Architectural Draftsman; Architect Assistant; Operational Inspector; Industrial Designer; Draftsman Assistant; Implementation Inspector
Business Administration	Claims Analyst; Tax Assistant; Administrative Assistant; Commercial Assistant; Warehouse Assistant; Receptionist; Security Inspector; Stockist; General Services Assistant; Office Assistant; Administrative Assistant; Human Resources Assistant
Commercial and Sales	Commercial Representative; Sales Representative; Sales Supervisor; Commercial Assistant; Store Supervisor; Sales Manager; Outside Sales Representative; Sales Assistant; Retail Sales Representative; Store Clerk; Telemarketing Sales Representative; Commercial Clerk; E-Commerce Assistant; Customer Service Analyst; Collection Assistant; After Sales Assistant; E-Commerce Relationship Assistant; Call Center Supervisor; Budget Sales Representative
Communications and Marketing	Web Designer; Video Editor; E-Commerce Assistant; Digital Marketing Assistant; Public Relations Assistant; Event Planner; Junior Marketing Analyst; Communications Assistant; Social Media Assistant; Event Assistant; Field Researcher
Engineering	Maintenance Mechanic; Mechanical Technician; Production Supervisor; Production Leader; Electronic Technician; Programmer; Industrial Production Assistant; Mechanical Maintenance Assistant; Mechanic Assistant; Engineering Assistant; Works Supervisor; Designer
Finance	Billing Clerk; Financial Assistant; Cashier; Financial Assistant; Administrative Assistant
Foreign Trade	Ship Inspector; Foreign Trade Assistant; Import Assistant; Export Assistant; Foreign Trade Assistant; Administrative Assistant
Hospitality and Tourism	Room Attendant; Receptionist; Reservations Attendant; Restaurant Server; Hotel Host; Host; Hotel Front Desk; Hospitality/Tourism Logistics Assistant; Reservations Assistant; Housekeeper; Event Assistant; Reception Supervisor; Reservations Assistant; Tour Guide; Hospitality and Events Administrative Assistant; Guest Services
Industrial	Installation Technician; Maintenance Technician; Foreman; Equipment Inspection Technician; Electromechanical Technician; Maintenance Mechanic; Production Assistant; Maintenance Technician; Inspection Assistant; Manufacturing Assistant; Automation Technician; Junior Production Operator; General Assistant; Industrial Maintenance Assistant
Information Technology	IT Solutions Analyst; Help Desk Support Analyst; IT Instructor; IT Support Technician; Systems Technician; User Support Technician; Typist; IT Coordinator; Project Assistant; External Technical Consultant; Database Administrator; Systems Engineer; Network Operator; Network Traffic Engineer
Legal	Office Assistant; Secretary; Junior Legal Assistant; Junior Tax Assistant; Junior Administrative Assistant; Legal Assistant; Corporate Tax Assistant; Corporate Assistant; Paralegal Assistant; Tax Assistant; Accounting Corporate Assistant; Junior Tax Analyst; Junior Attorney; Legal Administrative Assistant; Compliance Assistant
Social Services	Educational Advisor; Pastoral Assistant; Junior Educational Agent; Assistant Monitor; Child Caregiver; Social Project Assistant; Elderly Caregiver Junior Assistant; Junior Social Advisor; Social Responsibility Assistant; Social Educator; Project Coordinator; Sourcing Assistant; IMS Junior Assistant; Social Project Analyst
Technical	Electrical Assistant; Equipment Inspection Technician; Electromechanical Technical Assistant; Work Safety Technician Assistant; Technical Assistant; Junior Technical Sales Operations Analyst; Maintenance Coordinator; Production Supervisor; Operations Coordinator; Electrical Technician; Electromechanical Technician; Safety Technician
Telecommunications	Infrastructure Assistant; Support Assistant; Network Technician; Fiber Optic Technician; Junior Field Supervisor; Telecommunications Installer; Systems Assistant; Project Assistant; Operations Technician I; Junior Telecommunications Technician; Field Technician; Service Desk Assistant; Cabling Technician; Junior Technical Monitor; Junior IAT Technician; Junior Network Officer; Junior Infrastructure Analyst; Junior Network Analyst; Junior Telecommunications Analyst; Field Supervisor; Cloud Analyst; Junior Systems Analyst; Service Desk Coordinator; Project Analyst; Junior Operations Analyst; Network Systems Analyst I; IAT Technician I; Technical Support Supervisor; Cabling and Structured Process Analyst; Junior Network Analyst; Network Administrator; Infrastructure Consultant; Help Desk Supervisor; Senior Technician
Telemarketing	Telemarketer; Telesales Operator; Customer Service Operator; Service Assistant; E-Commerce Clerk; Social Clerk; Collection Assistant; Commercial Clerk; Public Clerk; Call Center Clerk; Collection Clerk; Customer Relationship Assistant Customer; Data Center Assistant Junior Customer Service Analyst; Telemarketing Supervisor; Customer Relationship Manager; Junior Relationship Analyst; Fulfillment Coordinator; Middle Office Junior Analyst; Junior Customer Service Analyst; Junior Customer Relationship Analyst; CDC Supervisor; SAP Manager; Full Bilingual Customer Service Agent; Back Office Analyst; Unit Customer Service Supervisor; Data Center Analyst
Transportation and Logistics	Delivery Driver; Transport Manager; Inventory Analyst; Project Consultant; Junior Buyer; Transport Analyst; Stockist; Warehouse Assistant; Distribution Assistant; Dispatch Assistant; Forklift Operator; Operations Assistant; Freight Analyst; Lecturer; Logistics Assistant; Tow Truck Driver; Parking Attendant; Transport Administrative Assistant; Loader Assistant; Junior Logistics Analyst; Warehouse Manager; Junior Strategy Analyst; Senior Logistics Analyst; Junior Operations Analyst; Purchasing Supervisor; Dispatch Supervisor; Supply Chain Analyst; Technology Administrative Assistant; Inventory Management Analyst; Inventory Analyst; Traffic Supervisor; Performance Analyst; Industrial Technical Buyer; Logistics Administrative Assistant

Note.—This table provides a list of job titles split by professional area shown to respondents that select "Completed High School" as their highest level of educational attainment. Respondents first select the professional area they would like to work in, and then select the job titles they would like to be shown. Respondents can select as many job titles within their chosen professional area as they like.

TABLE H2. Job Titles per Professional Area – Completed Technical School

Professional Area	Job Titles
Accounting	Accounting Technician; Accounting Assistant; Accounting Technical Support Analyst; Accounting Support Technician; Accounting Analyst; Tax Assistant
Automation, Audio and Video	Automation Technician; Audio Visual Technician; Audio and Lighting Technician; Radio Technician; Multimedia Audio Visual Technician
Chemistry	Chemistry Technician; Laboratory Assistant; Chemistry Technical Consultant; Chemical Technical Buyer
Construction	Building Technician; Building Maintenance Technician; Building Technician Assistant; Building Budget Technician; Pipeline and Track Integrity Technician
Electronics and Refrigeration	Electronics Technician; Electrical Technician; Electronics Technician; Electronics Technician Assistant; Electrical Maintenance Technician; Electrical Designer; Refrigeration Technician; Electronic Security Technician; Air Conditioning Supervisor
Information Technology and Systems Engineering	IT Technician; Systems Engineering Technician; Business Intelligence Specialist; Technical Support Analyst; Mobile Engineering Specialist
Mechanics, Electromechanics and Industrial Maintenance	Mechanical Technician; Electromechanical Technician; Industrial Maintenance Technician; Maintenance Mechanic; Industrial Mechanic; Plumber; Forklift Maintenance Technician; Preventive and Corrective Maintenance Technician; Instrumentation Technician
Networks and Telecommunications	Networks Technician; Telecommunications Technician; Assistant Telecommunications Technician; Fiber Optic Technician
Occupational Safety	Occupational Safety Technician; Occupational Safety Instructor; Training Instructor; Occupational Safety Technical Consultant

Note.—This table provides a list of job titles split by professional area shown to respondents that select "Completed Technical School" as their highest level of educational attainment. Respondents first select the professional area they would like to work in, and then select the job titles they would like to be shown. Respondents can select as many job titles within their chosen professional area as they like.

TABLE H3. Job Titles per Professional Area – Completed College

Professional Area	Job Titles
Agriculture, Livestock and Veterinary	Farm Manager; Agronomic Efficiency Assistant; Vet Assistant; Farm Material Maintenance Supervisor; Digital Agricultural Consultant; Agricultural Buyer; Seed Specialist; Senior Agronomist Sales Representative; Planning Supervisor; Vaccinator; Agricultural Partnership Manager; Manager Grain Sorter; Quality Analyst; Agricultural Monitor; Veterinarian; Agronomist; Market Development Manager; Agronomist; Breeding Manager; Biologist; Livestock Manager; Senior Digital Agricultural Analyst; Agricultural Manager; General Farm Manager; Soybean Production Manager; Market Development Agronomist; Agricultural Intelligence Specialist; Clinical Research Coordinator; Embryologist
Architecture and Design	Architect; Coordinating Architect; Architectural and Urban Planning Analyst; Designer; Planning and Budget Analyst; Architect Project Manager; Designer; Junior Architect; Industrial Designer; Data Architect; Project Developer; Junior Executive Project Supervisor; Specifier Architect; Senior Architect; Software Architect; Project Coordinating Architect
Business Administration	Commercial Representative; Financial Manager; Cost Supervisor; Commercial Manager; Business Analyst; Human Resources Coordinator; Quality Control Analyst; Quality Control Manager; Human Resources Analyst; Business Consultant; Administrative Manager; Store Manager; Merchandising Specialist; Project Manager; Project Analyst; Tax Accounting Supervisor; Financial Consultant; Accounting Assistant; Purchasing Analyst; Logistics Analyst
Commercial and Sales	Commercial Manager; Store Manager; Commercial Analyst; Commercial Supervisor; Sales Manager; Technical Sales Representative; Sales Supervisor; Commercial Consultant; Sales Executive; Account Manager; Product Manager; Business Development Representative; Customer Service Analyst
Communications and Marketing	Advertising Service; Trade Marketing Specialist; Digital Marketing Manager, Communications and Marketing Coordinator; E-Commerce Specialist; E-Commerce Consultant; Video Editor; Corporate Communications Advisor; Digital Marketing Specialist; Institutional Relations Analyst; Marketing Analyst; Graphic Designer; Digital Media Analyst
Engineering	Maintenance Manager; Project Engineer; Operations Manager; Production Scheduler; Planning and Budget Engineer; Control and Automation Engineer; Facilities Engineer; Engineering Coordinator; Product Engineer; Civil Construction Engineer
Finance	Banking Analyst; Tax Analyst; Financial Analyst; Insurance Technician; Financial Coordinator; Accounting Analyst; Accountant; Credit Analyst; Tax Analyst; Pricing Analyst; Treasurer; Internal Auditor; Financial Assistant; Administrative Analyst
Foreign Trade	Foreign Trade Assistant; Import and Export Analyst; Foreign Trade Analyst; Supply Chain Analyst; Exchange Trader; Export Supervisor; External Commercial Coordinator; Customs Clearance Analyst; Contract Lawyer; Foreign Trade Coordinator
Hospitality and Tourism	Travel Analyst; Tour Consultant; Head of Reception; Restaurant and Hotel Manager; Event Supervisor; Lounge Manager; Housekeeper; Hospitality Coordinator; Lodging Supervisor; Concierge; Hotel Administrator; Bar/Restaurant Manager; Agent Consultant; Leisure Travel Consultant; Hotel Event Producer; Hotel General Manager; Hotel Superintendent; Full Exchange Sales Consultant; Public Relations Analyst
Industrial	Industrial Coordinator; Print Coordinator; Operations Manager; Production Scheduler; Process Analyst; Industrial Automation Technician; Maintenance Manager; Senior Process Analyst; Full Process Analyst; Operations Manager; Mechatronics Technician; Production Leader
Information Technology	IT Coordinator; External Technical Consultant; Data Analyst; Programmer; Computer Support Analyst; Data Communication Technician; Telecommunications Network Engineer; SAP ABAP Analyst; SAP Business Consultant; Information Technology Consultant; IT Architect; Programmer; Big Data Analyst; Web Developer; Data Analyst; Full Stack Developer
Legal	Legal Assistant; Corporate Tax Advisor; Corporate Assistant; Tax Assistant; Administrative Assistant; Corporate Accounting Assistant; Legal Administrative Assistant; Senior Tax Advisor; Legal Supervisor; Legal Advisor; Labor Lawyer; Tax Lawyer; Civil Litigation Lawyer; Bidding Analyst; Consultant; Legalization Supervisor External; Supervising Lawyer; Real Estate Lawyer
Social Services	Full Partner Educational Advisor; Senior Pastoral Assistant; Social Educator; Educational Agent; Care Supervisor; Child Care Supervisor; Social Project Agent Senior Social Assistant; Full Social Advisor; Senior Project Coordinator; Full Sourcing Assistant; SGI Assistant; Elementary School teacher; Educational Project Coordinator; Social Responsibility Analyst; Full Social Project Analyst
Technical	Maintenance Coordinator; Production Supervisor; PCP Coordinator; Electronics Technician; Electromechanical Technician; Occupational Safety Technician; Civil Technician; Senior Technical Sales Representative; Heavy Machinery Maintenance Technician; Production Planner; Operations Manager; Production Engineer; Process Engineer; Factory Manager; Internal Controls Consultant; Industrial Designer
Telecommunications	Full Telecommunications Technician; Full Infrastructure Technician; Cloud Technician; Senior Network Technician; Systems Technician; Service Desk Coordinator; Technical Project Analyst; Operations Technician II; IAT Technician; Technical Support Supervisor; Cabling and Structured Processes; Network Administrator; Help Desk Supervisor; Expert Technician; Infrastructure Analyst; Senior Network Analyst; Network Engineer; Telecommunications Specialist; Technological Security Analyst; Full Stack Analyst; Full Systems Analyst; Senior Operations Analyst; Network Systems Analyst III; Full Project Manager; Full Network Analyst; Infrastructure Analyst - DevOps
Telemarketing	Full Service Analyst; Telemarketing Supervisor; Customer Relationship Manager; Junior Planning Analyst; CDC Supervisor; Relationship Analyst; Service Coordinator; Middle Office Analyst; Head of Customer Success; SAC Planning Consultant; SAP Manager; Billing Supervisor; Fulfillment Agent; Back Office Analyst; Full Customer Service Analyst; Full Planning Analyst; Senior CDC Supervisor; Billing Supervisor; Relationship Analyst; Producer Service Unit Manager; Business; Customer Service Project Manager; Full Middle Office Analyst; Full Customer Service Analyst; Full Investor Relations Analyst; Head of Customer Service; Senior SAP Manager
Transportation and Logistics	Junior Logistics Analyst; Purchasing Supervisor; Dispatch Supervisor; Transport Leader; Fleet Analyst; Full Buyer; Inventory Leader; Warehouse Manager; Transport Manager; Technology Administrative Assistant; Traffic Supervisor; Performance Analyst; Buyer Industrial Technician; Logistics Administrative Assistant; Supply Chain Director; Purchasing Manager; Planning Manager; Logistics Manager; Foreign Trade Coordinator; Production Planning and Control Coordinator; Logistics Controller; Senior Logistics Analyst; Logistics Distribution Center Manager; Operations Coordinator; Operational Excellence Supervisor; Operations Management Coordinator; Operations Manager; Reverse Logistics Manager; Fleet Manager

Note.—This table provides a list of job titles split by professional area shown to respondents that select "Completed College" as their highest level of educational attainment. Respondents first select the professional area they would like to work in, and then select the job titles they would like to be shown. Respondents can select as many job titles within their chosen professional area as they like.

Table H4. Geographic Location

States	City
Acre	Rio Branco
Alagoas	Maceió; Arapiraca
Amapá	Macapá; Santana
Amazonas	Manaus; Itacoatiara
Bahia	Salvador; Feira de Santana; Vitória da Conquista; Camaçari; Juazeiro; Itabuna; Lauro de Freitas; Teixeira de Freitas; Barreiras; Ilhéus; Jequié; Alagoinhas; Porto Seguro; Simões Filho; Paulo Afonso; Eunápolis; Santo Antônio de Jesus
Ceara	Fortaleza; Caucaia; Juazeiro do Norte; Maracanaú; Sobral; Crato; Itapipoca; Maranguape; Iguatu
Distrito Federal	Brasília
Espírito Santo	Serra; Vila Velha; Cariacica; Vitória; Cachoeiro de Itapemirim; Linhares; São Mateus; Guarapari; Colatina; Aracruz
Goiás	Goiânia; Aparecida de Goiânia; Anápolis; Rio Verde; Aguas Lindas de Goiás; Luziânia; Valparaíso de Goiás; Trindade; Formosa; Senador Canedo; Novo Gama; Catalão; Itumbiara; Jataí
Maranhão	São Luís; Imperatriz; São José de Ribamar; Timon; Caxias; Codó; Açailândia; Bacabal
Mato Grosso	Cuiabá; Várzea Grande; Rondonópolis; Sinop; Tangará da Serra
Mato Grosso do Sul	Campo Grande; Dourados; Três Lagoas; Corumbá
Minas Gerais	Belo Horizonte; Uberlândia; Contagem; Juiz de Fora; Betim; Montes Claros; Ribeirão das Neves; Uberaba; Governador Valadares; Ipatinga; Sete Lagoas; Divinópolis; Santa Luzia; Ibirité; Poços de Caldas; Patos de Minas; Pouso Alegre; Teófilo Otoni; Barbacena; Sabará; Varginha; Vespasiano; Conselheiro Lafaiete; Itabira; Araguari; Ubá; Passos; Coronel Fabriciano; Muriaé; Araxá; Nova Serrana; Ituiutaba
Pará	Belém; Ananindeua; Santarém; Marabá; Parauapebas; Castanhal; Abaetetuba; Cametá; Marituba; São Félix do Xingu; Bragança; Barcarena; Altamira; Tucuruí; Paragominas; Tailândia; Breves; Itaituba
Paraíba	João Pessoa; Campina Grande; Santa Rita; Patos
Paraná	Curitiba; Londrina; Maringá; Ponta Grossa; Cascavel; São José dos Pinhais; Foz do Iguaçu; Colombo; Guarapuava; Paranaguá; Araucária; Toledo; Apucarana; Campo Largo; Pinhais; Arapongas; Almirante Tamandaré; Piraquara; Umuarama; Cambé; Fazenda Rio Grande
Pernambuco	Recife; Jaboatão dos Guararapes; Olinda; Caruaru; Petrolina; Paulista; Cabo de Santo Agostinho; Camaragibe; Garanhuns; Vitória de Santo Antão; Igarassu; São Lourenço da Mata; Santa Cruz do Capibaribe; Abreu e Lima
Piauí	Teresina; Parnaíba
Rio de Janeiro	Rio de Janeiro; São Gonçalo; Duque de Caxias; Nova Iguaçu; Niterói; Belford Roxo; Campos dos Goytacazes; São João de Meriti; Petrópolis; Volta Redonda; Macaé; Magé; Itaboraí; Cabo Frio; Angra dos Reis; Nova Friburgo; Teresópolis; Barra Mansa; Mesquita; Maricá; Nilópolis; Rio das Ostras; Queimados; Itaguaí; Araruama; Resende; São Pedro da Aldeia; Japeri; Itaperuna; Barra do Piraí
Rio Grande do Norte	Natal; Mossoró; Parnamirim; São Gonçalo do Amarante
Rio Grande do Sul	Porto Alegre; Caxias do Sul; Canoas; Pelotas; Gravataí; Santa Maria; Viamão; Novo Hamburgo; São Leopoldo; Rio Grande; Alvorada; Passo Fundo; Sapucaia do Sul; Santa Cruz do Sul; Cachoeirinha; Uruguaiana; Bento Gonçalves; Bagé; Erechim
Rondônia	Porto Velho; Ji-Paraná; Ariquemes; Vilhena
Roraima	Boa Vista
Santa Catarina	Joinville; Florianópolis; Blumenau; São José; Chapecó; Itajaí; Criciúma; Jaraguá do Sul; Palhoça; Lages; Balneário Camboriú; Brusque; Tubarão
São Paulo	São Paulo; Guarulhos; Campinas; São Bernardo do Campo; São José dos Campos; Santo André; Ribeirão Preto; Osasco; Sorocaba; Mauá; São José do Rio Preto; Mogi das Cruzes; Santos; Diadema; Jundiaí; Piracicaba; Carapicuíba; Bauru; Itaquaquecetuba; São Vicente; Franca; Praia Grande; Guarujá; Taubaté; Limeira; Suzano; Taboão da Serra; Sumaré; Barueri; Embu das Artes; Indaiatuba; Cotia; São Carlos; Americana; Itapevi; Marília; Araraquara; Hortolândia; Jacareí; Presidente Prudente; Rio Claro; Araçatuba; Ferraz de Vasconcelos; Santa Bárbara d'Oeste; Itapecerica da Serra; Francisco Morato; Itu; Bragança Paulista; Pindamonhangaba; Itapetininga; São Caetano do Sul; Franco da Rocha; Mogi Guaçu; Jaú; Botucatu; Atibaia; Santana de Parnaíba; Araras; Valinhos; Cubatão; Sertãozinho; Jandira; Birigui; Ribeirão Pires; Caraguatatuba; Votorantim; Várzea Paulista; Itatiba; Tatuí; Barretos; Guaratinguetá; Catanduva; Salto; Poá; Ourinhos; Paulínia; Assis; Leme; Itanhaém; Caieiras; Mairiporã
Sergipe	Aracaju; Nossa Senhora do Socorro; Lagarto
Tocantins	Palmas; Araguaína

Note.—This table provides a list of all possible city location preferences selected by the respondent split by the 26 states and the Federal District in Brazil. Regardless of the state, we always display the following cities: São Paulo, Rio de Janeiro, Belo Horizonte, Fortaleza, Manaus, and Curitiba.

TABLE H5. Wages

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
Completed High School	1,300	1,550	1,850	2,055	2,387	4,000
Completed Technical School	1,500	2,450	3,000	3,000	3,550	4,500
Completed College:						
Agriculture, Livestock and Veterinary	2,000	2,400	2,800	3,755	4,562	12,500
Architecture and Design	2,000	2,400	2,800	3,755	4,562	12,500
Business Administration	2,000	2,400	2,725	3,639	4050	15,000
Commercial and Sales	2,000	2,450	2,900	3,920	4750	15,000
Communications and Marketing	2,000	2,387	2,700	3,520	3,850	15,000
Engineering	2,000	2,350	2,650	3,574	3,900	12,500
Finance	2,000	2,400	2,725	3,639	4,050	15,000
Foreign Trade	2,000	2,400	2,725	3,639	4,050	15,000
Hospitality and Tourism	2,000	2,288	2,550	3067	3,300	12,500
Industrial	2,000	2,300	2,600	3,222	3,462	12,500
Information Technology	2,000	2,600	3,350	4,737	6,112	15,000
Legal	2,000	2,400	2,800	3,492	3,900	15,000
Social Services	2,000	2,287	2,550	3,066	3,300	12,500
Technical	2,000	2,287	2,550	3,066	3,300	12,500
Telecommunications	2,000	2,350	2,650	3,574	3,900	12,500
Telemarketing	2,000	2,287	2,550	3,066	3,300	12,500
Transportation and Logistics	2,000	2,288	2,550	3,067	3,300	12,500

Note.—This table provides details on wage distribution by level of educational attainment. For each level of educational attainment, we report the minimum (Min.), the first quartile (1st Qu.), the median (Median), the mean (Mean), the third quartile (3rd Qu.), and the maximum (Max.). For level of educational attainment "Completed College," we report statistics for each professional area. The wage distribution was constructed based on the distribution of wages found on Catho's platform. We further refined the wage distribution by referencing the RAIS data and performing pilots.

TABLE H6. Industries

Categories	Industries
Agriculture, Livestock and Veterinary	Food; Chemistry; Beverages
Architecture and Design	Consulting; Real Estate Management and Development; Hotel, Restaurant, and Leisure; Transport and Infrastructure
Business Administration	Pharmaceuticals; Food, Paper and Forest Products; Consulting; Chemicals; Capital Markets and Investments; Banking and Insurance; Transport and Infrastructure; Beverages; Energy Equipment and Services; Real Estate Management and Development; Personal Services; Telecommunications; Metal and Mining; Automotive; Textiles and Apparel; Hotel, Restaurant, and Leisure
Commercial and Sales	Pharmaceuticals; Food, Paper and Forest Products; Consulting; Chemicals; Capital Markets and Investments; Banking and Insurance; Transport and Infrastructure; Beverages; Energy Equipment and Services; Real Estate Management and Development; Personal Services; Telecommunications; Metal and Mining; Automotive; Textiles and Apparel; Hotel, Restaurant, and Leisure
Communications and Marketing	Pharmaceuticals; Food, Paper and Forest Products; Consulting; Chemicals; Capital Markets and Investments; Banking and Insurance; Beverages; Personal Services; Telecommunications; Automotive; Textiles and Apparel; Hotel, Restaurant, and Leisure
Engineering	Transport and Infrastructure; Energy Equipment and Services; Real Estate Management and Development; Telecommunications
Foreign Trade	Pharmaceuticals; Food, Paper and Forest Products; Chemicals; Beverages; Energy Equipment and Services; Personal Services; Metal and Mining; Automotive; Textiles and Apparel.
Finance	Pharmaceuticals; Food, Paper and Forest Products; Consulting; Chemicals; Capital Markets and Investments; Banking and Insurance; Transport and Infrastructure; Beverages; Energy Equipment and Services; Real Estate Management and Development; Personal Services; Telecommunications; Metal and Mining; Automotive; Textiles and Apparel; Hotel, Restaurant, and Leisure
Hospitality and Tourism	Hotel, Restaurant, and Leisure; Real Estate Management and Development
Industrial	Pharmaceuticals; Food, Paper and Forest Products; Chemicals; Transport and Infrastructure; Beverages; Energy Equipment and Services; Personal Services; Metal and Mining; Automotive; Textiles and Apparel
Information Technology	Consulting; Capital Markets and Investments; Banking and Insurance; Energy Equipment and Services; Telecommunications; Automotive
Legal	Pharmaceuticals; Food, Paper and Forest Products; Consulting; Chemicals; Capital Markets and Investments; Banking and Insurance; Transport and Infrastructure; Beverages; Energy Equipment and Services; Real Estate Management and Development; Personal Services; Telecommunications; Metal and Mining; Automotive; Textiles and Apparel; Hotel, Restaurant, and Leisure
Social Services	Consulting; Healthcare
Technical	Pharmaceuticals; Food, Paper and Forest Products; Chemicals; Transport and Infrastructure; Beverages; Energy Equipment and Services; Personal Services; Metal and Mining; Automotive; Textiles and Apparel
Telecommunications	Capital Markets and Investments; Banking and Insurance; Energy Equipment and Services; Telecommunications
Telemarketing	Capital Markets and Investments; Banking and Insurance; Personal Services; Telecommunications
Transportation and Logistics	Pharmaceuticals; Food, Paper and Forest Products, Chemicals; Beverages; Energy Equipment and Services; Personal Services; Automotive; Textiles and Apparel

Note.—This table provides the range of industries by category. Column (1) presents the professional areas for respondents that select "Completed High School" and "Completed College" as their highest level of educational attainment. Upon selecting a professional area, we randomly select an industry for the synthetic job posting. Respondents that select "Completed Technical School" as their highest level of educational attainment are only shown industries within the "Technical" professional area.

TABLE H7. Firm Characteristics

Categories	Description
Establishment Year	*Unif*[1980,2012]
Number of Employees	*Unif*[500,20000] in intervals of 500.
Number of Countries	1 (50%), *Unif*[2,5] (50%)

Note.—This table provides details on firm characteristics. *Establishment Year* is drawn from a discrete uniform distribution ranging from 1980 to 2012. *Number of Employees* is drawn from a discrete uniform distribution ranging from 500 to 20,000 in intervals of 500. *Number of Countries* is the number of countries that the firm operates in. Firms can be domestic (i.e., operate only in Brazil) or multinational (i.e., operate in more than one country) with equal probability. We report the company descriptions of domestic (multinational) firms in Appendix Table H8 (H9).

TABLE H8. Introductory Sentence – Domestic Firm

Text
We are a [*sector*] company founded in [*year*] with more than [*number of employees*] employees across the country.
Founded in [*year*], our company operates in the [*sector*] sector. We have more than [*number of employees*] employees with operations across the country.
Since [*year*], our company has operated in the [*sector*] sector. Our [*number of employees*] employees are distributed across Brazil.
We operate in the [*sector*] sector from [*year*], with offices across Brazil, totaling [*number of employees*] employees.
Our company has operated across the country since [*year*]. We operate in the [*sector*] sector. Our company has more than [*number of employees*] employees.
We have more than [*number of employees*] employees with offices across Brazil. We have operated in the [*sector*] sector since [*year*].
Founded in [*year*], our company operates in almost every state in the country. We have [*number of employees*] employees who seek to transform the [*sector*] sector.
Founded in [*year*], we are a company in the [*sector*] sector with offices throughout Brazil and [*number of employees*] employees.
We are a company in the [*sector*] sector that was founded in [*year*]. We have more than [*number of employees*] employees to deliver excellent services.
Our company is a key player in the Brazilian market. With our [*number of employees*] employees, we have served as an exemplary company in the [*sector*] sector since [*year*].
Founded in [*year*], our company is one of the largest companies in the [*sector*] sector. We have [*number of employees*] employees across the country.
Our company has operated since [*year*] in the [*sector*] sector. Today, our more than [*number of employees*] employees are distributed across 20 states.
Our company operates in all states of the country. We employ the most qualified employees and already have more than [*number of employees*] employees representing our brand. Founded in [*year*], we are proud to be a key player in the country's [*sector*] sector.
Founded in [*year*], our company has offices in almost all major cities across the country. We are a company in the [*sector*] sector. Our team is made up of more than [*number of employees*] highly qualified employees.

Note.—This table provides the introductory sentences presented in the synthetic job postings of all domestic firms in Brazil. The sentences are randomly selected and each contains the following three variables: (1) *sector* indicates the given company's sector, (2) *year* indicates the establishment year of the given company, and (3) *number of employees* indicates the given firm's number of employees, as described in Table H7.

Table H9. Introduction Sentence – Multinational Firm

Text
We are a [*sector*] company, founded in [*year*], with more than [*number of employees*] employees across more than [*number of countries*] countries.
Founded in [*year*], our company operates in the [*sector*] sector. We have more than [*number of employees*] employees and we are present in more than [*number of countries*] countries.
Since [*year*], our company has operated in the [*sector*] sector. Our [*number of employees*] employees are distributed among [*number of countries*] countries.
We have operated in the [*sector*] sector since [*year*], with offices across more than [*number of countries*] countries, totaling [*number of employees*] employees.
Our company has a presence in more than [*number of countries*] countries and has operated since [*year*] in the [*sector*] sector. Our company has more than [*number of employees*] employees.
We have over [*number of employees*] employees across [*number of countries*] countries. Our company has operated in the [*sector*] sector since [*year*].
Present in more than [*number of countries*] countries, our company has [*number of employees*] employees. Since [*year*], we have operated in the [*sector*] sector.
The company was founded in [*year*] with the aim of bringing efficient solutions to the market. Our more than [*number of employees*] employees collaborate daily to put us at the top. Across all the [*number of countries*] that we operate in, we are a key player in the [*sector*] sector.
We have been working in the [*sector*] sector since [*year*]. Since then, we have expanded our operations to [*number of countries*] countries and serve as an international player in the [*sector*] sector.
Our team has [*number of employees*] employees across [*number of countries*] countries. Since [*year*], we have sought excellence in the [*sector*] sector.
Our more than [*number of employees*] employees, represented across more than [*number of countries*] countries, have positively contributed to the [*sector*] sector since [*year*].
Founded in [*year*], we have more than [*number of employees*] employees. We are a company in the [*sector*] sector that generates jobs across [*number of countries*] countries.
The company was founded in [*year*]. Since then, our growth in the [*sector*] sector has expanded globally. Last year, we opened new offices and now operate in [*number of countries*] countries, totaling [*number of employees*] employees.
Founded in [*year*], our company operates in the [*sector*] sector. Together with our [*number of employees*] employees, we work efficiently in all [*number of countries*] countries where we operate.
Founded in [*year*], our company operates in the [*sector*] sector. As of the end of last month, we have [*number of employees*] employees across [*number of countries*] countries.
Founded in [*year*], our company is currently a leader in the [*sector*] sector. Our [*number of employees*] employees are distributed across [*number of countries*] countries.
We have [*number of employees*] employees and we are looking to grow our team. Present in [*number of countries*] countries, our company was founded in [*year*] with the objective of transforming the [*sector*] sector.
A [*sector*] company that is proud of each of its [*number of employees*] employees. We are present in [*number of countries*] countries and, since our company's creation ([*year*]) we have been striving for quality.
We are a company founded in [*year*] and we are present in [*number of countries*] different countries. We have [*number of employees*] employees and we seek to be the best in the market and attract the best professionals who want to work in the [*sector*] sector.

Note.—This table provides the introductory sentences presented in the synthetic job postings of all multinational firms. The sentences are randomly selected and each contains the following four variables: (1) *sector* indicates the given company's sector, (2) *year* indicates the establishment year of the given company, (3) *number of employees* indicates the given firm's number of employees and (4) *number of countries* indicates the number of countries that the given company operates in, as described in Table H7.

TABLE H10. Financial Strength

Categories	Text
Profit	Our company prides itself on its strong financial performance. Last year, we recorded one of the highest profits in our history and among all companies operating within our sector in the last 5 years.
Credit Rating	Our company has a long history of financial responsibility, which is shown by our exceptional credit rating of *var_credit_rating*. This proves publicly that we are able to stay up to date with financial commitments made to our creditors.
Growth Outlook	Our company is proud of its sustained and substantial growth over the past year. We are highly optimistic about our future growth prospects and have committed to 30% annual growth over the next three years.
Bankruptcy Probability	Our company has a low risk of bankruptcy, as demonstrated by our exceptional credit rating of *var_credit_rating*. This publicly proves that we are able to keep up with the financial commitments made to our creditors.

Note.—This table presents sentences that signal financial strength through four channels: profit, credit rating, growth outlook, and bankruptcy probability. We randomize the employer's credit rating, designated by *var_credit_rating*, in the credit rating and bankruptcy probability sentences. For the credit ratings we randomize across, see Table H11. Synthetic job postings have a 0.1 probability of including sentences on financial strength.

TABLE H11. Credit Ratings

Option 1	Option 2	Option 3
Fitch AAA	S&P AAA	Moody's Aaa

Note.—This table presents employer credit ratings. If a signal of financial strength is included in the synthetic job posting, and the financial strength sentence chosen falls under the credit rating or bankruptcy probability category, we randomly select one of the three credit rating options presented above. See Table H10 for a list of financial strength sentences.

Table H12. ESG Signaling Sentences

Categories	Text
Environmental	Advancing environmental sustainability and reducing climate change are top priorities for our company. Our firm takes active measures to reduce our carbon footprint, and we pride ourselves on achieving carbon neutrality and on publishing our CO2 emissions every year.
	We are one of the largest Brazilian companies with a permanent investment program in environmental protection. Not only do we recycle 80% of the resources used in our activities, but also we maintain collection points for materials, such as batteries, in the cities where we operate.
	We are always looking for ways to give back to the natural environment. Our company is proud to take responsibility for our land footprint by committing to permanently protect and restore more land than we use by 2025.
	As part of our commitment to a zero waste future, our firm has eliminated single-use plastics in all product packaging and offices.
	In order to save resources and increase the preservation of nature, we are fully transparent about our internal policies on energy consumption and environmental protection. Our company issues monthly public reports showing our energy consumption and other actions taken to preserve nature.
	To demonstrate our commitment to improved transparency with respect to environmental impact, we annually disclose a detailed report on the types of materials used in our activities, as well as their estimated impact on the environment and associated recycling rates.
	We strive to promote increased transparency in our operations and environmental impact. Thus, we publish a detailed breakdown of our annual water usage on our website.
	We believe in the importance of better understanding how we impact our surrounding ecosystem. We publicly disclose a detailed annual report quantifying our land footprint and associated impact on surrounding areas.
Social	Whether related to gender, race, ethnicity, class, age, or accessibility, our firm is committed to improving representation through targeted recruitment, development, and retention of our employees. As part of our commitment, we aim to increase representation of women in senior roles to at least 50% in up to three years.
	We want our employees to be successful and reach their full potential. At our firm, we support employee growth and professional development through a dedicated mentorship program where new hires are matched with senior managers who provide guidance and training.
	We are dedicated to transparency and accountability in our diversity and inclusion efforts. To this end, our firm publishes an annual diversity, equity, and inclusion report on the demographics of our workforce.
Governance	Our company has an internal zero tolerance policy with executive political involvement. To this end, we ensure that our executives and directors are not formally associated with any political party or organization.
	Our company has an internal zero tolerance policy for all forms of bribery, whether involving a government official or entity or any commercial party, such as a customer or supplier. To this end, we maintain a system of internal controls to prevent any improper or corrupt payments.
	Our company supports greater transparency and integrity in its government relationships. To this end, our activities with the government are subject to robust internal procedures, designed to ensure that our efforts are aligned both with our public policy priorities and with the law.
	Our company prides itself on conducting business with integrity and responsibility. We have a formal anonymous whistleblowing process in place and an internal code of conduct to identify and prevent misconduct.

Note.—This table presents ESG signaling sentences split by environmental, social, and governance categories. ESG signaling sentences are included in synthetic job postings with a probability of 0.2 for approximately 80% of the respondents and 0.5 for approximately 20% of the respondents. If included, we randomly select between one and two signaling sentences.

TABLE H13. ESG Certification

Categories	Text
B Corporation	As a B Corporation® Certified company, we pride ourselves on meeting the highest verifiable standards of social and environmental performance. The certification reflects our commitment to positively impacting our employees, the environment and the wider community.
Great Place to Work	Our goal is to promote a fun, creative and inclusive work environment. Our company is proud to be certified by Great Place to Work™, the global authority on work culture and employee experience.
Green Business Bureau	We are proud to have been recognized as a "Green Business Bureau Certified" company by the Green Business Bureau, which distinguishes companies that demonstrate their commitment to sustainability. This certification reflects our commitment to prioritizing and implementing sustainable business practices.

Note.—This table presents descriptive sentences of ESG certifications. Certifications are randomized across B Corporation, Great Place to Work, and Green Business Bureau certifications. ESG certifications are included in synthetic job postings with a probability of 0.1. If included, only one certificate is selected per job posting.

TABLE H14. Auxiliary Sentences – Firm Description

Text
We strive for excellence and always aim to provide the best experience for our customers.
Our qualified team offers a suite of product and service solutions.
We are recognized for our innovation and for transforming our sector.
Our main objective is to bring solutions to the market that guarantee the satisfaction of our customers.
We are recognized for excellence in management and quality of services.
We value our employees' ability to guarantee quality at all stages of our internal operations.
Our goal is to guarantee customer satisfaction and deliver the best products with quality, innovation and efficiency.
We strive for constant growth and are passionate about innovation.
We strive to do our best every day and are proud to be credited as one of the largest companies in our sector.
Innovation is embedded in our culture and we constantly look for ideas to improve our operations.
As one of the top companies in our sector, success is in our DNA.
As leaders in innovation, we seek to improve the future for our employees.
We aim to make people's lives easier and strive to satisfy our customers.
We strive for growth, and our success has made our company a key player in the market.
We believe that there is always room to innovate new solutions to improve everyday life.

Note.—This table provides auxiliary sentences translated into English that provide additional structure to the synthetic job posting. These auxiliary sentences provide general information on the firm and its business. We include one auxiliary sentence in the job posting if the job posting has neither an ESG signaling sentence nor an ESG certification.

TABLE H15. On-The-Job Opportunities

Categories	On-The-Job Opportunities
Mentoring and Training	Global mentoring program; Mentorships; Training in project management, communication, and strategic business topics; Training for personal development and well-being; Technical training; Various training courses; Personalized mentoring program; Mentoring program.
Personal Development	Technical, cultural, and interpersonal skill improvement; Develop knowledge about organizational strategy; Lectures on experience and development; Lectures on career and business; Feedback for career development; Enhance business capabilities; Opportunity to perfect your skills; Career development plan; Network across various sectors to gain market insights and assess future trends; Technical knowledge development; Opportunity for growth and learning; Opportunity to apply skills learned in the classroom; Workshops for technical development; Access to an individualized development path; Constant practical learning through participation in challenging projects.
Company Culture	Free expression of opinions without fear; Team-building activities, Full immersion in our corporate culture; Rotational positions; Support from HR throughout your journey with our company; Working at a global company.

Note.—This table presents types of on-the-job opportunities split by *mentoring and training*, *personal development*, and *company culture* categories. Each synthetic job posting will display 2–5 randomly drawn on-the-job opportunities with probability 0.5.

TABLE H16. On-The-Job Activities – Completed High School (1/2)

Professional Area	On-The-Job Activities
Agriculture, Livestock and Veterinary	Implement measures to protect plant health; Help organize and plant seedlings; Help administer serums and vaccines; Assist in artificial insemination tasks; Perform slaughterhouse services; Tame and train animals; Conduct maintenance on used equipment; Prepare, bag, transport and distribute feed; Carry out pest control of cereals stored in silos and sacks; Help carry out experimental research projects; Grow, plant, sow and harvest crops using appropriate tools and equipment; Handle various equipment intended for agricultural production; Tend animal herds; Shear, clean, and groom animals for shows, instructional purposes or other reasons; Maintain the paddocks, fields, stables, cages and pens that house animals; Collect animals for milking.
Architecture and Design	Fill out project checklists; Record changes to original designs; Draft diagrams of project characteristics; Evaluate work progress; Conduct project meetings with architects and clients; Examine production-related documents and projects submitted and received from the commercial sector; Analyze problems presented and propose appropriate solutions; Assist managers in creating and presenting projects; Provide technical assistance and advice; Prepare budgets; Prepare documentation on materials; Manage payments to suppliers, prepare contracts, analyze projects and provide general support; Prepare designs according to customer preferences; Maintain potential and current customer relations and guarantee the quality of services provided; Design interior projects with varying complexity; Develop spreadsheets and presentations.
Business Administration	Provide support for operational needs; Communicate with customers; Various administrative tasks; Customer service; Prospect new customers; Organize documentation; Develop internal control spreadsheets; Perform administrative activities; Perform activities relevant to the sector; Provide general services requested by superiors; Support the operations department; Answer customer questions; Perform HR support tasks; Support the administrative manager; Manage email correspondences.
Commercial and Sales	Schedule meetings with potential partners; Visit establishments to present our products; Participate in weekly performance monitoring meetings; Active customer acquisition; Customer service; Sale of products and services; Draft budgets; Follow up on product delivery; Communicate via telephone; Manage purchases and evaluate necessary documentation; Provide support to the sales team; Register new customers; Perform telephone sales; Research competitors and market trends; Execute tasks.
Communications and Marketing	Create art for website and social networks; Develop marketing campaigns and management reports; Develop digital and printed communication materials; Answer customer questions regarding products and purchasing policies; Create templates and presentations; Strategic content planning for digital media such as Instagram, Facebook and TikTok; Create educational content for customers; Manage results, projects, trends and marketing strategies; Optimize content based on good SEO practices and strategies; Collaboratively explore different solutions to determine the most effective business strategy; Manage and coordinate e-commerce projects; Identify opportunities for new digital products, services, and other solutions; Solve strategic and creative problems involving the business; Generate engaging and interesting content that appeals to the public and drives engagement; Analyze customer purchasing behaviors.
Engineering	Perform duties to support employees, contractors, and service providers; Give technical guidelines to supervisors and other employees; Monitor projects and locations of teams and services; Carry out a review of completed services to ensure they meet quality standards; Carry out updates, monitor production plans, and collaborate on production programming; Actively contribute to the design and assembly of devices and equipment to enhance productivity and improve the quality of manufactured products; Prepare analysis of performance reports; Analyze and monitor indicators; Classify projects according to defined criteria to support workflow; Identify expected benefits of project implementation; Request and administer evaluations and tests; Carry out cost and material assessments; Set priorities for improvement and innovation; Plan tasks in accordance with specified requirements; Organize and oversee delivery schedules for materials and installations, taking into account printing, finishing, and installation times and coordinating with other concurrent projects.
Finance	Perform financial-support tasks; Make payments and manage the internal payments system; Register and update receipts; Issue invoices; Carry out closing tasks and calculate taxes; Issue payment slips; Make various payments; Issue reports; Prepare and update spreadsheets; Record accounting entries; Assist in tax planning; Assist in the preparation of trial balances; Register and monitor products and services; Update price list; Update entries in ERP software.
Foreign Trade	Compose import memos; Track load receipt; Perform import tasks; Analyze products received; Issue supply orders; Analyze and register documents; Launch accounts payable in the payments system; Manage relationships with carriers; Update spreadsheets; Assist in the preparation of weekly reports; Assess the accuracy of documentation received; Provide support to the administrative manager; Service customers; Execute freight-forwarding tasks; Receive customer orders.
Hospitality and Tourism	Receive and serve customers cordially; Provide accurate information about firm services; Coordinate clean-up activities; Ensure the quality of services provided; Handle reservations; Perform daily closing activities; Monitor services provided; Issue follow-up reports on the status of services; Fulfill customer orders; Purchase various materials; Close the cashier; File financial information daily; Replace materials when necessary; Answer customer complaints; Prepare customer satisfaction reports.

TABLE H16. On-The-Job Activities – Completed High School (2/2)

Professional Area	On-The-Job Activities
Industrial	Assist in manufacturing operations; Perform daily field inspections; Help operate machinery such as lathes, grinders, boring machines, milling machines, and others; Oversee and execute maintenance tasks; Oversee the daily schedules of technicians, ensuring efficient use of visit time; Conduct input surveys; Install replacements for defective parts.
Information Technology	Conduct data collection; Test systems; Monitor system performance; Inventory software and hardware; Guide support areas; Consult technical documentation; Consult alternative sources of information about the systems used; Provide third-party support; Install and configure software and hardware; Provide technical support for customers; Evaluate new technologies; Propose new methods and techniques; Organize forums on local networks; Provide training on company programs; Prepare reports of problems that may occur.
Legal	Write amendments and assignments; Arrange purchase and sale agreements, payments, exchanges, and the transfer of rights; Follow up on processes; Document organization; Prepare the documentation layout; Adjust the documentation to the established layout; Review documents; Classify and organize documents; Publish the documents in the system; Document meeting minutes; Assist in the preparation of contractual minutes; Request documents on demand; Assist in customer prospecting; Produce documents and reports for presentations; Prepare research on the registration status in public agencies; Regularize tax status and obtain certificates from relevant agencies; Conduct tasks related to updating and regularizing registration, including research, record-keeping, debt forgiveness, and updates from public agencies; Request certificates from public agencies; Manage administrative processes with public agencies.
Social Services	Provide social services; Assist in reintegration activities; Attend to the needs of people associated with the role; Attend to the welfare of individuals for whom one is responsible; Document the weekly activities carried out; Welcome, assist, oversee, and instruct cohabitants on the rules and usage of the shared environment; Prepare and submit reports to the local government, in collaboration with social services and other duties relevant to the role; Participate in activities within the corporate social responsibility sector; Guide and accompany participants according to established programming and technical guidelines; Perform socio-educational activities; Monitor and keep all equipment working; Provide guidance and direction to individuals for whom one is responsible; Assist in the assembly and disassembly of mobile social units; Social approach with people.
Technical	Test, adjust and regulate machines; Assist with relevant projects in development; Produce project progress reports; Prepare documentation for presentation at weekly project meetings; Prepare budget control spreadsheets; Request funds and equipment necessary for the role; Maintain inventory control; Enter the necessary information to complete each service in the designated system; Register technical information; Solve problems and oversee solutions for projects; Conduct tasks for short-term and long-term technical activities per internal requirements.
Telecommunications	Visit data centers; Perform server and service backups; Monitor application releases for security updates; Maintain and update IT inventory; Maintain and configure corporate wireless internet; Service company employees; Provide remote and on-site support; Network configuration; Analyze servers; Install and configure Office package; Assess equipment performance; Assess operating systems; Provide telephone support; Register and route calls to relevant departments; Monitor and update calls from the company; Create service procedures; Install equipment; Install CCTV systems; Operate telecommunication equipment and systems; Perform measurement tests and repairs; Answer phones according to priority and impact criteria; Follow up and resolve incidents and alerts through the monitoring tool; Offer technical assistance and support to partner teams to ensure optimal performance; Be aware of team timelines; Check personal protective equipment and tools; Assemble and install cables; Operate test equipment; Conduct surveys for project preparation; Monitor 24-hour service team to analyze productivity and output quality; Service high-priority customers.
Telemarketing	Answer calls and emails from company representatives; Active telemarketing for prospecting customers; Schedule inspections with customers; Archive and file relevant documents; Manage inventory; Answer inquiries through the company website; Provide information to customers; Evaluate requests for office visits; Provide solutions to issues presented by customers; Record service information; Update company website with photos and videos; Participate in e-commerce operations with an emphasis on customer service, utilizing tools such as WhatsApp, phone, and email; Attend to the reception desk; Certify service providers and suppliers as needed; Visit customers regularly to monitor the use of services.
Transportation and Logistics	Participate in the distribution of supplies; Organize stock and distribution; Monitor operational performance indicators; Prepare control sheets; Assist in the planning of goods storage; Separate, send and receive materials; Assess and plan deadlines and priorities; Issue invoices; Participate in the development of studies of logistics alternatives; Prepare a monthly report of the activities carried out in the month; Request necessary materials; Receive and serve customers; Provide requested information to customers; Create spreadsheets with indicators to be used in presentations by management; Assist with the organization of the distribution center.

Note.—This table provides types of on-the-job activities split by professional area shown to respondents that select "Completed High School" as their highest level of educational attainment. Each synthetic job posting will display 3–5 randomly drawn on-the-job activities with probability 0.6.

Table H17. On-The-Job Activities – Completed Technical School

Professional Area	On-The-Job Activities
Accounting	Customer service; Launch invoices; Calculate taxes and prepare financial documents; Perform accounting entries and reconciliations; Provide accounting audit advice; Prepare accounting documents; Bank reconciliation; Prepare payroll; Perform office tasks; Prepare expenses; Help prepare trial balances and balance sheets; Close the balance sheet; Provide technical advice; Review trial balances and management reports; Analyze financial statements.
Automation, Audio and Video	Assemble, supervise and review panel assemblies; Provide remote and in-person support to customers; Supervise the technological space; Perform preventive maintenance activities; Monitor and control technological resources; Configure and install hardware and software; Follow up with assembly team on mechanical completion; Execution of tests and maintenance procedures; Install projection screens and audio systems for corporate rooms; Customer service; Install, cabling and infrastructure for automation; Install speakers; Install receivers; Work with audio and video equipment; Provide technical services, equipment installation and sound testing.
Chemistry	Sample collection and analysis; Process control; Quality control; Industrial process control; Work in accordance with safety standards; Follow routine procedures; Keep the work area clean and organized; Preparation, separation and preservation of containers; Planning and execution of projects; Prepare solutions and measurement equipment; Material characterization analysis; Work in the laboratory for the development of new products; Customer service; Prepare documentation for audits; Perform physical-chemical tests.
Construction	Oversee new construction and residential and commercial renovation projects; Technical visits; Budgeting; Building maintenance; Manage internal and contracted employees; Prepare management reports; Review projects; Manage schedules; Issue and monitor purchase orders; Support and plan tasks; Monitor inventories and assess needs for replacement; Assist the construction team; Guide construction workers; Inspect the service performed; Develop project documents.
Electronics and Refrigeration	Repair and install electronic devices; Develop electronic circuit devices; Preventive and corrective maintenance of equipment; Identify the causes of device defects; Replace electronic components; Install electronic security systems; Develop and execute electrical projects; Contract labor; Configure audio, video, automation and network equipment; Develop grounding system projects; Maintenance of electric motors; Perform technical assistance activities; Perform equipment programming; Test equipment; Customer service; Maintenance repair of electronic systems; Prepare technical drawings; Carry out technical visits; Develop reports; Install refrigeration and ventilation equipment; Determine materials and accessories needed, and install equipment; Carry out various tests on the systems.
Information Technology and Systems Engineering	Perform repairs; Verify network protection; Provide technical support to users; Perform IT activities; Install and maintain computers and notebooks; Troubleshoot internal systems; Manage functional networks; Manage Production Part Approval Process (PPAP); Assist in software development; Ensure compliance with information security guidelines; Identify and correct problems with electronic devices; Develop websites; Implement computerized systems; Configure corporate VPN access; Provide audiovisual support.
Mechanics, Electromechanics and Industrial Maintenance	Perform technical maintenance; Prepare drawings of mechanical projects; Provide exceptional customer service; Maintain and calibrate equipment; Conduct and oversee inspection and maintenance services; Develop and test equipment and machines; Evaluate operating conditions in equipment installation; Preventive maintenance and corrective measures; Provide market price quotations; Select and install electronic and mechanical materials and components; Prepare technical reports; Assembly of mechanical devices; Assembly of electrical infrastructure; Install and assemble equipment; Replace electrical parts and components; Install electric motors; General building maintenance; Document repair actions; Implement and monitor small automation upgrades on existing equipment.
Networks and Telecommunications	Implement computer systems; Carry out technical visits; Report procedures; Work in the telecommunications operational area; Install cabling, ducts and piping for the data network; Carry out repairs, exchanges and maintenance; Install devices; Assist and troubleshoot customer service issues for staff; Carry out preventive maintenance; Participate in the preparation of telecommunication projects; Diagnose and troubleshoot issues with telephone systems and infrastructure; Technically verify processes and services; Perform telecommunications equipment integration; Test routers; Preventive and corrective maintenance and programming in PABX.
Occupational Safety	Perform risk management; Inspect locations; Equipment installations; Monitor works in order to identify risk factors; Accompany casualties with the emergency room; Promote safety training at work; Develop accident prevention programs; Guide activities developed by contracted companies regarding health and safety aspects; Integrate negotiation processes; Promote vaccination campaigns; Record the occurrence of accidents; Participate in the adoption of technologies and work processes; Assist in the implementation of PCMSO, PCMAT, PPRA, and LTCAT health and safety programs; Supervise prevention plans of environmental accidents; Guide the Internal Commission for Accident Prevention (CIPA); Implement the necessary safety signs.

Note.—This table provides types of on-the-job activities split by professional area shown to respondents that select "Completed Technical School" as their highest level of educational attainment. Each synthetic job posting will display 3–5 randomly drawn on-the-job activities with probability 0.6.

Table H18. On-The-Job Activities – Completed College (1/2)

Professional Area	On-The-Job Activities
Agriculture, Livestock and Veterinary	Plan rational and scientific experiments related to the company's agricultural projects; Conduct target audience research; Manage herd breeding tasks; Ensure the health and well-being of animals; Contribute to the genetic improvement of animals; Supervise sustainability projects, such as reducing water consumption and proper waste treatment; Manage the processing of renewable biomass; Monitor pests; Develop fertilization methods; Conducting regular animal health check-ups; Propose solutions to problems; Manage the execution of tasks; Implement objectives and actions for the area; Participate in meetings with the board of directors; Plan and negotiate large contracts.
Architecture and Design	Design visual materials for digital platforms, including presentations, logos, branding, and digital advertising graphics; Create and finalize designs for digital media, such as logos, visual identities, and advertising materials, by reviewing content and preparing files for production; Design and create furniture for display promotions in retail spaces, including stores and kiosks; Prepare project drawings, define their characteristics and determine the stages of execution and other technical elements; Monitor project progress, conduct analyses to prepare results; Create technical drawings for the design and implementation of augmented reality training programs; Operate printers, monitor and test prototypes; Develop technical solutions in order to improve the description of existing products; Develop quantitative projects; Assist in the development of projects for property regularization, obtaining operating permits, renovation and execution of construction works; Create and maintain a list of materials needed for the project, selecting appropriate specifications and standards based on guidelines provided; Supervise the assembly of prototypes, evaluate the performance of each component and evaluate the final product; Lead the design process for projects by consolidating information from feasibility studies and creating detailed project plans; Develop and design interior spaces, including residential, commercial and office spaces, create detailed plans, select finishes and furnishings, and manage budget and purchasing for the project.
Business Administration	Routine administrative activities; Assist in administrative processes; Prepare spreadsheets; Prepare management reports; Manage team; Create reports and performance indicators; Prospect new customers; Coordinate administrative activities; Follow up on HR activities; Participate in planning meetings; Plan, execute and optimize sector tasks; Participate in the implementation of corporate projects; Plan training; Validate project pricing; Analyze reports.
Commercial and Sales	Present our products and services to large customers; Manage sales team; Responsible for the commercial department; Lead the expansion process; Define performance metrics; Coordinate commercial events; Create, structure and improve KPIs to manage the operations and performance; Prepare the cost structure and price formation; Establish goals; Solve conflicts; Conduct feasibility studies; Delegate sector activities; Plan and carry out training; Develop and implement sales plans; Monitor the sales team.
Communications and Marketing	Develop analytical panels to generate visibility and identify solutions; Develop and monitor marketing campaigns through all channels (email, SMS, portal, app, etc.); Manage maintenance of digital platforms, ensuring content is always up-to-date and consistent across platforms; Implement new tools that enhance the customer experience (such as online chat, BOT, etc.); Coordinate all aspects of content, from agenda definition, briefing for writers, project management, editing, and platform publishing; Develop sponsored links ad strategies, especially in Google Ads, making necessary adjustments to campaigns to improve performance and monitor metrics through web analytics tools; Measure and improve the effectiveness of digital marketing campaigns and internal marketing efforts, ensuring brand appeal and reinforcing the dissemination of our brand and culture; Collaborate with the team to develop and implement efficient solutions and strategies in various areas; Develop sales methods and results; Assist sales in creating new materials and strategies, and track performance by creating monthly reports on marketing results; Ensure a seamless customer experience on the brand's proprietary platforms, and create graphic materials for point-of-sale, magazines, promotional campaigns, digital media, and training materials; Design and execute social media, e-commerce, and paid media campaigns; Coordinate and participate in events, meetings, and committees on behalf of the company to represent the company's interests and objectives; Maintain relationships with service providers, industries, and suppliers to present new projects, proposals and businesses.
Engineering	Plan, organize, execute, and oversee projects in the field of civil construction; Conduct investigations and technical surveys; Generate various reports to provide and forward information and data related to its area of activity to the direct supervisor; Manage areas of the company and departments in accordance with pre-established standards and procedures; Provide and participate in training programs within their area of expertise; Prepare reports and technical reports in their area of expertise; Implement management of preventive, corrective, predictive, and autonomous maintenance; Perform and monitor the execution of product validation tests; Interpret technical standards relevant to the product, its maintenance, clarify doubts in its application, and train multipliers for understanding and implementation; Implement continuous improvement efforts to optimize processes and resources; Analyze and design systems and highly complex applications, determining graphical interface, ergonomic navigation criteria, database structure, and program coding; Choose work resources, such as system development methodologies, programming languages, and development tools; Monitor the execution of services using performance indicators, assess performance and make decisions; Responsible for the development of new components and systems, from initial draft to launch, ensuring project requirements following engineering standards, procedures and techniques; Perform review of technical drawings; Carry out and supervise production plans, collaborate with production programming and control, and determine productivity index.
Finance	Apply statistical and procedural methods that contribute to the detection of fraud; Support the structuring of complex financial models; Financial analysis; Identify investment strategies; Prepare cash flow; Prepare billing projections; Carry out tax planning; Analyze options for new investments; Prepare financial statements; Prepare daily balance sheets; Manage the accounting department; Accounting and tax analysis; Prepare and review management reports; Financial modeling; Monitor changes in legislation.
Foreign Trade	International freight contracting; Monitor and control import processes; Monitor customs clearance; Conduct meetings in English; Follow up regularly to ensure deliveries within the stipulated deadlines; Evaluate the feasibility of import costs; Quoting international freight; Plan import activities and export; Analyze international market trends; Negotiate the best commercial conditions; Perform monthly control of indicators; Manage the import and export process; Ensure compliance with current legislation in the countries in which we operate; Evaluate the necessary framework according to Brazilian legislation; Define pricing and supply policies.

TABLE H18. On-The-Job Activities – Completed College (2/2)

Professional Area	On-The-Job Activities
Hospitality and Tourism	Establish the best rate policy according to supply and demand; Manage expenses; Perform corrective actions when budgets are not met; Monitor and manage the routine and compliance of all departments; Take care of marketing and advertising actions; Assist in hiring employees; Supervising and guiding the work of managers in each area; Managing contact with suppliers; Manage customer satisfaction; Cooperate with customers to determine their needs and provide advice; Use promotion techniques and prepare promotional materials for services; Participate in travel seminars to stay up to date with tourism trends; Enter data into our software and maintain client files; Maintain statistical and financial records; Assist in hiring and firing employees.
Industrial	Implement maintenance management; Contribute to improvement measures and investigate customer complaints and internal issues; Adopt corrective action for issues identified internally and externally, prioritizing customer complaints; Monitor risk-prone equipment, processes and operations; Collaborate with internal teams to assess risks and opportunities; Monitor productivity, quality, and people-management indicators; Conduct daily evaluations of the quality of work, providing positive and constructive feedback to employees; Develop and manage sector indicators; Inspect the company's facilities; Manage and oversee the team performing maintenance tasks, assembling and installing machines, components, and equipment; Create technical drawings to support projects; Monitor the quality of services using performance indicators, evaluate performance and make decisions; Perform critical analysis of processes to introduce improvements to management; Balance the needs of customers, employees, shareholders and suppliers; Ensure compliance with environmental protection regulations, and promote adherence to standards of hygiene and safety in the workplace.
Information Technology	Manage IT strategic processes; Plan and manage IT processes; Ensure that IT supports business processes; Manage the IT team, defining goals and deadlines; Prioritize and support IT projects; Define IT governance measures; Ensure mechanisms and tools for continuity of IT services in case of emergencies (e.g., a cybersecurity attack or system crash); Monitor and present IT indicators; Ensure information security; Request and manage IT resources; Develop programs in demand; Develop user documentation of company programs; Refine the architecture of existing systems; Generate new versions of company programs; Create unit tests.
Legal	Participate in projects to analyze legal and juridical regulations; Prepare opinions and tax reviews; Serve national and international clients; Direct field work and keep leadership updated on the development of projects; Participate in technical discussions to strengthen the company's stance on controversial matters; Support leadership in identifying new business opportunities; Support the sales team in prospecting meetings; Monitor the sector's internal processes; Draft service contracts; Plan, control and direct activities from the legal sector; Monitor notary processes; Distribution of processes; Draft contracts; Close contracts; Attend hearings.
Social Services	Responsible for educational assistance and guidance; Plan and execute workshops and social projects; Create plans for collaborating with relevant social service organizations; Responsible for actions that promote referral and reintegration into the labor market; Family monitoring; Preparation of reports and monitoring of communities in socially vulnerable situations; Develop and implement socio-educational activities, promoting recreational activities, stimulating community participation and ensuring proper use of equipment and materials; Mediate conflicts; Set priorities; Participate in internal and external audit processes, contribute to information availability or collection and recording of evidence; Act with integrity; Plan, prepare and manage educational projects; Manage deadlines and create schedules.
Technical	Consult on the implementation, management, and ongoing improvement of internal control processes; Develop projects; Provide technical support by monitoring project contracting processes; Prepare presentations on departmental projects; Manage the planning and execution of tasks related to operational activities, including development, support, and additional activities; Perform corrective and preventive maintenance; Develop processes and procedures for all technical activities; Develop documentation to support all field activities; Develop metrics and customer service activities; Work on medium and long-term projects according to internal demand; Support the commercial team in the execution of proposals, presentations and training.
Telecommunications	Monitor tasks on Windows servers; Support virtual servers; Support DLP (Data Loss Prevention) solutions; Map sensitive business data; Create DLP (Data Loss Prevention) control policies for mapped data; Ensure full operation of the infrastructure of company solutions; Create and review environmental policies, extract reports following the company's pre-established guidelines and best practices; Communicate with international support via email or conference call; Be aware of data security protocols; Responsible for supporting and maintaining physical and cloud servers, the company's network and communication infrastructure; Configure web servers with HTTPS; Manage system functions and take action to resolve issues by suggesting efficient solutions to restore service; Identify risks and implement solutions to secure technical environments and devices; Implement, support and monitor server resources, data centers, network assets and IT services; Manage computing environments and build architecture technology for information security; Analyze systems, identify vulnerabilities, map risks and implement solutions for the security of technical environments and devices; Prepare documentation on related operational procedures; Monitor delivery and quality; Creation and delivery of KPIs.
Telemarketing	Manage company representatives; Serve the customer reception department; Handle all company documents; Collect and manage necessary documents for customer purchase and processing; Propose products; Research and suggest the most commonly used technical terms in the market; Implement proposals in a service area by incorporating the relevant technical indicators for central operations; Develop strategies and create a portfolio for customers; Track and evaluate performance indicators for the region; Conduct market research and identify opportunities for expansion in the region; Prospect customers; Monitor revenue indicators; Analyze internal processes and implement solutions to improve productivity and quality; After-sales management; Organize the establishment of a call center for proactive customer prospecting; Carry out strategic planning to meet customer demands; Provide training on solutions and respond to customer inquiries as needed.
Transportation and Logistics	Plan all the company's logistics, from transport to the acquisition and storage of products; Manage material acquisition to meet company needs, considering factors such as quality, cost and service; Develop and control all stages of the production cycle, with a focus on financials; Assess and manage transportation resources, with the goal of maintaining efficiency; Coordinate the company's import and export team; Monitor production-related tasks; Prepare, plan and analyze the company's costs; Coordinate the execution of logistics plans; Choose new technologies for implementation; Make final hiring and employment decisions; Evaluate employee productivity; Coordinate programs to improve work efficiency; Participate in weekly meetings with management; Lead services provided to customers; Pricing services and projects.

Note.—This table provides types of on-the-job activities split by professional area shown to respondents that select "Completed College" as their highest level of educational attainment. Each synthetic job posting will display 3–5 randomly drawn on-the-job activities with probability 0.6.

Table H19. On-The-Job Activities – Sentences

Professional Area	On-The-Job Activities
Agriculture, Livestock and Veterinary	We seek a candidate who excels in a collaborative workplace, utilizing technology to bridge the gap between human progress and the natural world's performance, with knowledge of current agriculture and animal husbandry practices; In this role, you will use your knowledge and technique in order to contribute with solutions to improve productivity in the field.
Architecture and Design	We are looking for a multifaceted candidate with phenomenal design sensibilities who is highly imaginative and embraces experimentation, customers, and has a passion for bringing ideas to life; In this role, you will explore your knowledge and creativity alongside a team passionate about new ideas and constant innovation, contributing to the consolidation of our company's image and reputation.
Business Administration	In this role, you will support our company's day-to-day operations and partner with corporate and regional teams to help create reliable, scalable, and sustainable solutions; In this role, you will play a critical role on a strong and growing team within our company and leverage your team's strengths to support company initiatives and advance your career in a dynamic, fast-paced environment; In this position, you will join a team that creates value by utilizing best business practices to formulate and execute solutions to various challenges.
Commercial and Sales	We are looking for a passionate candidate to represent our company and its external partners through developing and implementing the sales team's day-to-day strategy and activities; Our company is looking for candidates to help us provide exceptional customer service; In this role, you will represent our company in customer interactions, providing services focused on understanding their needs and solving their problems.
Communications and Marketing	In this role, you will partner with our in-house communications and marketing departments to help develop and execute turnkey campaigns that address our target customers and their different personalities, pain points and needs; We are looking for a creative communications lead, dynamic and effective that can help implement complex and high-impact marketing programs and campaigns that change the consumer's mindset and drive action; Our ideal candidate is passionate about marketing strategies that aim to convey the values and proposals of our company, positively impacting our competitiveness through clear and honest communication.
Engineering	The ideal candidate for this role has a strong technical background and enjoys working on a team dedicated to the smooth running of our business in partnership with our engineering department to help coordinate all relevant technical and commercial initiatives; In our company, you will be part of a team of engineers focused on finding solutions for the implementation of systems aimed at improving productivity in order to offer our services to an ever-growing customer base.
Finance	Our company is looking for a qualified candidate to work alongside our finance team and perform a variety of accounting and finance tasks; We are looking for analytical professionals with a broad strategic vision to join our financial team and implement solutions.
Foreign Trade	In this role, you will be responsible for helping to ensure compliance with import and export rules and regulations, as well as helping to facilitate the timely release of goods through customs and related government agencies. We are looking for a candidate with outstanding organizational skills who can act as a liaison between factories, forwarders, brokers, and customers; Our ideal candidate is passionate about the challenges posed by e-commerce in an increasingly dynamic and complex world and is capable of understanding scenarios and the determining factors for our competitiveness.
Hospitality and Tourism	We are looking for a warm, welcoming, and articulate candidate to ensure that each client's experience is relaxing and effortless. If you love meeting new people, providing services and helping to build a great brand, we'd love to talk; In this role, you'll have the opportunity to work with a team dedicated to providing the best experience for our customers and contributing to solutions and services that enable them to get the most out of the time they invest in you.
Industrial	We are looking for a candidate to help manage production schedules and coordinate services to maximize productivity; In this role, you will help maintain a clean and orderly production environment and drive efficiency; Our ideal candidate is an enthusiast of search for organization and constant improvement of the production environment, ensuring the fulfillment of goals and the best safety practices.
Information Technology	We are looking for a talented IT professional with a strong technical background to help with our company's systems integrations and improve our customer-facing applications; In your role, you will help support our day-to-day IT operations and provide timely resolution to service requests and issues; Our ideal candidate is passionate about technology solutions and has the knowledge to implement solutions that improve the efficiency of internal systems and the quality of the products offered to our customers.
Legal	In this role, you will work closely with the legal and compliance team to provide legal and strategic support across the business; Join our team in developing more effective ways to ensure regulatory tasks and projects are completed with high level of efficiency and optimism to achieve the objectives of the entire company; We are looking for dedicated professionals who can use their technical knowledge in favor of creating legal protocols that reduce costs and increase the predictability of the impact of external changes on production of our company.
Social Services	In this role, you will be responsible for providing, coordinating and ensuring comprehensive care for patients; We are looking for a candidate to provide individual or group counseling services to help individuals and their families achieve effective personal development and adjustment; Our ideal candidate is someone who understands the complexities of human relationships and is passionate about listening, advising and proposing solutions for patients, always focused on ethics and improving individual well-being.
Technical	We are looking for a candidate with a strong ability to think creatively and solve technical challenges and limitations, as well as demonstrate excellent communication skills at all levels, both in the technical and creative dimensions; In this role, you will be responsible for providing technical support to users and internal customers of the company; In this role, you will be part of a team focused on the creation and application of technical solutions aimed at the efficiency and robustness of our production system.
Telecommunications	If you're ready to hone your customer service skills while gaining professional knowledge in the telecommunications industry, our team looks forward to working with you! conversation with potential customers in a transparent manner; In our company, you will work with a team specialized in creating and implementing innovations in telecommunication that increase the performance of our activities and contact with customers.
Telemarketing	A strong candidate will demonstrate excellent communication and customer service skills; Individuals in this role are responsible for placing business to business (B2B) phone calls to clients and scheduling meetings for outside sales teams; In this role, you will work seamlessly with the customer solutions team, from the initial service to the final contact with the proposals and solutions for your demand.
Transportation and Logistics	This position will serve as a link between carriers and our distribution network, ensuring efficient movement of inbound and outbound supply and the consistent implementation of our operational excellence; We are looking for professionals who do their best to develop and improve our supply area.

Note.—This table provides sentences describing on-the-job activities split by professional area. Each synthetic job posting will display one randomly drawn sentence describing on-the-job activities with probability 0.5.

TABLE H20. Workload Requirements

Workload Requirements
Monday to Friday from 8am to 6pm
Monday to Friday from 9am to 7pm
Monday to Friday from 10am to 8pm
Monday to Friday from 10am to 6pm
Monday to Friday from 9am to 5pm
Monday to Friday from 1pm to 9pm
40 hours a week, 5 days a week
Tuesday to Saturday from 10am to 6pm
Tuesday to Saturday from 9am to 5pm

Note.—This table provides types of estimated workload requirements in a given synthetic job posting. Each job posting displays one of the workload options with probability 0.7.

TABLE H21. Work-from-Home

Work-from-Home
Hybrid Work Model - You choose how many days of the week you want to work from the office.
Work model: 3 x 2 (3 days at home and 2 days at the office).
Work model: 2 x 3 (2 days at home and 3 days at the office).
In-person and remote work model (hybrid).

Note.—This table presents work-from-home arrangements. One work-from-home arrangement is randomly drawn from this list and included in the job posting with probability 0.3 for respondents that select "Completed High School" or "Completed College" as their highest level of educational attainment.

TABLE H22. Auxiliary Sentences – Job Opening

Text
Our company's recruitment process seeks to hire qualified professionals to join our team.
Our company is seeking innovative professionals to fill our open positions.
We are seeking top-notch professionals to fill our open positions.
Our hiring process targets professionals for our office in Brazil.
We seek to attract outstanding professionals.

Note.—This table provides auxiliary sentences translated into English that provide additional structure to the synthetic job posting. These auxiliary sentences provide information on the company's job opening and hiring process. One auxiliary sentence is always included in the job postings and is randomly selected.

TABLE H23. Job Prerequisites

Categories	Job Prerequisites
Work-Related Prerequisites	Make a real impact in the business world; Align actions and behaviors with the company's values and mission; Take initiative; Collaborative; Independent; Proactive and eager to work; Collaborative and flexible; Can-do attitude; Eager to learn and contribute to company culture; Team player; Dedication to completing projects successfully
Personal Attributes	Desire for real challenges; Energetic; Willingness to make it happen; Humble and ethical; Curious and inquisitive

Note.—This table presents job prerequisites split by work-related characteristics and respondents' personal attributes. Each synthetic job posting will display 2–3 randomly drawn job prerequisites with probability 0.5, with prerequisites drawn from *Work-Related Prerequisites* and *Personal Attributes* each with probability 0.5.

TABLE H24. Required Majors – Completed College

Professional Area	Required Majors
Agriculture, Livestock and Veterinary	Business Administration; Agricultural Science; Agricultural Engineering; Food Engineering; Fisheries Engineering; Forest Engineering; Veterinary Engineering
Architecture and Design	Architecture and Urbanism; Visual Arts; Cinema and Audiovisual; Social Communication; Design; Advertising
Business Administration	Business Administration; Actuarial Sciences; Accounting; Economics; Production Engineering; Statistics; Mathematics; Psychology; Advertising and Marketing
Commercial and Sales	Business Administration; Actuarial Sciences; Accounting; Economics; Production Engineering; Statistics; Mathematics; Advertising and Marketing
Communications and Marketing	Business Administration; Visual Arts; Social Sciences; Cinema and Audiovisual; Media; Design; Journalism; Language Arts; Advertising; Radio; TV; Internet (Audiovisual and Multimedia Communication); International Relations; Public Relations
Engineering	Biological Sciences; Aeronautical Engineering; Agricultural Engineering; Environmental and Sanitary Engineering; Cartographic and Surveying Engineering; Civil Engineering; Food Engineering; Bioengineering; Computer Engineering; Control and Automation Engineering; Fortification and Construction Engineering; Materials Engineering; Mining Engineering; Fisheries Engineering; Petroleum Engineering; Production Engineering; Telecommunications Engineering; Electrical Engineering; Electronic Engineering; Forest Engineering; Mechanical Engineering; Metallurgical Engineering; Naval Engineering; Chemical Engineering; Textile Engineering; Physics
Finance	Business Administration; Accounting; Economics; Engineering; Statistics; Mathematics
Foreign Trade	Business Administration; Economics; Production Engineering; Advertising and Marketing; International Relations; Public Relations
Hospitality and Tourism	Business Administration; International Relations; Public Relations; Advertising and Marketing; Social Communication
Information Technology	Computer Science; Economics; Computer Engineering; Network Engineering; Telecommunications Engineering; Statistics; Informatics; Mathematics; Information Systems; Engineering
Industrial	Food Engineering; Control and Automation Engineering; Materials Engineering; Production Engineering; Electrical Engineering; Electronic Engineering; Metallurgical Engineering; Textile Engineering
Legal	Law
Social Services	Social Services; Economics; Social Sciences; Psychology
Technical	Agricultural Science; Biological Sciences; Civil Engineering; Food Engineering; Control and Automation Engineering; Materials Engineering; Production Engineering; Electrical Engineering; Electronic Engineering; Metallurgical Engineering; Textile Engineering; Statistics; Physics; Geography; Geology; Mathematics; Meteorology; Information Systems
Telecommunications	Computer Engineering; Telecommunications Engineering; Network Engineering; Electrical Engineering; Electronic Engineering; Information Systems
Telemarketing	Business Administration; Agricultural Science; Actuarial Sciences; Accounting; Economics; Natural Sciences; Social Sciences; Production Engineering; Statistics; Physics; Mathematics; Psychology; Advertising; International Relations; Public Relations
Transportation and Logistics	Business Administration; Agricultural Engineering; Food Engineering; Control and Automation Engineering; Materials Engineering; Production Engineering; Electrical Engineering; Electronic Engineering; Mechanical Engineering; Metallurgical Engineering; Chemical Engineering; Textile Engineering

Note.—This table provides a list of majors split by seventeen categories, presented to respondents that select "Completed College" as their highest educational level.

TABLE H25. Hiring Stages

Categories	Hiring Stages
Stage 1 - Application	Application; Online application
Stage 2 - Online Assessments	Online assessment; Exam; English test; Logic test
Stage 3 - Other Assessments	Group interview; Group case study interview; Online interview; Business challenge; Dynamics; Business hack
Stage 4 - Final Interview	Final interview with managers; Final interview; Interview panel with our managers; Panel interview; Final interview with managers and human resources department

Note.—This table presents hiring stages split by four categories. Respondents who select "Completed High School" or "Completed Technical School" as their current educational level are only presented with synthetic job postings with two stages, *Stage 1 - Application* and *Stage 4 - Final Interview*. Respondents who select "Completed College" as their highest level of educational attainment are presented with synthetic job postings with all four hiring stages.

TABLE H26. Nonwage Amenities

Categories	Amenities and Benefits
Amenities	In-office gym; Training platform; Wellness program; Physical activity support; Gym membership; Educational assistance program; Personal support program; Office bike storage and changing room
Benefits	Meal allowance; Food allowance; Medication allowance; Transportation allowance; Pharmacy allowance; Pregnancy allowance; Medical assistance; Internet allowance; Daycare allowance; Dental assistance; Partnerships with educational organizations; Language course allowance; Private pension plan

Note.—This table presents nonwage amenities split by two categories. *Amenities* are nonmonetary, physical work environment incentives that improve employee experience. *Benefits* are nonwage compensations to support employee financial, health, and personal needs. Each synthetic job posting will display 2–4 randomly drawn amenities or benefits from *Amenities* and *Benefits*, with a probability of 0.3 from *Amenities* and 0.7 from *Benefits*.

Table H27. Criteria for Strong Performance by E(SG) Practice

Practice	Criteria
Water usage and conservation	High ESG performance in water usage and conservation involves adopting practices to conserve and manage water resources. This includes: – Regularly monitoring and recording water usage. – Conducting an analysis of the company's value chain, including suppliers, services, and materials, to identify material areas of water usage. – Setting specific reduction targets for reducing water footprint relative to previous performance (e.g., a 5% reduction of water usage from baseline year). – Implementing water conservation methods at the majority of the company's corporate offices and facilities, such as installing low-flow water fixtures. – Managing nonhazardous wastewater through on-site watershed management, wastewater reuse or recycling, on-site partial-reclamation, or off-site water treatment.
Hazardous waste management	High ESG performance in hazardous waste management involves adopting practices to ensure proper storage, treatment, and disposal of dangerous waste. This includes: – Eliminating hazardous waste and materials. If elimination is not feasible: – Regularly monitoring and tracking hazardous waste production (e.g., batteries, paint, electronic equipment, etc.) throughout the entire value chain. – Setting a target of zero hazardous waste. – Implementing written procedures for the safe storage, use, and responsible disposal of each hazardous material (e.g., chemicals, pesticides, and fertilizers). – Properly sealing, labeling, and storing hazardous materials in a locked area separate from regular business activities.
Greenhouse gas emissions management	High ESG performance in greenhouse gas emissions management involves adopting practices to monitor, record, and reduce greenhouse gas (GHG) emissions. This includes: – Regularly monitoring and recording GHG emissions. – Setting specific reduction targets relative to previous performance (e.g., a 5% reduction of GHGs from baseline year). – Conducting an analysis of the company's value chain, including suppliers, services, and materials, to identify material risk contributions of greenhouse gas emissions. – Purchasing certified carbon credits to offset some or all of the greenhouse gas emissions generated by the company and supply chain.
Energy efficiency	High ESG performance in energy efficiency involves adopting practices to optimize energy consumption. This includes: – Monitoring energy usage and setting targets based on monitored intensity (e.g., energy use relative to revenue or volume produced). – Conducting regular energy audits to identify areas of inefficiency. – Implementing energy-efficient equipment and lighting solutions in buildings and facilities. – Sourcing electricity from renewable energy sources, such as on-site renewables, other clean or renewable-based generators, or a municipal power grid that generates at least 10% of power from renewable sources. – Implementing energy-saving strategies, such as energy use monitoring, efficient appliances and lighting, and the use of renewable energy, in the majority of company buildings and facilities.
Sustainable packaging	High ESG performance in sustainable packaging involves adopting practices to minimize the environmental impact of packaging materials. This includes: – Conducting a formal assessment of packaging design and materials to identify opportunities to minimize environmental impact. – Using nontoxic, recyclable packaging designed to have less overall environmental impact than common alternatives. – Providing clear recycling instructions on packaging to promote responsible disposal.
Supply chain emissions management	High ESG performance in supply chain carbon emissions management involves adopting practices to regularly monitor and record greenhouse gas (GHG) emissions throughout the supply chain. This includes: – Establishing communication channels with suppliers for tracking and reporting GHG emissions. – Evaluating suppliers based on their commitments to reducing emissions. – Auditing and supporting suppliers in completing corrective actions to reduce emissions. – Conducting a comprehensive analysis of the value chain to identify material risk contributions of greenhouse gas emissions. – Setting concrete GHG emission reduction targets throughout the supply chain.
Waste management and recycling	High ESG performance in waste management and recycling involves adopting practices to regularly monitor, record, and reduce waste production. This includes: – Setting specific waste reduction targets relative to previous performance (e.g., a 5% reduction of waste to landfill from a baseline year). – Implementing comprehensive, facility-wide recycling programs with ongoing collection of at least all standard materials in the area in the majority of company facilities. – Promoting recycling and reuse of materials on-site with clearly marked bins for proper use. – Posting a written recycle/reduce/reuse policy in at least 80% of facilities. – Implementing a product/packaging reclamation and recycling or reuse program, either designed internally or in collaboration with a third party.
Sustainable transportation policies	High ESG performance in sustainable transportation policies involves adopting practices to prioritize the use of sustainable and low-emission vehicles for product transportation and distribution. This includes: – Utilizing clean and low-emission vehicles (e.g., hybrid, LPG, electric) to transport and distribute products. – Using strategic planning software to minimize fuel usage and optimize transportation routes. – Implementing a written shipping or distribution policy that prioritizes environmentally-efficient practices. – Prioritizing shipping methods with lower environmental impacts, such as sea or rail transportation over air shipment.

Note.—This table presents the criteria for achieving strong ESG performance split by selected environmental practice in the firm survey. Respondents first select two environmental practices and then are presented with the corresponding criteria for strong ESG performance.

Table H28. Criteria for Strong Performance by (E)S(G) Practice

Practice	Criteria
Inclusive hiring practices	High ESG performance in inclusive hiring practices involves adopting practices that promote inclusivity and equity throughout the hiring process. This includes: – Including a commitment to diversity, equity, and inclusion in all job postings. – Conducting anonymous or 'blind' reviews of applications or resumes without attaching names or identifiable characteristics. – Actively recruiting from organizations that serve underrepresented populations. – Conducting regular reviews of job description language and requirements to ensure they are inclusive and equitable.
Management of diversity, equity, and inclusion	High ESG performance in managing diversity, equity, and inclusion involves adopting practices to foster an inclusive and equitable workplace. This includes: – Implementing a formal, written nondiscrimination policy that covers at a minimum: gender, race, disability, political opinion, sexual orientation, age, religion. – Designating an individual or group explicitly responsible for overseeing diversity, equity, and inclusion efforts (i.e., a Diversity Manager or Inclusion Committee). – Offering comprehensive training to all employees on diversity-related topics. – Establishing voluntary employee resource or affinity groups. – Tracking workforce diversity through anonymous surveys. – Setting measurable diversity improvement goals that are reviewed by senior executives. – Conducting pay equity analyses by gender, race/ethnicity, or other demographic factors and implementing equal compensation improvement plans if necessary.
Professional development policies	High ESG performance in professional development policies involves implementing policies and opportunities to support employee professional growth. This includes: – Establishing a formal onboarding process for new employees. – Providing employees with regular ongoing training on core job responsibilities at least annually. – Implementing a formal policy to encourage internal promotions and hiring for advanced positions (e.g., posting job openings internally first). – Facilitating external professional development opportunities (e.g., conference attendance, online training). – Supporting employees in pursuing intensive continuing education credentials (e.g., college degrees, professional licenses) through reimbursements or programs.
Employee health and safety practices	High ESG performance in employee health and safety practices involves adopting practices that promote protection from work-related safety and health hazards. This includes: – Conducting annual safety and health training for all employees. – Transparently recording and sharing data internally on injuries and accidents. – Establishing a formal safety reporting system for employees to submit safety concerns. – Appointing a safety program representative or committee reporting to senior management. – Conducting thorough investigations into accidents and incidents according to documented standard procedures. – Conducting annual evaluations of the safety and health system with active senior management involvement.
Civic engagement & giving	High ESG performance in civic engagement and giving involves adopting practices that actively contribute to the development of the local community through philanthropic efforts. This includes: – Making financial or in-kind donations to nonpolitical causes. – Investing in the local community or providing pro-bono services. – Establishing corporate partnerships with charitable organizations. – Having a formal commitment to donations (e.g., 1% for the planet). – Offering programs to match individual employees' charitable donations. – Implementing screening practices for charitable contributions or impact measurement mechanisms for community investments.
Employee engagement and satisfaction	High ESG performance in employee engagement and satisfaction involves adopting practices to monitor and evaluate employee satisfaction and engagement levels. This includes: – Calculating and comparing employee attrition rates against industry benchmarks to gauge workforce retention. – Conducting regular surveys, at least annually, to assess employee satisfaction and engagement levels and comparing the results to industry benchmarks. – Outperforming industry benchmarks on employee attrition and satisfaction. – Implementing formalized feedback and complaint mechanisms that go beyond direct reporting lines to address employee concerns and improve company practices. – Conducting comprehensive reviews of employee feedback and complaints, at least every other year, with active involvement from employees.
Local sourcing and spending policies	High ESG performance in local sourcing and spending policies involves adopting practices that prioritize economic growth and sustainability within the local community. This includes: – Incorporating a written preference for purchasing from local suppliers into procurement policies. – Establishing formal targets or goals for the amount of local purchasing. – Maintaining lists of preferred local vendors for specific facilities. – Ensuring a significant portion (80%+) of the Cost of Goods Sold is spent within the country of operations, specifically from in-country registered companies or national citizens. – Allocating a substantial portion (60%+) of annual company expenses to independent suppliers local to the company's headquarters or relevant facilities.

Note.—This table presents the criteria for achieving strong ESG performance split by selected social practice in the firm survey. Respondents first select two social practices and then are presented with the corresponding criteria for strong ESG performance.

Table H29. Criteria for Strong Performance by (ES)G Practice

Practice	Criteria
Anti-corruption reporting and prevention	High ESG performance in anti-corruption reporting and prevention involves adopting practices to promote integrity and address potential related concerns. This includes: – Establishing a comprehensive written employee whistle-blowing policy with confidentiality provisions. – Communicating the anti-corruption reporting and prevention systems at least annually to relevant internal and external stakeholders. – Implementing mechanisms for continuous monitoring and internal assessments (e.g., internal employee self-evaluations, automated controls monitoring). – Ensuring that senior managers promptly implement necessary changes based on the findings of internal and external reviews. – Providing annual training on the anti-corruption system.
Financial controls	High ESG performance in financial controls involves adopting practices that promote sound financial management. This includes: – Establishing robust control activities, such as password protection systems that are periodically changed and tailored to different access levels based on employee positions. – Conducting fraud risk assessments at least annually and reporting internal control deficiencies to senior management. – Documenting lines of financial reporting, responsibilities, and limits for the authorization, approval, and verification of disbursements in writing. – Implementing and documenting financial controls that, at a minimum, cover controls for cash disbursement, accounts receivable, accounts payable, and inventory management.
Executive compensation and responsibility	High ESG performance in executive compensation and responsibility involves adopting practices to promote sustainable practices and accountability within the leadership team. This includes: – Integrating social and environmental considerations into executive roles and job descriptions. – Aligning executive compensation with social and environmental goals. – Reviewing social and environmental performance at the Board of Directors level. – Formally incorporating contributions to social and environmental goals in performance reviews.
Code of Ethics	High ESG performance in relation to a company's Code of Ethics involves adopting a formal written Code of Ethics that comprehensively documents policies around various governance-related topics. This includes: – Prohibiting bribes in any form, including kickbacks or gifts, on any portion of contract payments or soft dollar practices. – Establishing formal oversight policies and committing to public disclosure of direct or indirect contributions to political parties, politicians, lobby groups, charitable organizations, and advocacy groups. – Providing regular training on the Code of Ethics to the Board of Directors, newly hired employees, managers, and non-managerial employees, and other relevant stakeholders. – Promptly communicating any changes to the code to all relevant stakeholders.
Stakeholder communication and commitment	High ESG performance in stakeholder engagement and commitment involves adopting practices that emphasize proactive communication with stakeholders. This includes: – Implementing a formal stakeholder engagement plan or policy that includes identification of relevant stakeholder groups. – Establishing formal and regular processes to gather information from stakeholders (focus groups, surveys, community meetings). – Implementing formal procedures to address results from stakeholder engagement, with a designated individual or team responsible for appropriate follow-ups. – Reporting stakeholder engagement results to the highest level of oversight in the company.
Compliance management	High ESG performance in compliance management involves adopting practices to formally address potential material compliance breaches. This includes: – Promptly reporting breaches, including case details, to the Board of Directors. – Conducting thorough investigations into compliance breaches through independent parties. – Taking appropriate actions, such as dismissing or disciplining involved employees and terminating contracts with business partners in breach. – Making improvements to compliance documents and programs based on reported cases.
Mission statement	High ESG performance in relation to a company's mission statement involves: – Having a written corporate mission statement that is formally shared with employees or publicly available. – Including a commitment to a specific positive social impact (e.g., poverty alleviation, sustainable economic development). – Including a commitment to a specific positive environmental impact (e.g., reducing waste sent to landfills through upcycled products). – Including a commitment to serve a target beneficiary group in need (e.g., low-income customers).

Note.—This table presents the criteria for achieving strong ESG performance split by selected governance practice in the firm survey. Respondents first select two governance practices and then are presented with the corresponding criteria for strong ESG performance.